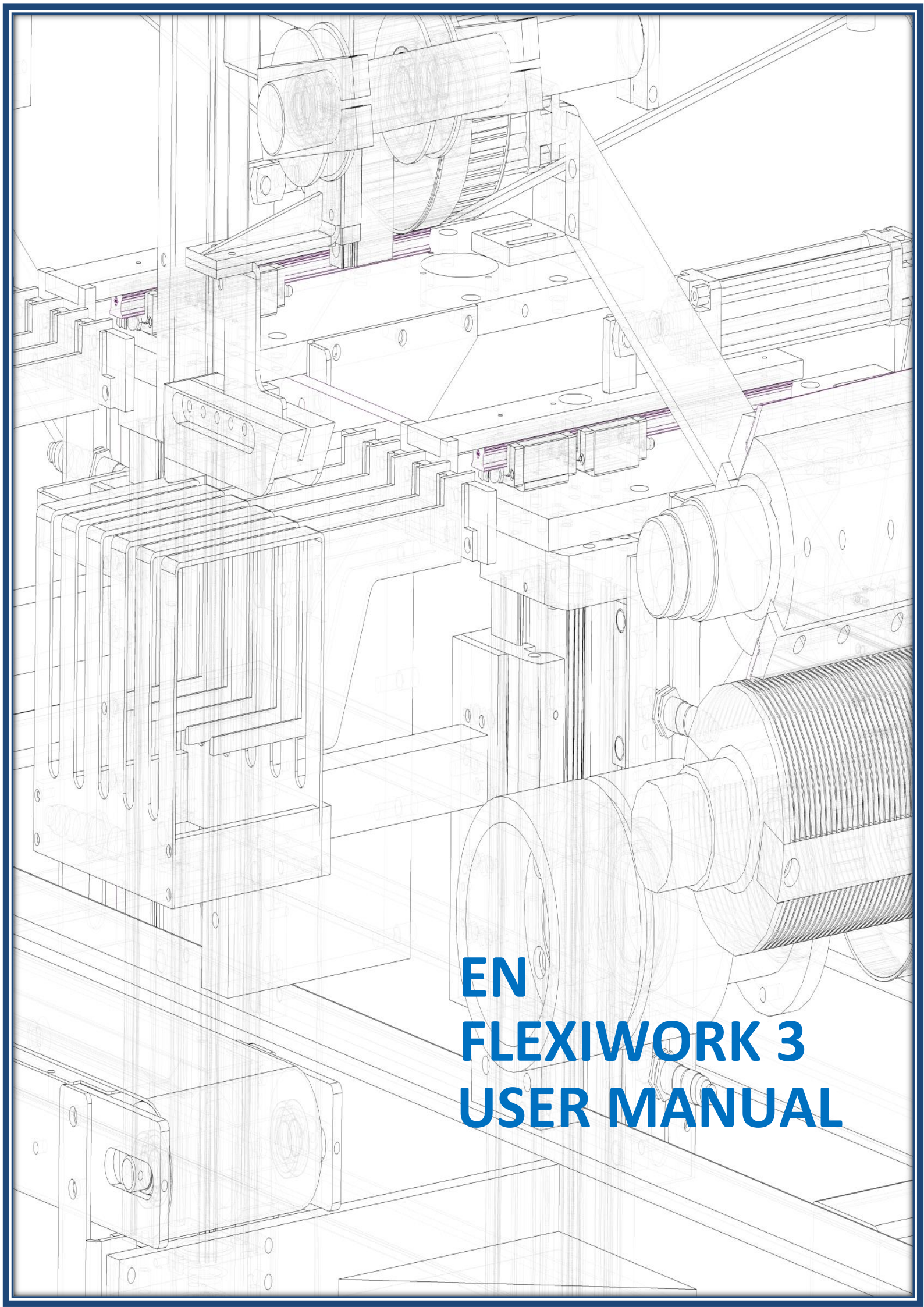


A detailed technical line drawing of a mechanical assembly, likely a textile machine. The drawing shows various components including a large motor at the bottom right, a complex frame with multiple horizontal and vertical beams, and several cylindrical rollers or spindles. A series of vertical bars or guides are visible on the left side. The drawing is rendered in a clean, technical style with blue lines on a white background.

EN FLEXIWORK 3 USER MANUAL

EC declaration of conformity***EG Konformitätserklärung***

We, as **KANSAN MAKINA VE TIC. AS** are a manufacturer settled at the address Capak Mah. 2594 Sok. No:6 Torbali - Izmir 35860 Turkey.

We hereby declare that the machine, whose name, serial number and model are indicated below, are in accordance with below mentioned Standards.

Product Details: Wet Wipe Converting Machine , model FLEXIWORK 3
Serial Number: 2021-

Machine is in conformity with the following standards and normative documents.

Machinery Directive 2006/42/EC; *Maschinenrichtlinie 2006/42/EG*

EMC-Directive 2014/30/EU; *EMV-Richtlinie 2014/30/EU*

Equipment and protective systems intended for use in potentially explosive atmospheres 2014/34/EC;

Low Voltage Directive 2014/35/EU;

Applied harmonized norms:

EN ISO 12100 Safety of machinery - General principles for design - Risk assessment and risk reduction

EN 60204-1 Safety of machinery - Electrical equipment of machines - Part 1: General requirements

EN ISO 13850 Safety of machinery - emergency stop - design principles

Documentation Officer: *name*... signature: ...

Place, Date:

Signatory and details of Signatory:

Signature:

USER MANUAL

This user manual belongs to **FLEXIWORK 3** / Automatic Wet Wipes Machines. It is written to provide adjusting machine healthy and effective,productive and safe usage of it.

Please directly contact with us if any technical problem is not clear in this user manual.

SALES DEPARTMENT

TEL : +90 (232) 853 9634

E-mail : info@kansanmak.com

AFTER-SALES SUPPORT

E-mail : support@kansanmak.com



Adres: Çapak Mah. 2594 sk. No. 6

Torbalı - İzmir 35860 Türkiye

Tel: +90 232 853 9634

Faks: +90 232 853 9107

E-posta: info@kansanmak.com

URL: www.kansanmak.com

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KANSAN

YOUR SINGLE SOURCE
FOR ALL YOUR WET WIPE MANUFACTURING NEEDS



Use the right equipment and realize your potential

Let your experience speak up with Kansan's
advanced machinery systems and turnkey
production lines for wet wipes



www.kansanmak.com

Who are we?

Kansan is the leading supplier of complete wet wipe manufacturing solutions. The story started in a little workshop back in 1992; Kansan has been proudly taking firm steps forward ever since. The key component to healthy and continuous growth has been not only producing state-of-the-art machines; but also, generating, providing unconditional support to all customers worldwide on their way to success. In Kansan we take pride and consider our customers' success our very own.

Global business is constantly evolving and forcing players from all levels to adapt. With proactive attitude and strategies, Kansan constantly monitors the dynamics, requirements and trends of global trade. This gives the opportunity to redesign our services and products accordingly. Kansan has always been able to provide its customers with flexible and tailor made solutions and always will be.

Decades of technical and commercial know-how, having a highly energetic and motivated team, Kansan is always ready to take on new challenges. We profoundly analyze our customers' demands and put them into practice. We strive to incorporate each and every bit of feedback and suggestion from our clients with our ever developing R&D process.



Your single source for all your wet wipe manufacturing needs...

One of Kansan's main competitive advantages is to offer complete and compact wet wipe production lines since 2008. By obtaining each machine from the same supplier, our customers enjoy receiving after sales service and technical support from a single source. This flexibility, adaptability and coordination make Kansan a reliable solution provider for any wet wipe producer, may it be a multinational organization or a new comer of wet wipe industry.

What does quality mean for Kansan?

We combine our distinguished craftsmanship with top quality components from premium suppliers around the world. For Kansan, quality is a comprehensive concept involved in all stages; ranging from material and component selection to after sales services. To ensure the utmost quality level; every single component is examined by different teams to determine if they are suitable for operation. We apply necessary technical modifications depending on our clients' special requests.

Machines are also tested for their performance, speed and productivity in harsh conditions by final quality control teams before delivery. An average test may take up to 5-7 days in accordance with the specifications of the machines. Moreover, our component providers run additional test procedures for their products on our machines.

Providing the best quality in a timely manner is the objective of whole Kansan team, hence every single project is handled like a new challenge to create a difference for the client as well as developing our skills and technical know-how.

Kansan owes its reputation also to its innovative approach and passion for high technology. In order to offer state-of-the-art machines, we put a heavy emphasis on R&D and strict quality control regulation

What we offer?



High Capacity: 4800 to 20000 wipes per minute according to model of the machines.

600 to 1000 cuts per minute for flat pack lines

Up to 700 cuts per minute for cross fold lines.

100 to 240 stacks per minute

Up to 130 packs per minute with single packing machine

Up to 120 lids per minute by single lid applicator machine.

Full Automation: Automatic splicing on single and parent roll unwind, full automated and synched machine operation both up and downstream

Online Production Quality Control: Folding control, clip cutting control, wetting precision control, die-cut punch detection-rejection, label position control detection-rejection, batch-code detection-rejection, on-lid label control and rejection, lid position control and rejection.

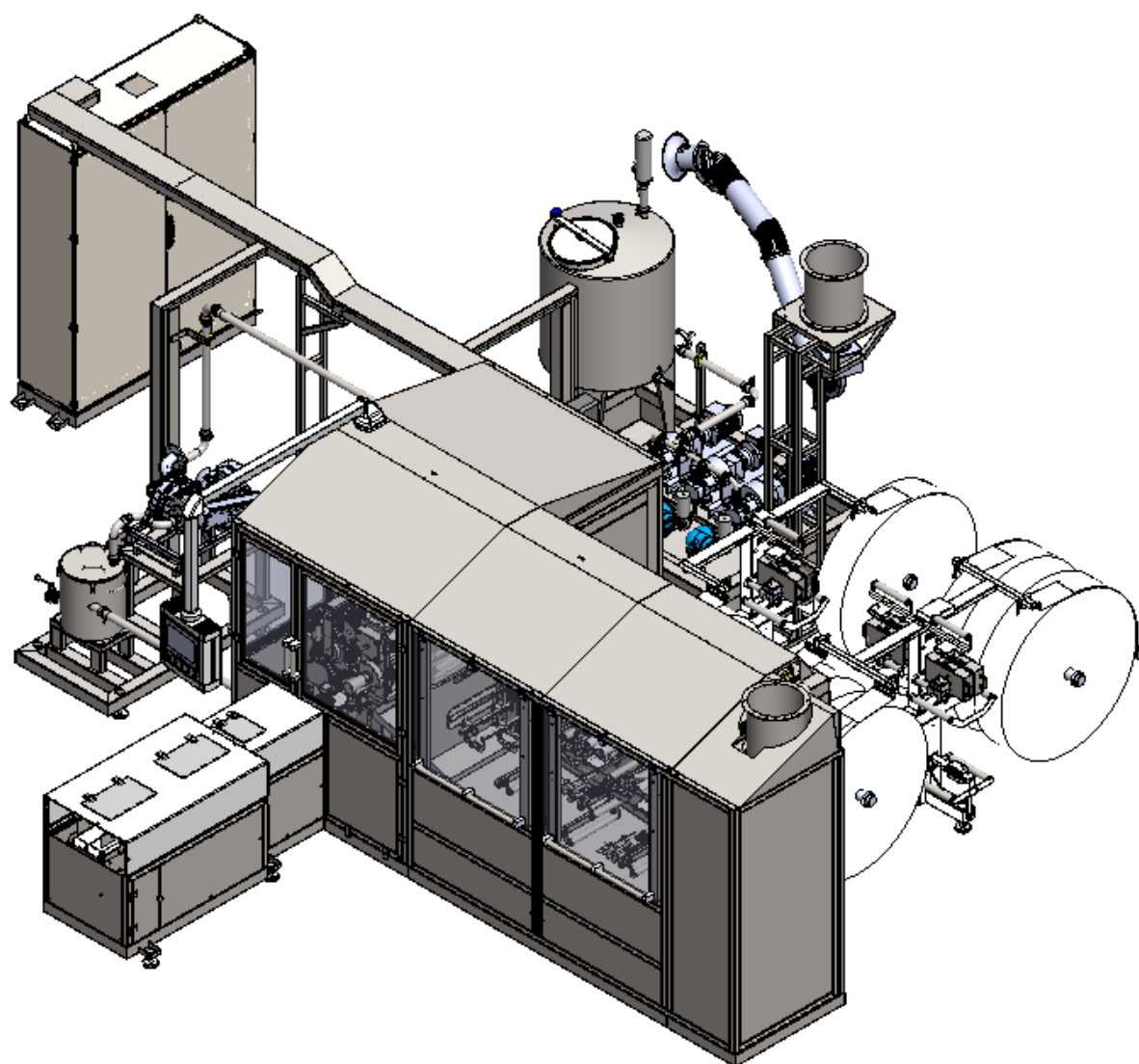
Comfort: Web guiding systems, user friendly HMI screens, product recipes in PLC, rulers, hand wheelers in all mechanical parts, auto rejection systems of unfit products and splice joints, automatic bobbin splice units. Shop floor reporting system and Kansan's "Wipes in-Vision" detailed production reporting system for wet wipe manufacturers.

Safety:, automatic interlocking system on doors, multi-level alarm and control systems, full CE compliance.

Economy: Reasonable cost of ownership, minimum manpower required machine operation.

Hygiene: All FDA approved sanitary lotion delivery system and conveyor belts, GMP design, food grade and pharma grade components

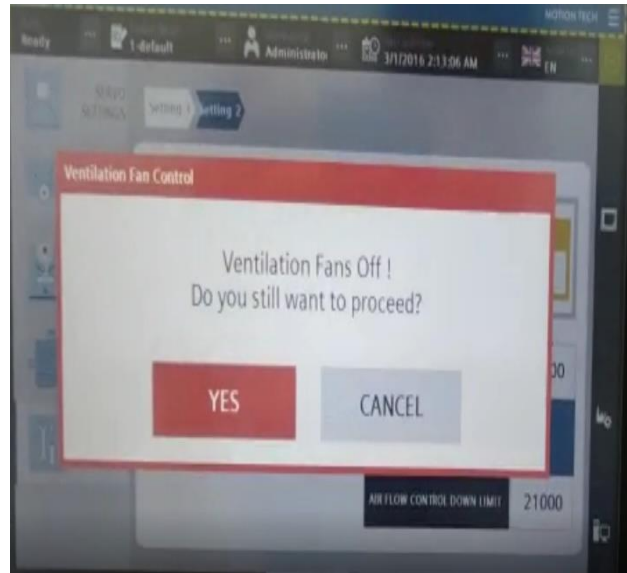
GENERAL VIEW OF MACHINE



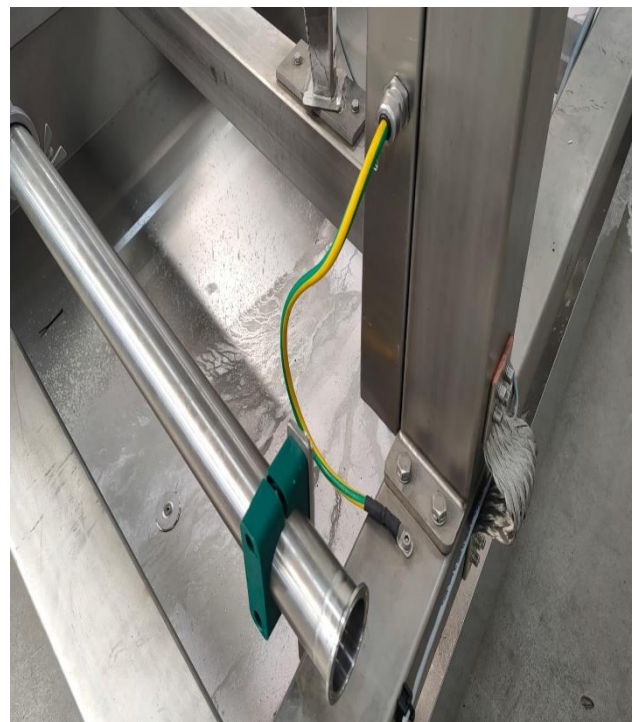
GENERAL WARNING ABOUT WORKING WITH FLAMMABLE PRODUCTS

- This user manual contains basic information about the product. Assembly, installation, use and maintenance must be carried out and guaranteed in accordance with the nominal specifications of the manual.
- **All misuse and all illegal actions, including non-compliance with the provisions of this manual, are not the responsibility of the manufacturer.**
- Before installation, please read all instructions.
- Provide adequate training for responsible personnel.
- Ensure that responsible personnel fully understand the contents of this manual.
- **Ensure that all obligations and national regulations and all special safety precautions are fully followed.**
- Consult the manufacturer on matters that are not understood in the content of the user manual or during the production process.
- Ensure that these application instructions and other user manuals are accessible to responsible personnel at all times.
- **Check that these instructions and all other safety instructions are followed.**
- The machine is designed to work with 70% isopropyl alcohol solution. Do not work with other flammable liquids in any cases.
- **The machine may only be used by trained and certified personnel.**
- Do not operate the device if the ventilation system indicates "**blocked**".
- Don't clean the machine windows with dry materials.
- **Do not plug or unplug any electrical outlet or plug** while the machine is working with flammable liquids.
- **Do not make any changes to the original design without the manufacturer's permission.**
- Before starting the machine, **make sure** that the excess fluid supply and **ventilation system is active.**
- Do not neglect the regular maintenance and precautionary measures given in this user manual.
- Replace the original component with the **originals only.**
- Only use **anti-static conveyor belts.**
- Do not leave doors open for more than 10 minutes when working with flammable liquids.

- Make sure that the **sensors in the ventilation system are connected** before the machine starts.



- Make sure that the **general controls of the grounding cables** are made between **3-6 months**. If there is oxidation, breakage or abrasion in the grounding cables, the cables should be replaced.



- **Carbon brushes should be checked regularly, if there is any wear, the brushes should be replaced.**



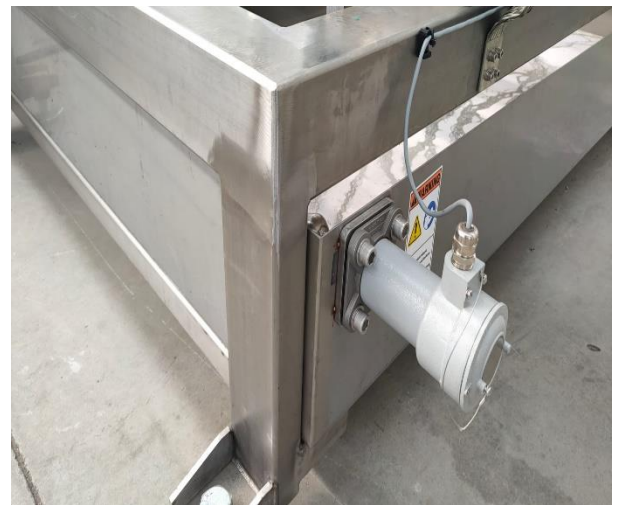
- **The ventilation system should be checked before the machine is started. Any factor that will reduce the capacity of the ventilation system should be eliminated.**



- Anti-static window films on machine windows should never be removed.



- Personnel who will work on the machine must wear antistatic gloves and shoes.
- It should be checked that **the additional tank switch is working against the possibility of the tank overflowing.**



- The surface where the machine will be installed must have an epoxy coating to indicate antistatic properties.
- The liquids accumulating in the reservoir pan drain must be transferred to a special treatment system or an atex compatible tank with an atex pump. The customer should never be kept this liquid open or transfer it to the sewer. The disposal and collection of these liquids is the responsibility of the customer.



- The hose and hose nozzle shown in the image must be replaced with exactly the same brand and model parts.



ATEX FW MACHINE GUIDE



In the connection you see in the photo below, you can see the ventilation fan junction box and ex records. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for Atex. The machine should not be operated with the junction box cover open. The tightness of the unions should be checked from time to time. The cables must not move out or in when pulled.



In the connection you see in the photo below, you can see the ventilation fan junction box and ex records. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for Atex. The machine should not be operated with the junction box cover open. The tightness of the unions should be checked from time to time. The cables must not move out or in when pulled.



In the connection you see in the photo below, you can see the ventilation fan junction, pi air difference sensor and ex unions. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. The machine should not be operated with the junction box cover open. The tightness of the unions should be checked from time to time. Cables must not move out or in when pulled.



BUREAU VERITAS **Ex**

EU - Type Examination Certificate

(1) Equipment and protective systems intended for use in potentially explosive atmospheres – Directive 2014/34/EU

(3) EU - Type Examination Certificate Number: **EPS 14 ATEX 1 789 X** Revision 1

(4) Equipment: **TR.Ex Transmitter and IV.Ex Sensors**

(5) Manufacturer: **petz industries GmbH & Co. KG**

(6) Address: **Mühlenweg 2
96358 Teuschnitz / Hallstadt
Germany**

(7) This equipment and any acceptable variation thereto are specified in the annex to this certificate and the documentation therein referred to.

(8) Bureau Veritas Consumer Products Services Germany GmbH, notified body No. 2004 in accordance with Article 21 given in the Directive 2014/34/EU of the European Parliament and of the Council of 26 February 2014, certifies that this equipment has been found to comply with the essential health and safety requirements relating to the design and construction of equipment and protective systems intended for use in potentially explosive atmospheres, given in Annex II of the Directive. The examination and test results are recorded in the confidential documentation under the reference number 15TH0087.

(9) Compliance with the essential health and safety requirements has been assured by compliance with:

**EN 60079-0:2012+A11:2013 EN 60079-7:2007 EN 60079-11:2012
EN 60079-18:2015 EN 60079-26:2007 EN 60079-31:2014**

(10) If the sign "X" is placed after the certificate number, it indicates that the equipment is subject to special conditions for safe use specified in the annex to this certificate.

(11) This EU - Type Examination Certificate relates only to the design and examination of the specified equipment in accordance with Directive 2014/34/EU. Further requirements of this Directive apply to the manufacture of this equipment and its placing on the market. Those requirements are not covered by this certificate.

(12) The marking of the equipment shall include the following:

II 2110 Ex eb mb Ia [Ga] IIC T4 Gb Transmitter
 II 2110 Ex Ia [Ga] IIC T130°C Db Sensor
 II 2110 Ex ia IIC T0/T5/T4 Ga/Gb
 II 2110 Ex ia IIC T130°C Da/Db Explosion protection

Nuremberg, 2016-06-30

Page 1 of 3

Certificates without signature and seal are void. This certificate is allowed to be distributed only if not modified. Extracts or modifications must be authorized by Bureau Veritas Consumer Products Services Germany GmbH. EPS 14 ATEX 1 789 X, Revision 1.

BUREAU VERITAS Consumer Products Services Germany GmbH Thurn und Taxis Straße 16, 90411 Nuremberg, Germany
Phone: +49 90 76001-0 ccs-members@bbs.bureauveritas.com
www.bureauveritas.de/bbs

BUREAU VERITAS **Ex**

Annex

(13) **EU - Type Examination Certificate EPS 14 ATEX 1 789 X** Revision 1

(15) **Description of equipment:**

The transmitter TR.Ex and the associated sensors type IV.Ex are used for acquisition and analysis of different measurements, e.g. temperature, humidity and difference pressure. The integrated terminal box protected by kind of ignition protection Ex is provides a direct electrical connection in explosive areas.

The transmitter TR.Ex shall only be installed and operated within zones 1 and 2. The sensors are protected by kind of ignition protection Ex ia and can be installed and used within the zones 0 and 1 and at the border of areas requiring Ga/Gb and Da/Db.

Rated data:

TR.Ex Transmitter:

Maximum transmitter ambient temperature range:	T4	-40°C to +70°C
--	----	----------------

24 Vac/dc: ± 20%, U_n = 30 V, U_i = 4.5 V, I_n = 0.107 A (thermal), I_{thermal} = 0.821 A (short circuit)
P_n = 0.428 W, Intrinsic characteristic

Maximum external inductances and capacitances:

L _{ext} [mH]	0.100	0.050	0.020	0.010	0.005	0.002	0.001
C _{ext} [pF]	5.30	6.10	15.00	19.00	30.00	63.00	200.00

IV.Ex Sensors:

Maximum sensor ambient temperature range:	T6	-40°C to +65°C
	T5	-40°C to +60°C
	T4	-40°C to +115°C
	T3-T1	-40°C to +125°C

U_i = 4.5 V, I_n = 0.107 A (thermal), I_n = 0.821 A (short circuit), P_n = 0.428 W

Maximum internal inductances and capacitances:

	IV EX.R	IV EX.P	IV EX.D
Ci	330 nF	1034 nF	1034 nF
Li	negligible		

Besides these sensors also alternative sensors can be used providing that the intrinsic safe values of the [ia] output are met.

Page 2 of 3

Certificates without signature and seal are void. This certificate is allowed to be distributed only if not modified. Extracts or modifications must be authorized by Bureau Veritas Consumer Products Services Germany GmbH. EPS 14 ATEX 1 789 X, Revision 1.

BUREAU VERITAS Consumer Products Services Germany GmbH Thurn und Taxis Straße 16, 90411 Nuremberg, Germany
Phone: +49 90 76001-0 ccs-members@bbs.bureauveritas.com
www.bureauveritas.de/bbs

BUREAU VERITAS **Ex**

(16) **Reference number:** 15TH0087

(17) **Special conditions for safe use:**

The transmitter TR.Ex shall be protected from excessive UV light exposure.

The transmitter TR.Ex shall be protected from mechanical impact.

After installing the sensors in areas requiring Ga/Gb and Da/Db the operator shall verify the leak tightness of the facility to achieve a zone separation.

The enclosure of the transmitter TR.Ex shall never be opened while circuits are alive.

(18) **Essential health and safety requirements:**

Met by compliance with standards.

Nuremberg, 2016-06-30

Page 3 of 3

Certificates without signature and seal are void. This certificate is allowed to be distributed only if not modified. Extracts or modifications must be authorized by Bureau Veritas Consumer Products Services Germany GmbH. EPS 14 ATEX 1 789 X, Revision 1.

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Phone: +49 90 76001-0 ccs-members@bbs.bureauveritas.com
www.bureauveritas.de/bbs

An Atex armature is used for lighting inside the machine. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for Atex. The tightness of the unions should be checked from time to time. The cables must not move out or in when pulled.



CERTIFICATO DI ESAME CE DEL TIPO (AII, III)
EC-TYPE EXAMINATION CERTIFICATE (Annex II)

Apparecchio o Sistema di Protezione inteso per l'uso in atmosfere potenzialmente esplosive, Direttiva 94/9/CE
Equipment or Protective System intended for use in potentially explosive atmospheres, Directive 94/9/EC

[3] Certificato di Esame CE del Tipo numero CEC 13 ATEX 144
EC-Type Examination Certificate number 13222-AT1145

[4] Apparecchio o Sistema di Protezione Apparecchi di illuminazione / lighting fixtures
Equipment or Protective System Type I Type EVF - EVFE -

[5] Costruttore CO.SI.ME S.R.L.
Manufacturer Via Asiago, 51
Indirizzo 20128 Milano - Italy
Address 20128 Milano - Italy

[7] Questo apparecchio o sistema di protezione ed ogni sua variante approvata è descritto nell'allegato al presente certificato e nei documenti descrittivi in esso richiamati.
This equipment or protective system and any acceptable variation therein is specified in the schedule to this certificate and the documents therein referred to.

[8] Il CEC, organismo notificato n° 1131, in conformità all'articolo 9 della Direttiva 94/9/CE del Consiglio dell'Unione Europea del 23 Marzo 1994, certifica che questa apparecchiatura o sistema di protezione è conforme ai Requisiti Essenziali di Sicurezza e Salute per il progetto e la fabbricazione di apparecchiature e sistemi di protezione destinati ad essere utilizzati in atmosfere potenzialmente esplosive, definiti nell'Allegato I della Direttiva.
CEC, notified body No. 1131, in accordance with Article 9 of the Council Directive 94/9/EC of 23 March 1994, certifies that this equipment or protective system has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment and protective systems intended for use in potentially explosive atmospheres given in Annex I to the Directive.

I risultati dell'esame e dei test sono descritti nel rapporto confidenziale elencato nella sezione 16.
The examination and test results are recorded in confidential report listed in section 16.

[9] La conformità ai Requisiti Essenziali di Sicurezza e Salute è assicurata dalla conformità alle:
Compliance with the Essential Health and Safety Requirements has been assured by compliance with:
EN 60079-0:2012, EN60079-1:2007, EN60079-31:2009
In the case in which the normal harmonized rules have been verified, in conformity with the Essential Health and Safety Requirements, if applicable, not listed in the list of Annex Harmonized Standards are used, compliance with the Essential Health and Safety Requirements is certified by the CEC.

[10] Il simbolo "A" posto dopo il numero del certificato indica che l'apparecchiatura o il sistema di protezione è soggetto a condizioni speciali per un utilizzo sicuro, specificate nell'allegato al presente certificato.
If the sign "A" placed after the certificate number, it indicates that the equipment or protective system is subject to special conditions for safe use specified in the schedule to this certificate.

[11] Questo Certificato di esame CE del Tipo è relativo soltanto al progetto, agli esami ed alle prove dell'apparecchio o sistema di protezione specificato in accordo con la Direttiva 94/9/CE. Ulteriori requisiti di questa Direttiva si applicano al processo di produzione e fornitura dell'apparecchiatura o sistema di protezione. Questi requisiti non sono oggetto del presente certificato.
This EC-Type Examination Certificate relates only to the design, examination and tests of the specified equipment or protective system in conformity with the Directive 94/9/EC. Further requirements of the Directive apply to the manufacturing process and supply of this equipment or protective system. These are not covered by this certificate.

[12] L'apparecchiatura o sistema di protezione deve riportare i seguenti contrassegni:
The marking of the equipment or protective system shall include the following:
Ⓢ I 2 G Ex d IIC T6 Gb
Ⓢ I 2 D Ex tb IIIC T80°C Db IP66/67

CONSORZIO EUROPEO CERTIFICAZIONE
L'ORGANO DELIBERANTE
Legnano, 2 dicembre 2017
Il Direttore Tecnico (A. P. 10/17)
Il Direttore Generale (L. T. 10/17)

This certificate may only be reproduced in its entirety and without any change, schedule included.

CEC - CONSORZIO EUROPEO CERTIFICAZIONE S.C.A.R.L.
Via Pissone, 45 - 20020 LEGNANO (MI) Italy - Tel. +39 0331 442 200 - Fax +39 0331 440 054
www.consortiocec.com - info@consortiocec.com - C.F.P. IVA 13073160155 - Reg. Imp. MI 12073160155 - R.E.A. 1812104
ART. 107 - rev. 1 - 09/2013

You can see atex junction box used in many parts of the machine. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. The machine should not be operated with the junction box cover open. The tightness of the unions should be checked from time to time. The cables must not move out or in when pulled.



You can see atex junction box used in many parts of the machine. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. The machine should not be operated with the junction box cover open. The tightness of the unions should be checked from time to time. Cables must not move out or in when pulled.



Approved certificates of Atex junction boxes used in the machine.



CERTIFICATO DI ESAME CE DEL TIPO (All. III)
EC-TYPE EXAMINATION CERTIFICATE (Annex III)

[2] **Apparecchio o Sistema di Protezione inteso per l'uso in atmosfere potenzialmente esplosive, Direttiva 94/9/CE**
Equipment or Protective Systems intended for use in Potentially Explosive Atmospheres, Directive 94/9/EC

[3] **Certificato di Esame CE del Tipo numero** : CEC 10 ATEX 158 - Rev.2
EC-Type Examination Certificate number : 102080-AET778

[4] **Apparecchio o Sistema di Protezione** : Unità di misura, controllo ed interruttori UCS
Equipment or Protective System : Measuring, control and interrupter assembly UCS
Morsettiere Tipo AJB, GWC, GWE, GWM, GWR, GWY, GWS
Terminal Boxes type: AJB, GWC, GWE, GWM, GWR, GWY, GWS

[5] **Costruttore** : CO.SIME srl
Manufacturer :

[6] **Indirizzo** : Via Asiago, 51
Address : 20128 Milano – Italy

[7] **Questo apparecchio o sistema di protezione ed ogni sua variante approvata è descritto nell'allegato al presente certificato e nei documenti descrittivi in esso richiamati.**
This equipment or protective system and any acceptable variation thereto as specified in the schedule to this certificate and the documents therein referred to.

[8] **Il CEC, organismo notificato n° 1131, in conformità all'articolo 9 della Direttiva 94/9/CE del Consiglio dell'Unione Europea del 23 Marzo 1994, certifica che questa apparecchiatura o sistema di protezione è conforme ai Requisiti Essenziali di Sicurezza e Salute per il progetto e la fabbricazione di apparecchiature e sistemi di protezione destinati ad essere utilizzati in atmosfere potenzialmente esplosive, definiti nell'Allegato II della Direttiva.**
CEC, notified body No. 1131, in accordance with Article 9 of the Council Directive 94/9/EC of 23 March 1994, certifies that this equipment or protective system has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment and protective systems intended for use in potentially explosive atmospheres given in Annex II to the Directive.
I risultati dell'esame e dei test sono descritti nel rapporto confidenziale elencato nella sezione 16.
The examination and test results are recorded in confidential reports listed in section 16.

[9] **La conformità ai Requisiti Essenziali di Sicurezza e Salute è assicurata dalla conformità alle:**
Compliance with the Essential Health and Safety Requirements has been assured by compliance with:
EN60079-0:2012, EN 60079-0:2012/A1:2013, EN60079-7:2007, EN60079-11:2012, EN60079-31:2009
Nel caso in cui tra le norme tecniche citate fossero presenti norme non armonizzate, la conformità ai Requisiti essenziali in materia di Sicurezza e Salute è comunque stata verificata.
If standards not listed in the list of Atex Harmonised Standards are used, compliance to the Essential Health and Safety Requirements is verified anyway.

[10] **Il simbolo "X" posto dopo il numero del certificato indica che l'apparecchiatura o il sistema di protezione è soggetto a condizioni speciali per un utilizzo sicuro, specificate nell'allegato al presente certificato.**

CEC – CONSORZIO EUROPEO CERTIFICAZIONE
Certificato di Esame CE del Tipo
EC-Type Examination Certificate

Organo Notificato n. 1131

If the sign "X" is placed after the certificate number, it indicates that the equipment or protective system is subject to special conditions for safe use specified in the schedule to this certificate.

[11] Questo Certificato di esame CE del Tipo è relativo soltanto al progetto, agli esami ed alle prove dell'apparecchio o sistema di protezione specificato in accordo con la Direttiva 94/9/CE. Ulteriori requisiti di questa Direttiva si applicano al processo di produzione e fornitura dell'apparecchiatura o sistema di protezione. Questi requisiti non sono oggetto del presente certificato.
This EC-Type Examination Certificate relates only to the design, examination and tests of the specified equipment or protective system in accordance to the Directive 94/9/EC. Further requirements of the Directive apply to the manufacturing process and supply of this equipment or protective system. These are not covered by this certificate.

[12] **L'apparecchiatura o sistema di protezione deve riportare i seguenti contrassegni:**
The marking of the equipment or protective system shall include the following:

UCS	II 2G	Ex e IIC T6/T4 Gb
	II 2(1) G	Ex ia/ib IIC T6/T4 Gb
	II 2D	Ex e [ib] IIC T6/T4 Gb
	II 2(1)D	Ex e [ia] IIC T6/T4 Gb
		Ex tb IIC T80C/T130°C Db IP66
		UCS GWE Ex tb IIC T80C/T130°C Db IP65
		Ex tb IIC T80C/T130°C Db IP66
		UCS GWE Ex tb IIC T80C/T130°C Db IP65

Morsettiere	II 2G	Ex e IIC T6/T4 Gb
	II 2(1) G	Ex ia/ib IIC T6/T4 Gb
	II 2D	Ex e [ib] IIC T6/T4 Gb
	II 2(1)D	Ex e [ia] IIC T6/T4 Gb
		Ex tb IIC T80C/T130°C Db IP66
		UCS GWE Ex tb IIC T80C/T130°C Db IP65
		Ex tb IIC T80C/T130°C Db IP66
		UCS GWE Ex tb IIC T80C/T130°C Db IP65

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L'ORGANO DELIBERANTE

Il Direttore Tecnico (A. FUGAZZI) Il Direttore Generale (L. TIMASSI)

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CERTIFICATO DI ESAME CE DEL TIPO n° CEC 10 ATEX 158-Rev.2
to EC-TYPE EXAMINATION CERTIFICATE no. CEC 10 ATEX 158 Rev.2

[15] **Descrizione - Description**

L'apparecchio è composto da una o più custodie di poliestere rinforzato con fibra di vetro (GW), alluminio (AJB) ed acciaio inox (GWY) con caratteristiche meccaniche indicate rispettivamente nelle note tecniche ACH2047N, certificato di esame del componente KEMA (01ATEX2431, 01ATEX228U) e 02ATEX095U) con modo di protezione a Sicurezza aumentata "e". All'interno delle custodie sono installati dispositivi di misura, controllo, commutatori ed interruttori automatici ed/o valvole dell'esterno mediante attuatori, spie e finestre di ispezione. Le custodie potranno inoltre contenere morsettiere per circuiti a sicurezza intrinseca e circuiti non a sicurezza intrinseca. L'ingresso dell'aria è realizzato mediante l'utilizzo di un dispositivo Ex a certificato secondo le Norme EN 60079-0, EN 60079-7, EN 60079-31:2009.

The apparatus consists of one or a number of enclosures which are made from polyester glass reinforced (GW), aluminium (AJB) and stainless steel (GWY) with mechanical features indicated in the technical note 01ATEX2431, 01ATEX228U and 02ATEX095U. The enclosures are fitted with automatic switching, control and interlocking assembly, as well as terminals for intrinsically safe and non-intrinsically safe circuits and may be fitted with actuators elements, indicator lights and inspection windows. Connection to external is made by means of Ex e cable entries certified according to Standards EN 60079-0, EN 60079-7, EN 60079-31:2009.

Marcatore Markings:
La marcatura dell'apparecchio può essere completata utilizzando i codici "e", "m", "m", "Tb" e "TbT", in quanto applicabili, a seconda degli apparecchi e componenti installati. Per circuiti a sicurezza intrinseca, l'area per i terminali è contrassegnata da un colore azzurro.
The apparatus marking is completed by using the codes of "e", "m", "Tb" and "TbT", in applicable, depending on the built-in apparatus and components. For intrinsically safe circuits, the area for the terminals is marked by a light blue colour.

Temperatura ambiente standard di utilizzo / Standard operating temperature
UCS GWC - GWM - GWE - TAMB -20°C .. +40°C
Temperatura ambiente speciale di utilizzo / Special operating temperature
UCS GWC - GWM - GWE - TAMB -40°C .. +40°C
Temperatura ambiente standard di utilizzo / Standard operating temperature
UCS - AJB - GWY - GWS - TAMB -20°C .. +40°C
Temperatura ambiente speciale di utilizzo / Special operating temperature
UCS - AJB - GWY - GWS - TAMB -50°C .. +80°C

Classe di temperatura UCS/UCS: Temperature class
I componenti installati all'interno delle custodie avranno una temperatura superficiale massima inferiore ed uguale alla temperatura massima ammessa dalla costruzione elettrica : T 80°C per classe di temperatura T6. Nel caso in cui la temperatura del componente non sia dichiarata o sia superiore a quella ammessa, la costruzione elettrica sarà contrassegnata con la seguente avvertenza: "Non aprire in presenza di un'atmosfera di polvere esplosiva".
The components installed inside the enclosures shall have a maximum surface temperature equal or below to the maximum permitted surface temperature of the apparatus : T 80°C for temperature class T6. If the surface temperature of the component isn't indicated by the manufacturer or is more major than the maximum surface temperature allowed, the apparatus shall be marked with the following warning: "do not open when an explosive dust atmosphere is present".

Caratteristiche elettriche e tecniche UCS:
UCS: Electrical and thermal characteristics.
Le caratteristiche elettriche dell'apparecchio sono funzione delle caratteristiche dei componenti a prova di esplosione installati all'interno delle custodie.
The UCS electrical and thermal characteristics depend on the type of explosion-proof components installed inside the enclosures.

Tensione nominale di isolamento / Rated insulation voltage: fino a / up to 1000V
Tensione nominale / Rated voltage: fino a / up to 1000V ac - 250V dc
Corrente nominale / Rated current:
- Circuito principale / main circuit: max 20A
- Circuiti di controllo ed ausiliari / auxiliary and control circuits: max 20A
Sezione nominale morsettiere / Terminal cross section
- Circuito principale / main circuit: max 240 mm²
- Circuiti di controllo ed ausiliari / auxiliary and control circuits: max 6 mm²

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Organo Notificato n. 1131

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Il Direttore Tecnico (A. FUGAZZI) Il Direttore Generale (L. TIMASSI)

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Approved certificates of atex junction boxes used in the machine.

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Certificato di Esame CE del Tipo
EC-Type Examination Certificate

ALLEGATO – SCHEDULE
Originali Notifica n. 131

CEC
[13] CERTIFICATO DI ESAME CE DEL TIPO n° CEC 10 ATEX 158-Rev.2
[14] EC-TYPE EXAMINATION CERTIFICATE no. CEC 10 ATEX 158-Rev.2

Codice custodia per UCS Enclosure code for UCS	Potenzaq Max. dissipata [W] / Max. power dissipated [W]		Classe di temperatura / Temperature class	
	Ta max +40°C	Ta max +80°C	Ta max +40°C	Ta max +80°C
AJB 91... - AJB 1111	8	4	T6/T80°C	T4/T130°C
AJB 11... - AJB 1414	16,5	8,5	T6/T80°C	T4/T130°C
AJB 111... - AJB 1111	16,5	8,5	T6/T80°C	T4/T130°C
AJB 12... - AJB 2014	16,5	8,5	T6/T80°C	T4/T130°C
AJB 13... - AJB 2020	16,5	8,5	T6/T80°C	T4/T130°C
AJB 14... - AJB 2014	16,5	8,5	T6/T80°C	T4/T130°C
AJB 2... - AJB 2518	33	17,5	T6/T80°C	T4/T130°C
AJB 3... - AJB 3324	67,5	35,5	T6/T80°C	T4/T130°C
AJB 4... - AJB 4030	83,5	50	T6/T80°C	T4/T130°C
AJB 41... - AJB 4831	83,5	50	T6/T80°C	T4/T130°C
AJB 5... - AJB 6230	200	100	T6/T80°C	T4/T130°C

Codice custodia per UCS Enclosure code for UCS	Potenzaq Max. dissipata [W] / Max. power dissipated [W]		Classe di temperatura / Temperature class	
	Ta max +40°C	Ta max +80°C	Ta max +40°C	Ta max +80°C
GWC 99...	7	3,5	T6/T80°C	T4/T130°C
GWC 99...	7	3,5	T6/T80°C	T4/T130°C
GWC 87...	7	3,5	T6/T80°C	T4/T130°C
GWC 117...	9	4,5	T6/T80°C	T4/T130°C
GWC 111...	9	4,5	T6/T80°C	T4/T130°C
GWC 121...	9	4,5	T6/T80°C	T4/T130°C
GWC 141...	11	5	T6/T80°C	T4/T130°C
GWC 141...	11	5	T6/T80°C	T4/T130°C
GWC 167...	9,5	5	T6/T80°C	T4/T130°C
GWC 178...	9,5	5	T6/T80°C	T4/T130°C
GWC 178...	9,5	5	T6/T80°C	T4/T130°C
GWC 197...	9,5	5	T6/T80°C	T4/T130°C
GWC 171...	9,5	5	T6/T80°C	T4/T130°C
GWC 171...	13	7	T6/T80°C	T4/T130°C
GWC 171...	13	7	T6/T80°C	T4/T130°C
GWC 212...	13	7	T6/T80°C	T4/T130°C
GWC 281...	13	7	T6/T80°C	T4/T130°C
GWC 252...	14	7,5	T6/T80°C	T4/T130°C
GWC 252...	14	7,5	T6/T80°C	T4/T130°C
GWC 301...	14	7,5	T6/T80°C	T4/T130°C
GWC 361...	14	7,5	T6/T80°C	T4/T130°C
GWC 361...	15	7,5	T6/T80°C	T4/T130°C
GWC 302...	17	8	T6/T80°C	T4/T130°C
GWC 302...	26,5	19	T6/T80°C	T4/T130°C
GWC 363...	21	11	T6/T80°C	T4/T130°C
GWC 363...	21	11	T6/T80°C	T4/T130°C
GWC 402...	21	11	T6/T80°C	T4/T130°C
GWC 581...	21	11	T6/T80°C	T4/T130°C
GWC 602...	21	11	T6/T80°C	T4/T130°C
GWC 454...	21	11	T6/T80°C	T4/T130°C
GWC 473...	46,5	34,5	T6/T80°C	T4/T130°C
GWC 6230...	50	28	T6/T80°C	T4/T130°C

(*) Certificato di componente KEMA 02 ATEX 2055 U (*) Component certificate KEMA 02 ATEX 2055 U

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Certificato di Esame CE del Tipo
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ALLEGATO – SCHEDULE
Originali Notifica n. 131

CEC
[13] CERTIFICATO DI ESAME CE DEL TIPO n° CEC 10 ATEX 158-Rev.2
[14] EC-TYPE EXAMINATION CERTIFICATE no. CEC 10 ATEX 158-Rev.2

Codice custodia per UCS Enclosure code for UCS	Potenzaq Max. dissipata [W] / Max. power dissipated [W]		Classe di temperatura / Temperature class	
	Ta max +40°C	Ta max +80°C	Ta max +40°C	Ta max +80°C
GWC - GWS 99...	8	4	T6/T80°C	T4/T130°C
GWC - GWS 149...	8	4	T6/T80°C	T4/T130°C
GWC - GWS 111...	8	4	T6/T80°C	T4/T130°C
GWC - GWS 141...	8	4	T6/T80°C	T4/T130°C
GWC - GWS 209...	12	6	T6/T80°C	T4/T130°C
GWC - GWS 289...	12	6	T6/T80°C	T4/T130°C
GWC - GWS 161...	13	7	T6/T80°C	T4/T130°C
GWC - GWS 201...	13	7	T6/T80°C	T4/T130°C
GWC - GWS 202...	14	7,5	T6/T80°C	T4/T130°C
GWC - GWS 251...	14	7,5	T6/T80°C	T4/T130°C
GWC - GWS 252...	18	9,5	T6/T80°C	T4/T130°C
GWC - GWS 323...	21	11	T6/T80°C	T4/T130°C
GWC - GWS 303...	20	10,5	T6/T80°C	T4/T130°C
GWC - GWS 303...	20	10,5	T6/T80°C	T4/T130°C
GWC - GWS 343...	20	10,5	T6/T80°C	T4/T130°C
GWC - GWS 402...	20	10,5	T6/T80°C	T4/T130°C
GWC - GWS 403...	28,5	15	T6/T80°C	T4/T130°C
GWC - GWS 454...	28,5	15	T6/T80°C	T4/T130°C
GWC - GWS 505...	28,5	15	T6/T80°C	T4/T130°C
GWC - GWS 604...	28,5	15	T6/T80°C	T4/T130°C
GWC - GWS 758...	150	75	T6/T80°C	T4/T130°C
GWC - GWS 109...	200	100	T6/T80°C	T4/T130°C

Istruzioni per l'installazione di UCS...
Le istruzioni d'uso, come previsto dal costruttore deve essere seguite nel dettaglio per assicurare il funzionamento corretto e sicuro delle attrezzature.
UCS. Installation instructions.
The operating instructions as provided by the manufacturer shall be followed in detail to assure proper and safe operation of the equipment.

Temperatura ambiente standard di utilizzo morsettiere / Standard operating temperature terminal boxes
Custodie / Enclosure GWC - GWM - GWR: Tamb -20°C... +40°C
Temperatura ambiente speciale di utilizzo morsettiere / Special operating temperature terminal boxes
Custodie / Enclosure UCS GWC - GWM - GWR: Tamb -40°C... +80°C
Temperatura ambiente standard di utilizzo morsettiere / Standard operating temperature terminal boxes
Custodie / Enclosure AJB - GW - GWS: Tamb -20°C... +40°C
Temperatura ambiente speciale di utilizzo morsettiere / Special operating temperature terminal boxes
Custodie / Enclosure AJB - GW - GWS: Tamb -50°C... +80°C

Caratteristiche elettriche e termiche morsettiere / Terminal boxes electrical and thermal characteristics
Tensione nominale / Rated voltage: max. 800 V
Corrente nominale / Numero di conduttori / Sezione conduttore nominale: vedi manuale di installazione e manutenzione. Rated current / Number of conductors / Nominal conductor cross section: to be taken from the installation and maintenance instructions.

La relazione tra la massima temperatura ambiente, classe di temperatura e la massima temperatura superficiale "T" è riportata nella tabella sottostante.
The relation between maximum ambient temperature, temperature class and maximum surface temperature "T" is shown in the table below.

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ALLEGATO – SCHEDULE
Originali Notifica n. 131

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[13] CERTIFICATO DI ESAME CE DEL TIPO n° CEC 10 ATEX 158-Rev.2
[14] EC-TYPE EXAMINATION CERTIFICATE no. CEC 10 ATEX 158-Rev.2

Temperatura ambiente Maximum ambient temperature	Classe di temperatura Temperature class	Temperatura superficiale massima "T" Maximum surface temperature "T"
+40°C	T6	80°C
+60°C	T5	95°C
+80°C	T4	130°C

L'attuale massima temperatura ambiente consentita può essere inferiore a quella sopra indicata a seconda della temperatura massima di funzionamento dei morsetti utilizzati.
The actual allowed maximum ambient temperature may be lower than specified above depending on the maximum operating temperature of the terminals used.

Istruzioni per l'installazione di morsettiere:
Terminal Boxes Installation instructions.
Il grado di protezione di almeno IP66 (IP65 per GWE) si ottiene solo se vengono utilizzati pressacavi e elementi di chiusura certificati Ex e, che sono adatti per l'applicazione e installati correttamente.
The degree of protection of at least IP66 (IP65 for GWE) is only achieved if certified Ex e cable entries and blanking elements are used that are suitable for the application and correctly installed.

Test di Routine / Routine tests
Ogni apparecchio deve essere sottoposto a esami di routine secondo le norme EN 60079-7 §6.1 e/o EN 60079-11 §11.2 a seconda dei casi, applicando la tensione e il tempo di prova come previsto dalla norma, senza che accada nessuna scarica elettrica.
Each apparatus shall be subjected to routine tests according to EN 60079-7, clause 6.1 and/or EN 60079-11, clause 11.2, as applicable, using the applicable test voltage and test time, without breakdown.

[16] Rapporto numero / Report Number: CEC 142018 - RET 003
Condizioni speciali per un utilizzo sicuro - Special conditions for safe use
Nessuna - None
L'efficacia e l'affidabilità di questi apparecchi sono garantite seguendo le istruzioni del Manuale d'uso. Non sono ammesse modifiche non autorizzate rispetto al fascicolo tecnico agli atti.
The effectiveness and reliability of these apparatuses are guaranteed following the manufacturer's manual. Further modification are not allowed.
Requisiti Essenziali di Sicurezza e Salute - Essential Health and Safety Requirements
Nessuno - None
Riguardo ai Requisiti Essenziali di Sicurezza e Salute questo documento verifica la conformità solo agli standard Ex. La dichiarazione di Conformità del Produttore dichiara la conformità con altre Direttive pertinenti.
Concerning EN603 the schedule verifies the compliance with the Ex standards only. The manufacturer's Declaration of Conformity declares compliance with other relevant Directives.

[19] I documenti di riferimento listati di seguito costituiscono la documentazione tecnica dell'apparecchio o sistema di protezione oggetto di questo certificato. Questi documenti sono confidenziali e sono a disposizione delle sole autorità competenti.
The documents listed below constitute the technical documentation of the equipment or protective system, subject of this certificate. These documents are confidential and they are available only to the authorities.
Una copia di questi documenti è conservata presso l'archivio del CEC.
One copy of all documents is kept in CEC files.
- AC10024TN

L'ISPEZIONE CARICATA
Dott. Ing. Giuseppe TERZANI

Organo deliberante Antonio FUGAZZI Data: 16/03/2015

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CO.SI.ME. EQUIPMENT FOR ATEX

CO.SI.ME. srl Via Asago, 51 - 20128 Milano - ITALY
E-mail: info@cosime.com
Tel. +39 02 25 78 510 Fax +39 02 25 52 385
C.F./P.IVA 02323870160

Dichiarazione UE di Conformità
Declaration EU of Conformity

N° 14/20

Dichiaro sotto la nostra esclusiva responsabilità che i prodotti:
Declare under our sole responsibility that the products:

Morsettiere
Terminal boxes

Codice / Code AJB-GW ..

Serial number 20- 743 + 20- 752

al quale questa dichiarazione si riferisce sono conformi a:
to which this attestation relates are in conformity with:

Direttiva ATEX 2014/34/UE: apparecchi e sistemi di protezione
Directive ATEX 2014/34/UE: equipment and protective systems
destinati ad essere utilizzati in atmosfera potenzialmente esplosiva
intended for use in potentially explosive atmospheres
e sono costruiti in accordo alle seguenti Norme:
and they are manufactured in conformity with the following Standards:

EN 60079-0 : 2012+A11:2013 (*)
EN 60079-7 : 2015 (*)
EN 60079-31 : 2014 (*)

Con i seguenti modi di protezione:
with the following protection mode:

Ex e IIC T4 + T6 Gb IP66
Ex tb IIC T80°C + T130°C Db

ATEX certificate: CEC 10ATEX158

Nome dell'Organismo incaricato della sorveglianza della produzione: CESI
Name of notified body involved in the surveillance of production:

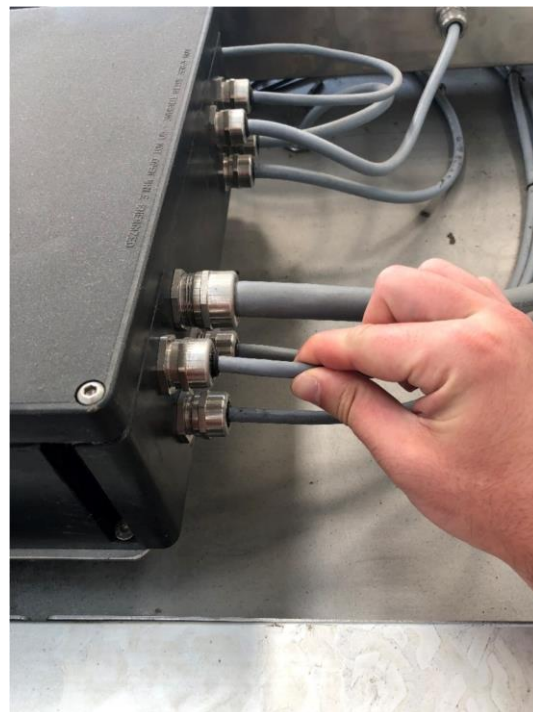
Numero dell'Organismo notificato: 0722
Number of notified body

Massimo Tonelli
(nome e firma o impronta equivalente della persona autorizzata)
(name and signature or equivalent marking of authorized person)

Milano 22/01/2020
(luogo e data di emissione)
(place and date of issue)

(*) Note: Norme aggiornate e controllate dal Costruttore in conformità alle ultime edizioni
(*) Note: EN standards updated and analyzed by Manufacturer according to last issue

You can see the ex records in the photo. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. The tightness of the unions should be checked time to time. Cables must not move out or in when pulled.



Approved certificates for Ex unions.

IEP ENERGY PETROLEUM INSTITUTE

Supplement No: 2

EU-Type Examination Certificate

(2) Equipment or Protective Systems Intended for use in Potentially Explosive Atmospheres

Directive 2014/34/EU

(3) EU – Type Examination Certificate Number: **IEP 14 ATEX 0217 X**

(4) Product: **Cable Glands UAG and UG Series**

(5) Firm Name: **Ortaçlar Elektrik Sanayi ve Ticaret Limited Şirketi**

(6) Firm Address: **Hadımköy Mah. İbnü Sinâ Cad. No:9 Arnavutköy / İSTANBUL - TURKEY**

(7) This product and any of acceptable variation thereto is specified in the schedule to this certificate and the documents therein referred to.

(8) The IEP Ulaştırma Enerji Petrol Gözetim, Sertifikasyon ve Teknik Hizmetler Organizasyonu Tic. Ltd. Şti., notified body number 2284 in accordance with Article 17 of the Directive 2014/34/EU of European Parliament and of the Council, dated 26 February 2014, certifies that this product has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of products intended for use in potentially explosive atmospheres, given in Annex II to the Directive. The examination and test results are recorded in confidential Report Nr: IEP.Rp.Ex.10-291-2 date 13.03.2018.

(9) Compliance with Essential Health and safety requirements has been assured by compliance with:

EN 60079-0:2013, EN 60079-1:2014, EN 60079-7:2015, EN 60079-31:2014

(10) If the sign "X" is placed after the certificate number, it indicates that the product is subject to Specified Conditions of Safe Use specified in the schedule to this certificate.

(11) This EU-Type Examination Certificate relates only to the design and construction of the specified product in accordance to the directive 2014/34/EU. Further requirements of the directive apply to the manufacturing process and supply of this product. These are not covered by this certificate.

(12) The marking of the equipment or protective system shall include the following:

Ex I M2 Ex db I Mb / I M2 Ex db I Mb
II 2G Ex db IIC Gb / II 2G Ex db IIC Gb
II 2D Ex db IIC Db

Responsible Person:
Nurgün Terzioğlu
Head of Certification Body

Supplement Date of Issue: 20.03.2018

IEP Ulaştırma Enerji Petrol Gözetim, Sertifikasyon ve Teknik Hiz. Org. Tic. Ltd. Şti.
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IEP ENERGY PETROLEUM INSTITUTE

(13) Schedule

(14) Certificate Nr: **IEP 14 ATEX 0217 X**

(15) Technical Details: **UAG series unarmour Cable Gland Details, IP 66/68**

Cable Gland Size	Entry Thread	d type GL mm	e type GL mm	Cable outer sheath Ø mm	
				Min.	Max.
12	M12x1,5	15	6	3,0	6,5
16	M16x1,5	15	7	4,0	8,0
20	M20x1,5	15	8	6,0	12,0
25	M25x1,5	15	8	10,0	14,0
32	M32x1,5	15	9	13,0	18,0
40	M40x1,5	15	9	18,0	25,0
50	M50x1,5	15	9	22,0	32,0
63	M63x1,5	15	14	34,0	44,0
63	M63x1,5	15	15	45,0	55,0
75	M75x1,5	20	20	45,0	55,0
75	M75x1,5	20	20	50,0	63,0
80	M80x2,0	20	20	50,0	63,0
90	M90x2,0	20	20	60,0	70,0
100	M100x2,0	20	20	70,0	80,0
115	M115x2,0	20	20	80,0	90,0
130	M130x2,0	20	20	90,0	100,0

UAG series Unarmour Cable Gland Details , IP 66/68

Cable Gland Size	Entry Thread	d type GL mm	e type GL mm	Cable outer sheath Ø mm	
				Min.	Max.
16	M16x1,5	15	7	6,1	11,6
20	M20x1,5	15	8	6,1	13,2
25	M25x1,5	15	8	9,5	15,9
32	M32x1,5	15	9	12,5	20,5
40	M40x1,5	15	9	18,2	26,2
50	M50x1,5	15	9	23,7	33,9
63	M63x1,5	15	15	27,9	40,4
63	M63x1,5	15	15	40,4	51
63	M63x1,5	15	15	47,2	58,9
75	M75x1,5	15	15	52,8	59
75	M75x1,5	15	15	59,1	67,9
90	M90x2,0	20	20	60	70
100	M100x2,0	20	20	70	80
115	M115x2,0	20	20	80	90
130	M130x2,0	20	20	90	100

Responsible Person:
Nurgün Terzioğlu
Head of Certification Body

IEP Ulaştırma Enerji Petrol Gözetim, Sertifikasyon ve Teknik Hizmetler Organizasyonu Ticaret Limited Şirketi
5746/1 Sok. No:9 K:2 Bornova - İZMİR / TURKEY Tel: +90 232 431 17 43 - 46 Fax: +90 232 431 17 30 E-mail: ertep@ertep.com.tr
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(16) Certificate Nr: **IEP 14 ATEX 0217 X**

(17) Description of Equipment

Cable Gland Series Type UAG and UG made from according to brass, stainless steel and aluminium. The ranges of cable glands are metallic and intended to terminate circular unarmoured and braided cables (as defined type designations) into a threaded entry point within associated flameproof, increased safety or dust tight enclosures (as defined by their coding). Cable sealing ring, rubber washer is made from silicone rubber. Climbing ring is made from PA V0.

The UAG and UG type range of cable glands are intended to terminate tape cables into enclosures without compromising the explosion protection provided by the enclosures in accordance with relevant codes of practice. They consist of a male-threaded front entry component, a front seal, a main body component, a rear seal, an actuating nut and a rear running coupling. The front entry component is intended to screw into an entry point of its associated enclosure. The seals are compressed onto the cable when the body component and actuating nut are tightened. Continuity diaphragm and skid washer is fitted behind the front seal.

Cable Gland Series Type UAG and UG has been evaluate in the contents of IP 66/68 with by cable.

Materials of Manufacture:
Brass to EN12168:1998 Grade CuZn39Pb (CW614N)
Mild steel to EN 10088-3:2005 Grade 220M07PS
Stainless steel to EN 10088-3:2005 Grade 316S31, 316S33, 316S31 or 316S33
Aluminium alloy not inferior to grade 6082 to EN755-3:1996 or LM25 to EN 1676:2010 (Not Group I)

Temperature of isolation:
NBR sealing ring : (- 50 ; + 120) °C
Neoprene sealing ring : (- 40 ; + 100) °C
Silicon sealing ring : (- 60 ; + 180) °C

(18) List of documentation:

Cable Gland Series Type UAG and UG operating manual date 08.12.2015 [8 pages]

Drawing Nr :	Drawing Name :	Date :
DCG-000	Exploded picture	18.12.2013
UAG-000	Dimensions	15.11.2013
UAG-001	Metric thread washer	01.03.2013
UAG-002	Metric thread washer	01.03.2013
UAG-003	Metric thread sizes	01.03.2013
UAG-004	Metric thread sealing sizes	01.03.2013
UAG-005	Metric thread body sizes	01.03.2013
UAG-006	Union dimensions	01.03.2013
UAG-007	Seal fitting sizes	01.03.2013
UAG-008	Fitting lid sizes	01.03.2013
UAG-009	Body sizes	01.03.2013

Responsible Person:
Nurgün Terzioğlu
Head of Certification Body

IEP Ulaştırma Enerji Petrol Gözetim, Sertifikasyon ve Teknik Hizmetler Organizasyonu Ticaret Limited Şirketi
5746/1 Sok. No:9 K:2 Bornova - İZMİR / TURKEY Tel: +90 232 431 17 43 - 46 Fax: +90 232 431 17 30 E-mail: ertep@ertep.com.tr
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IEP ENERGY PETROLEUM INSTITUTE

(19) Certificate Nr: **IEP 14 ATEX 0217 X**

Drawing Nr :	Drawing Name :	Date :
UAG-010	Gas thread size	01.03.2013
UAG-011	Gas threaded upper cap	01.03.2013
UAG-012	Gas threaded cap	01.03.2013
UAG-013	Gas threaded gasket	01.03.2013
UAG-014	Gas threaded body	01.03.2013
UAG-015	NPT threaded upper gasket	01.03.2013
UAG-016	NPT threaded cover	01.03.2013
UAG-017	NPT threaded gasket	01.03.2013
UAG-018	NPT threaded body	01.03.2013
UAG-019	PG threaded O-ring	01.03.2013
UAG-020	PG threaded cover	01.03.2013
UAG-021	PG threaded gasket	01.03.2013
UAG-022	PG threaded body	01.03.2013
UAG-023	PG threaded upper gasket	01.03.2013
UAG-024	NPT threaded body	08.12.2015
UAG-025	Metric threaded body	04.12.2015
UAG-026	Gas threaded body	14.12.2015
UAG-027	PG threaded body	04.12.2015
UG-014	Bellevue seal	03.09.2013
UG-015	Metric threaded body	23.08.2013
UG-016	PG threaded body	23.08.2013
UG-017	Gas threaded body	23.08.2013
UG-018	Cover part	12.08.2013
UG-026	Sealing pressure washer	01.11.2013
UG-027	Washer	05.11.2013
UG-028	Sealing pressure washer	01.11.2013
UG-029	NPT threaded body	23.08.2013

Cable Gland Series Type UAG and UG are indicated in the piece list Table 1 and date 08.12.2015.
This certificate is in the contents of standards that mentioned in item 9. It has been accepted that Cable Gland series Type UAG and UG are manufactured according to the producer instructions and the standards mentioned above.

Certificate History:

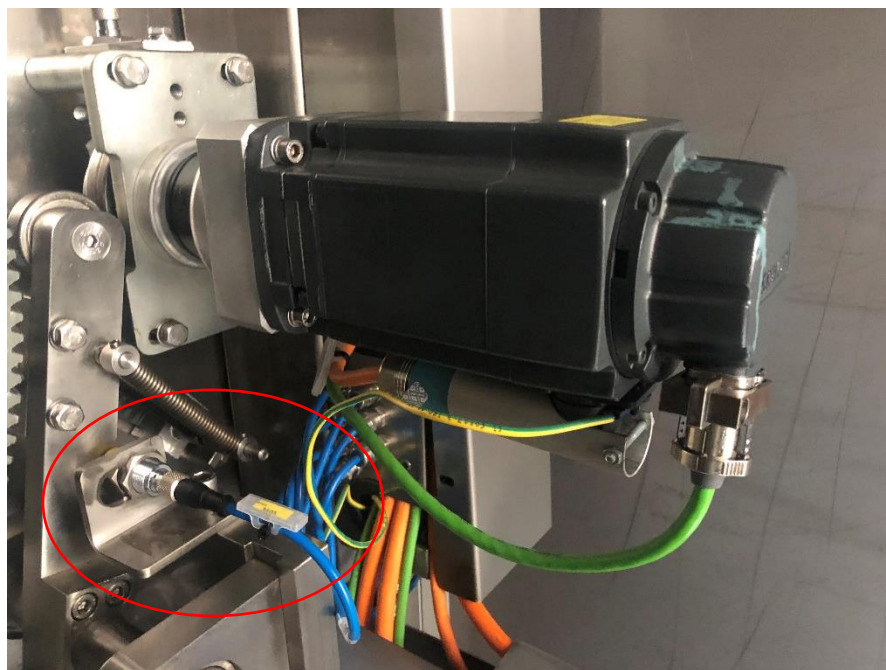
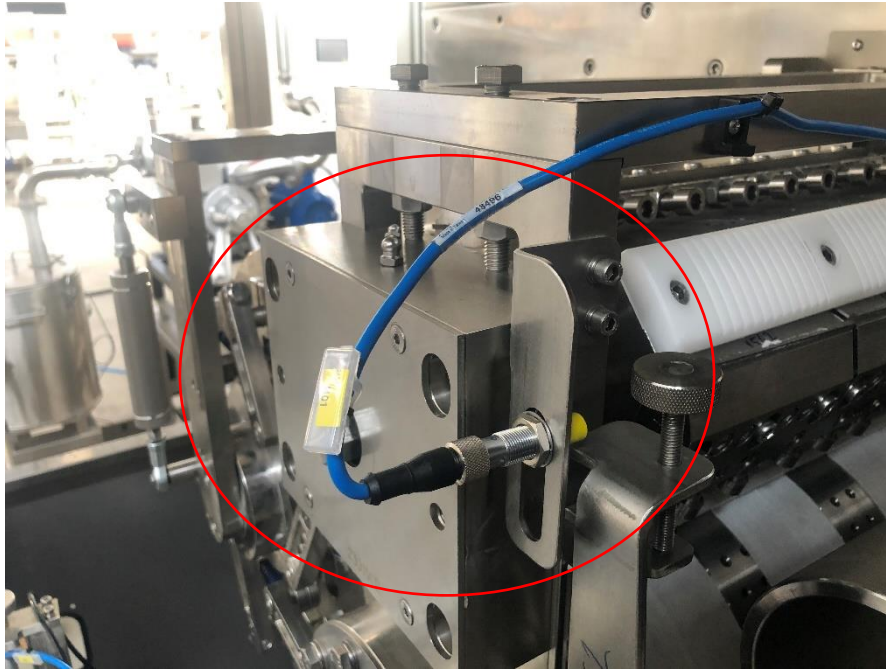
Supplement N°	Issue Date	Summary Description of Variation
02	20.03.2018	Update to new edition of EN 60079-0:2013, EN 60079-1:2014, EN 60079-7:2015, EN 60079-31:2014 Upgrade to ATEX directive 2014/34/EU Change of address
01	17.02.2016	New area addition: Mining
00	02.06.2014	First issue of certificate

Responsible Person:
Nurgün Terzioğlu
Head of Certification Body

Supplement Date of Issue: 20.03.2018

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5746/1 Sok. No:9 K:2 Bornova - İZMİR / TURKEY Tel: +90 232 431 17 43 - 46 Fax: +90 232 431 17 30 E-mail: ertep@ertep.com.tr
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The sensors used are 1 G and in case of any intervention, the same product or equivalent products should be used. After the intervention, it should be made suitable for atex.



The sensors used are 1 G and in case of any intervention, the same product or equivalent product should be used. After the intervention, it should be made suitable for Atex.

CERTIFICATE

(1) **EC-Type Examination**

(2) **Equipment and protective systems intended for use in potentially explosive atmospheres - Directive 94/9/EC**

(3) EC-Type Examination Certificate Number: **KEMA 02ATEX1090 X** Issue Number: **7**

(4) Equipment: **Two Wire Proximity Sensors Type ...Y1... / ...**

(5) Manufacturer: **Hans Turck GmbH & Co. KG**

(6) Address: **Witzlebenstrasse 7, 45472 Mülheim an der Ruhr, Germany**

(7) This equipment and any acceptable variation thereto is specified in the schedule to this certificate and the documents therein referred to.

(8) DEKRA Certification B.V., notified body number 0344 in accordance with Article 9 of the Council Directive 94/9/EC of 23 March 1994, certifies that this equipment has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment and protective systems intended for use in potentially explosive atmospheres given in Annex II to the directive.

The examination and test results are recorded in confidential test report number **NL/KEMEXTR06.0032104**.

(9) Compliance with the Essential Health and Safety Requirements has been assured by compliance with:

EN 60079-0: 2012 + A11 **EN 60079-11: 2012**

(10) If the sign "X" is placed after the certificate number, it indicates that the equipment is subject to special conditions for safe use specified in the schedule to this certificate.

(11) This EC-Type Examination Certificate relates only to the design, examination and tests of the specified equipment according to the Directive 94/9/EC. Further requirements of the directive apply to the manufacturing process and supply of this equipment. These are not covered by this certificate.

(12) The marking of the equipment shall include the following:

 **II 1G Ex ia IIC T4 ... T6 Ga or
II 2G Ex ia IIC T4 ... T6 Gb and / or
II 1D Ex ia IIC T36 °C or T115 °C Da**

This certificate is issued on 8 April 2016 and, as far as applicable, shall be revised before the date of cessation of presumption of conformity of (one of) the standards mentioned above as communicated in the Official Journal of the European Union.

DEKRA Certification B.V.


R. Schuller
Certification Manager

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T +31 88 96 83000 F +31 88 96 83100 www.dekra-certification.com Registered Arnhem 09085396

EU-Konformitätserklärung Nr. 5021-1M

TURCK

Wir, **HANS TURCK GMBH & CO KG**
WITZLEBENSTR. 7, D – 45472 MÜLHEIM A.D. RUHR

erklären in alleiniger Verantwortung, dass die Produkte
declare under sole responsibility that the products
Zweidraht Näherungsschalter Typ ...Y1... (gemäß EN 60947-5-6 NAMUR)
Two Wire Proximity Sensors Type ...Y1... (according to EN 60947-5-6 NAMUR)

auf die sich die Erklärung bezieht, den Anforderungen der folgenden EU-Richtlinien durch Einhaltung der
folgenden Normen genügen:
to which this declaration relates are in conformity with the requirements of the following EU-directives by compliance with the following
standards:

EMV – Richtlinie / EMC Directive 2014 / 30 / EU 26. Feb. 2014
EN 60947-5-6:2000

Richtlinie / Directive ATEX 2014 / 34 / EU 26. Feb. 2014
EN 60079-0:2012/A11:2013 EN 60079-11:2012

Weitere Normen, Bemerkungen
additional standards, remarks

Zusätzliche Informationen:
Supplementary information:

Angewandtes ATEX-Konformitätsbewertungsverfahren / ATEX - conformity assessment procedure applied:
Modul B + Modul D / E / module B + module D / E

EU-Baumusterprüfbescheinigung (Modul B) KEMA 02 ATEX 1090 X / EU-type examination certificate (module B):
ausgestellt von / issued by: DEKRA Certification B.V., Kenn-Nr. / number 0344,
Utrechtseweg 310, NL-6812 AR Arnhem

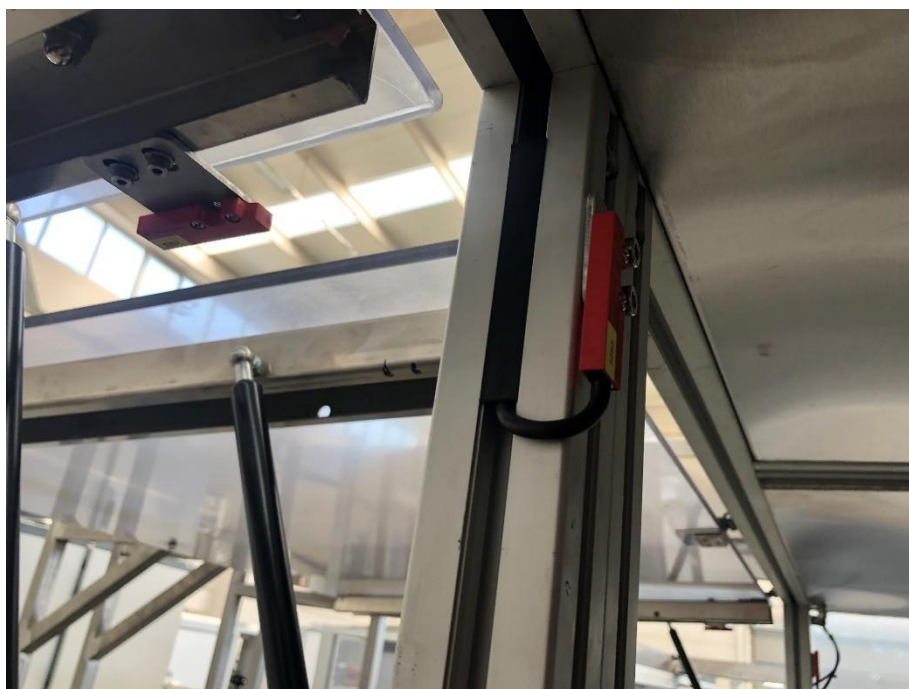
Zertifizierung des QS-Systems gemäß Modul D durch:
certification of the QS-system in accordance with module D by:
Physikalisch Technische Bundesanstalt, Kenn-Nr. / number 0102,
Bundesallee 100, D-38116 Braunschweig

Mülheim, den 27.01.2017

Ort und Datum der Ausstellung /
Place and date of issue


i.V. Dr. M. Lindo, Leiter Zulassungen / Manager Approvals
Name, Funktion und Unterschrift des Befehlten /
Name, function and signature of authorized person

You can see the door switches in the following photos. After any intervention, make it suitable for atex with suitable products. Otherwise, you may be putting your life at risk.



Approved certificate of the safety switch used in the machine.

 **INERIS**

(2) Equipment and protective systems intended for use in potentially explosive atmospheres
Directive 94/9/EC

(1) **EC-TYPE EXAMINATION CERTIFICATE**

(3) Number of the EC type examination certificate: **INERIS 07ATEX0036**

(4) Equipment or protective system:
SAFETY SWITCH TYPE XCS 2C...EX

(5) Manufacturer: **SCHNEIDER**

(6) Address: **ZI N°3
F - 16340 L'Isle d'Espagnac**

This equipment or protective system and any other acceptable alternative of this one are described in the annex of this certificate and the descriptive documents quoted in this annex.

(8) INERIS, notified body and identified under number 0080, in accordance with article 9 of Council Directive 94/9/EC of the 23rd March 1994, certifies that this equipment or protective system fulfils the Essential of Health and Safety Requirements relating to the design and construction of equipment and protective systems intended for use in potentially explosive atmospheres, described in annex II of the Directive.

The examinations and the tests are consigned in confidential report No P92475/07.

(9) The respect of the Essential Health and Safety Requirements is ensured by:

- conformity with:
EN 60 079-0 : 2006
EN 60 079-11 : 2007
EN 60 079-18 : 2004
EN 60 079-26 : 2004
EN 61241-0 : 2006
EN 61241-1 : 2004
EN 61241-11 : 2007
- specific solutions adopted by the manufacturer to meet the Essential Health and Safety Requirements described in the descriptive documents.

Only the entire document including annexes may be reprinted. IM1337AB

Sheet 1 / 4

Parc Technologique Alada BP 2 F-60550 Verneuil-en-Halatte
tél + 33(0)3 44 55 66 77 fax + 33(0)3 44 55 66 99 Internet www.ineris.fr
Institut national de l'environnement industriel et des risques
Etablissement public à caractère industriel et commercial - RCS Seine B 381 984 921 - Siret 381 984 921 00019 - APE 743B

EC-Type Examination Certificate N° INERIS 07ATEX0036

(10) Sign X, when it is placed following the Number of the EC type examination certificate, indicates that this equipment and protective system is subjected to the special conditions for safe use, mentioned in the annex of this certificate.

(11) This EC type examination certificate relates only to the design, examination and tests of the specified equipment or protective system in accordance to the directive 94/9/EC. Further requirements of the Directive apply to the manufacturing process and supply of this equipment or protective system, these are not covered by this certificate.

(12) The marking of the equipment or the protective system will have to contain:

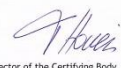
 **Ex II 1 G D**

 **Ex II 2 G D**

Ex Ia IIB T4
Ex IaD 20 T135°C
Ex mb IIB T4
Ex tD A21 IP67 T135°C

Verneuil-en-Halatte, 2007 11 19

 **INERIS**
T. BELBAERE
Project Manager at the ATEX
Equipment Evaluation Laboratory


Director of the Certifying Body,
By delegation
T. HOUËIX
Certification Officer
Certification Division

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Sheet 2 / 4

EC-Type Examination Certificate N° INERIS 07ATEX0036

(13) **ANNEX**

(14) **EC TYPE EXAMINATION CERTIFICATE N° INERIS 07ATEX0036**

(15) **DESCRIPTION OF THE EQUIPMENT OR THE PROTECTIVE SYSTEM**

The safety switch XCS 2C...EX is a proximity detector constituted with potted interrupters (ILS).

PARAMETERS RELATING TO THE SAFETY

The input characteristics are:

Ui (V)	Ii (A)	CI (pF)	LI (pH)
60	0,1	0 + 200 pF per meter of cable	0 + 1pH par meter of cable

MARKING

Marking has to be readable and indelible; it has to include the following indications:

SCHNEIDER
F - 16340 L'Isle d'Espagnac
XCS 2C...EX
INERIS 07ATEX0036
(Serial number)
(Year of construction)
 **Ex II 2 G D**
Ex mb IIB T4
Ex tD A21 IP67 T135°C
Tamb : -25°C to +85°C
Ue = 60 V ; Ie = 0,1 A or Pe = 3 VA

or  **Ex II 1 G D**
Ex Ia IIB T4
Ex IaD 20 T135°C

Marking may be carried out in the language of the country of use.
The protective system or equipment has also to carry the marking normally stipulated by its construction standards.

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EC-Type Examination Certificate N° INERIS 07ATEX0036

ROUTINE EXAMINATIONS AND TESTS

Each apparatus defined above has to have successfully passed the following individual tests before delivery:

- In accordance with § 9.1 from EN 60079-18 standard, a visual examination of encapsulation.
- In accordance with § 9.2 from EN 60079-18 standard, a 500 V AC dielectric test between the circuit and the case during at least one second.

(16) **DESCRIPTIVE DOCUMENTS**

The descriptive documents quoted hereafter constitute the technical documentation of the equipment, subject of this certificate.

- Technical file revision B of 2007.07.19 (8 pages) of 2007.09.26
- Instructions (2 pages) of 2007.10.22

(17) **SPECIAL CONDITIONS FOR SAFE USE**

None.

(18) **ESSENTIAL SAFETY AND HEALTH REQUIREMENTS**

The respect of the Essential Health and Safety Requirements is ensured by:

- Conformity to the European standard EN 60079-0, EN 60079-11, EN 60079-18, EN 60079-26, EN 61241-0, EN 61241-1 and EN 61241-11.
- All provisions adopted by the manufacturer and defined in the descriptive documents.

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Sheet 4 / 4

You can see the ex button and atex sensors on the bobbins in the following photos. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex.



[illegible]



Erklärung

Ident.-No.: 9174732_YUF7

im Sinne der Richtlinie 2014/34/EU

Hiermit erklären wir, dass die Geräte (siehe Seite 4),

der Produktfamilie W27X-3..... (ATEX)

mit den grundlegenden Anforderungen der auf Seite 2 aufgeführten EG-Richtlinien übereinstimmen. Bei einer nicht um uns abgestimmten Änderung eines auf der Seite 4 aufgeführten Gerätes verliert diese Erklärung, für dieses Gerät, ihre Gültigkeit.

Folgendes Prüfergebnis wurde für die aufgeführten Geräte erzielt:



830 EX nA op in IB T4 Gc X
830 EX to IB T130°C Dc X
 -20°C < Ta < +50°C

Damit diese Schutzklasse wirksam sein muss, wird vor der Inbetriebnahme die Vollständigkeit des Schutzgehäuses geprüft und gegebenenfalls durch den Betreiber angebracht und befestigt werden.

Ausnahmen der auf Seite 2 aufgeführten Normen sind der Betriebsanleitung zu entnehmen.

Die Sicherheitshinweise der mitgelieferten Produktokumentationen sind zu beachten.

Declaration

In Compliance with the Directive 2014/34/EU

We hereby declare that the devices (see page 4)

of the product family W27X-3..... (ATEX)

comply with the basic requirements of the EC Directive specified under page 2. If an item of equipment listed on page 4 is modified without our approval then this declaration loses its validity for this equipment.

Following inspection result have been achieved for the listed devices:



830 EX nA op in IB T4 Gc X
830 EX to IB T130°C Dc X
 -20°C < Ta < +50°C

To validate this protection class, the completeness of the protective housing has to be verified prior to commissioning and, if necessary, the protective housing has to be attached and installed by the operator.

Exceptions to the standards listed on page 2 can be found in the operating instructions.

The safety instructions contained in the product documentation must be observed.

2021-01-04

9108441.pdf

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Pos	Pos_text	Short_desc	Document_No.	Issued	SICK_Id-No.	Pos_Exch
0110	QJEU L 96	ENM DIRECTIVE 2014/30/UE	RL 2014/30/UE	2014-02-26	8606191	
0120	QJEU L 96	RHS DIRECTIVE 2011/65/UE	RL 2011/65/UE	2011-06-08	8606079	
0150	QJEU L 96	ATEX DIRECTIVE 2014/34/UE	RL 2014/34/UE	2014-06-06	8606190	
1230	LVD	LOW-VOLT. SWITCH. A. CONTROL. SWITCH. ELEM.	EN 60974-5	2007-12-01	8605900	
1240	LVD	LOW-VOLTAGE SWITCHGEAR AND CONTROLLER	EN 60974-5-2/A1	2012-11-01	8606124	
3090	ATEX	HAZARDOUS SUBSTANCES	EN 60581	2010-09-08	8606085	
4070	ATEX	EXPLOSIVE ATMOSPHERES, EQUIPMENT PROTECT.	EN 60079-5	2010-05-01	8606045	
4170	ATEX	EXPLOSIVE ATMOSPHERES, EQUIPMENT	EN 60079-10	2010-08-01	8606139	
4175	ATEX	EXPLOSIVE ATMOSPHERES	EN 60079-0/A11	2011-07-01	8606140	
4180	ATEX	EXPLOSIVE ATMOSPHERES, PROTECT. EQUIP. OPTIC. RAD.	EN 60079-18	2011-07-01	8606141	
4190	ATEX	EXPLOSIVE ATMOSPHERES, DUST INSUIT. PROTECTION	EN 60079-31	2014-07-01	8606187	

as of 2021-01-04

List of devices (count: 10)	
SICK-ID-No.	Device-type
1027988	WTB27X-3P1811
1027989	WL27X-3P1831
1027991	WESE27X-3P1830
1029955	WL27X-3P3431
1066386	WTE27X-3P1861
1069183	WLB27X-3P3411
2033627	WE27X-3P1830 EINWEG-LICHTSCHRANKE
2033628	WS27X-3D1830 EINWEG-LICHTSCHRANKE
2040634	WE27X-3P1830 EINWEG-LICHTSCHRANKE
2040635	WS27X-3D1830 EINWEG-LICHTSCHRANKE

as of 2021-01-04

Approved certificate of atex valves used in the machine.



CERTIFICATE

(1) EC-Type Examination

(2) **Equipment and protective systems intended for use in potentially explosive atmospheres - Directive 94/9/EC**

(3) EC-Type Examination Certificate Number: **DEKRA 11ATEX0273 X** Issue Number: 2

(4) **Equipment:** Solenoid valves, SY5000, SY7000, SY9000, SYJ300, SYJ500, SYJ700, SYJ3000, SYJ5000, SYJ7000 and VQZ2000 series, with 52-SY1.6-..., 52A-SY1.6-... and 57A-SY1.6-... series pilot valves

(5) **Manufacturer:** SMC Corporation

(6) **Address:** 4-14-1 Sotokanda, Chiyoda-ku, Tokyo 101-0021, Japan

(7) This equipment and any acceptable variation thereto is specified in the schedule to this certificate and the documents therein referred to.

(8) DEKRA Certification B.V., notified body number 0344 in accordance with Article 9 of the Council Directive 94/9/EC of 23 March 1994, certifies that this equipment has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment and protective systems intended for use in potentially explosive atmospheres given in Annex II to the directive.

The examination and test results are recorded in confidential test report number NL/DEK/ExTR15.0073/00.

(9) Compliance with the Essential Health and Safety Requirements has been assured by compliance with:

EN 60079-0 : 2012 + A11 : 2013 EN 60079-11 : 2012 EN 13463-1 : 2009

(10) If the sign "X" is placed after the certificate number, it indicates that the equipment is subject to special conditions for safe use specified in the schedule to this certificate.

(11) This EC-Type Examination Certificate relates only to the design, examination and tests of the specified equipment according to the Directive 94/9/EC. Further requirements of the directive apply to the manufacturing process and supply of this equipment. These are not covered by this certificate.

(12) The marking of the equipment shall include the following:



II 1 G Ex Ia IIC T6 ... T4 Ga

This certificate is issued on 9 December 2015 and, as far as applicable, shall be revised before the date of cessation of presumption of conformity of (one of) the standards mentioned above as communicated in the Official Journal of the European Union.

DEKRA Certification B.V.



R. Schuller
Certification Manager

Page 1/2



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DEKRA Certification B.V. Meander 1051, 6825 MJ Arnhem P.O. Box 5185, 6802 ED Arnhem The Netherlands
T +31 88 96 83000 F +31 88 96 83100 www.dekra-certification.com Registered Arnhem 09085396

Atex transmitter and float valve.



Approved certificates of Atex transmitter and float valve.

[1] **EU-TYPE EXAMINATION CERTIFICATE**

[2] **Equipment or Protective System intended for use in Potentially Explosive Atmospheres**
Directive 2014/34/EU

[3] EU-Type Examination Certificate Number: **DEMKO 19 ATEX 2222X Rev. 0**

[4] Product: **VEGABAR 28 (T) 29 (T) 38 (T) 39 (T) Pressure Transmitter**

[5] Manufacturer: **VEGA Grieshaber KG**

[6] Address: **Am Hohenstein 113, Schiltach, 77761 Germany**

[7] This product and any acceptable variation thereto are specified in the schedule to this certificate and the documents therein referred to.

[8] UL International Demko A/S, notified body number 0539 in accordance with Article 17 of the Council Directive 2014/34/EU of 26 February 2014, certifies that this product has been found to comply with the Essential Health and Safety Requirements relating to design and construction of products intended for use in potentially explosive atmospheres given in Annex II to the Directive. The examination and test results are reported in confidential report no. **4788060674.4.1**

[9] Compliance with the Essential Health and Safety Requirements has been assured by compliance with:

EN IEC 60079-0:2018	EN 60079-11:2012	EN 60079-26:2015
----------------------------	-------------------------	-------------------------

[10] If the sign "C" is placed after the certificate number, it indicates that the product is subject to special conditions for safe use specified in the schedule to this certificate.

[11] This EU-Type Examination Certificate relates only to the design and construction of the specified product. Further requirements of the Directive apply to the manufacturing process and supply of the product. These are not covered by the certificate.

[12] The marking of the product shall include the following:

 II 1 G	Ex ia IIC T4 Ga or
 II 1/2 G	Ex ia IIC T4 Ga/Gb or
 II 2 G	Ex ia IIC T4 Gb

This is to certify that the conformity of this Product (equipment or system) (Certificate Product) has been investigated and found to conform with the standards indicated on this Certificate, in accordance with the ATEX Product Certification Program Requirements. This certificate and any marks obtained apply only to the product (sample(s)) submitted by the Manufacturer. UL does not warrant the conformity or otherwise of the product (sample(s)) provided nor responsibility of other manufacturers' product. UL has not supervised either the Design or other manufacturing of the product (sample(s)) and this responsibility for conformity of product to all applicable standards, specifications, requirements or Directives. The test results may not be used, in whole or in part, in any other document without UL's prior written approval.

Certification Manager
Jan-Erik Storgaard

Jan-Erik Storgaard

Date of issue: 2019-12-10

Notified Body
UL International Demko A/S, Borupvang 5A, 2750 Ballerup, Denmark
Tel. +45 44 85 65 65, info.dk@ul.com, www.ul.com





UL-UL-0555-1 - Issue 21.0

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Page 1 of 3
02005-EX-191210

 **IEP ENERGY PETROLEUM INSTITUTE** 

[1] **EC-Type Examination Certificate**

[2] **Equipment or Protective Systems Intended for use in Potentially Explosive Atmospheres**
Directive 94/9/EC

[3] EC - Type Examination Certificate Number : **IEP 11 ATEX 073**

[4] Equipment : **F1 Ex type Level Switch**

[5] Manufacturer : **KAYSE Endüstriyel Mamüller San. Ve Tic. A.Ş.**

[6] Address : **Orhanlı Orta Mah. Aktepe Sanayi Sitesi No:12/B Tuzla / İstanbul - TURKEY**

[7] This equipment or protective system and any of acceptable variation thereto is specified in the schedule to this certificate and the documents therein referred to.

[8] The IEP Uluslararası Enerji Petrol Gazetimi, Sertifikasyon ve Teknik Hizmetler Organizasyonu Tic. Ltd. Sti. , notified body number 2284 in accordance with of the Council Directive 94/9/EC of 23 March 1994 certifies that this equipment or protective system has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment and protective systems intended for use in potentially explosive atmospheres given in Annex II to the Directive. The examination and test results are recorded in confidential Report N° IEP- RP.Ex - 10.388 date 21.12.2011

[9] Compliance with Essential Health and safety requirements has been assured by compliance with :

EN 13463-1:2009 , EN 13463-5:2003 , EN 60079-0:2007

[10] If the sign " X " is placed after the certificate number , it indicates that the equipment or protective system is subject to special conditions for safe use specified in the schedule to this certificate.

[11] This EC-Type Examination Certificate relates only to the design examination and testing of the specified equipment or protective system in accordance to the directive 94/9/EC. Further requirements of the directive apply to the manufacturing process and supply of this equipment or protective system. These are not covered by this certificate.

[12] The marking of the equipment or protective system shall include the following :

 II 1 G c IIA T6

Responsible person :

Nurettin Terzioğlu

Head of certification body

Date of issue : 22 December, 2011



IEP Uluslararası Enerji Petrol Gazetimi, Sertifikasyon ve Teknik Hizmetler Organizasyonu Tic. Ltd. Sti.
MTK Sitesi 5746/1 Sok. No:9 K:3 Camialtı - IZMİR / TURKEY Tel : +90 232 431 17 45 46 & Fax : +90 232 431 17 30
Number of Pages : 1 / 3

This certificate is granted subject to the general conditions of the IEP Energy Petroleum Institute.
This certificate may only be reproduced in its entirety and without any change , schedule included.

You can see the ex buttons used on the machine screen. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex.



Approved certificate of the HMI used in the machine.



You can see the ex buttons and button boxes used on the machine. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex.



Approved certificates of buttons and button boxes used in the machine.

Certificate Number	Issue	
CML 14ATEX3073X	10	
1 Equipment intended for use in Potentially Explosive Atmospheres Directive 2014/34/EU		
2 Component 07-351-***** ComEx Control and Indicating Stations		
3 Manufacturer BARTEC GmbH		
4 Address Max-Eyth-Strasse 16, 97980 Bad Mergentheim, Germany		
5 The equipment is specified in the description of this certificate and the documents to which it refers.		
6 CML B.V., Chamber of Commerce No 6738671, Hoogvordendreef 15, Amsterdam, 1101 BA, The Netherlands, Notified Body Number 2776, in accordance with Article 17 of Directive 2014/34/EU of the European Parliament and of the Council, dated 26 February 2014, certifies that this equipment has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment intended for use in potentially explosive atmospheres given in Annex II to the Directive.		
7 The examination and test results are recorded in the confidential reports listed in Section 12.		
8 If an 'X' suffix appears after the certificate number, it indicates that the equipment is subject to conditions of safe use (affecting correct installation or safe use). These are specified in Section 14.		
9 This EU Type Examination certificate relates only to the design and construction of the specified equipment or component. Further requirements of Directive 2014/34/EU Article 13 apply to the manufacture of the equipment or component and are separately certified.		
10 Compliance with the Essential Health and Safety Requirements, with the exception of those listed in the confidential report, has been demonstrated through compliance with the following documents:		
EN 60079-0:2012+A11:2013	EN 60079-1:2015	EN 60079-31:2014
EN 60079-1:2014	EN 60079-11:2012	
10 The equipment shall be marked with the following:		
 II 2 G D	 II 2 G D	
Ex db eb IIC T6 Gb	Ex db eb Ia IIC T6 Gb	
Ex tb IIIC T80°C Db	Ex tb IIIC T80°C Db	
Up to -55°C ≤ Ta ≤ +60°C	Up to -55°C ≤ Ta ≤ +60°C	
This certificate shall only be copied in its entirety and without change www.CMLEx.com		
1 of 6		
R C Marshall Certification Officer		

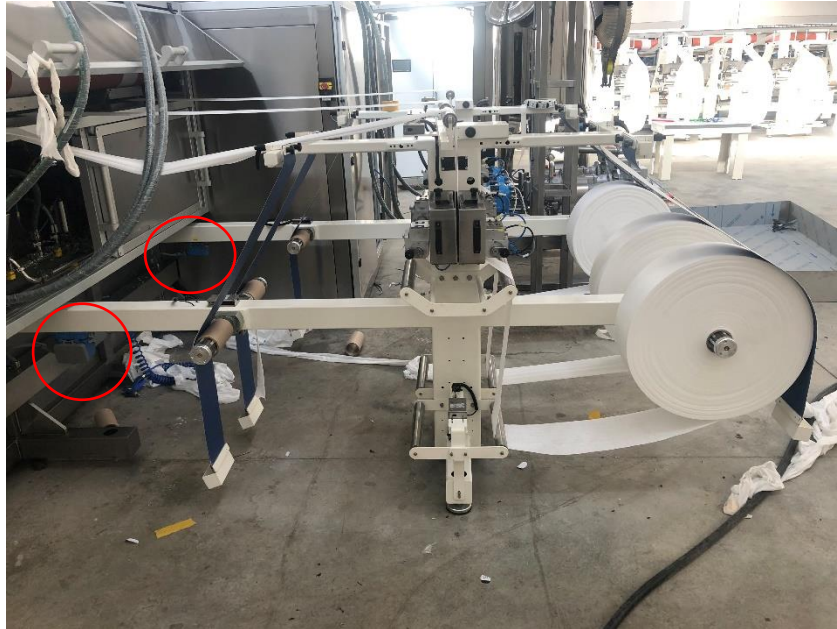
Certificate Number	Issue	
CML 17ATEX1105U	3	
1 Equipment intended for use in Potentially Explosive Atmospheres Directive 2014/34/EU		
2 Component Switch Module Type 07-332-***** Control Switch Type 07-333-***** Switch Module, 4-pole Type 07-3381-*****		
3 Manufacturer Bartec GmbH		
4 Address Max-Eyth-Strasse 16, 97980 Bad Mergentheim, Germany		
5 The component is specified in the description of this certificate and the documents to which it refers.		
6 CML B.V., Chamber of Commerce No 6738671, Hoogvordendreef 15, Amsterdam, 1101 BA, The Netherlands, Notified Body Number 2776, in accordance with Article 17 of Directive 2014/34/EU of the European Parliament and of the Council, dated 26 February 2014, certifies that this component has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment intended for use in potentially explosive atmospheres given in Annex II to the Directive.		
7 The examination and test results are recorded in the confidential reports listed in Section 12.		
8 The 'U' suffix after the certificate number indicates that the component is subject to conditions of safe use (affecting correct installation or safe use). These are specified in Section 14.		
9 This EU Type Examination certificate relates only to the design and construction of the specified equipment or component. Further requirements of Directive 2014/34/EU Article 13 apply to the manufacture of the equipment or component and are separately certified.		
10 Compliance with the Essential Health and Safety Requirements, with the exception of those listed in the confidential report, has been demonstrated through compliance with the following documents:		
EN 60079-0:2012+A11:2013	EN 60079-1:2014	EN 60079-7:2015
10 The component shall be marked with the following:		
 M 2	 II 2 G	
Ex db eb I Mb	Ex db eb IIC Gb	
Ta = -55°C ≤ Ta ≤ +40°C (16 A)	Ta = -55°C ≤ Ta ≤ +40°C (16 A)	
Ta = -55°C ≤ Ta ≤ +60°C (11 A)	Ta = -55°C ≤ Ta ≤ +60°C (11 A)	
Ts = -55°C ≤ Ts ≤ +85°C	Ts = -55°C ≤ Ts ≤ +85°C	
This certificate shall only be copied in its entirety and without change www.CMLEx.com		
1 of 5		
R C Marshall Certification Officer		

Certificate Number	Issue	
CML 17ATEX1119U	1	
1 Equipment intended for use in Potentially Explosive Atmospheres Directive 2014/34/EU		
2 Component Control and Switching Unit, Type 07-337-*****		
3 Manufacturer BARTEC GmbH		
4 Address Max-Eyth-Strasse 16, 97980 Bad Mergentheim, Germany		
5 The component is specified in the description of this certificate and the documents to which it refers.		
6 CML B.V., Chamber of Commerce No 6738671, Hoogvordendreef 15, Amsterdam, 1101 BA, The Netherlands, Notified Body Number 2776, in accordance with Article 17 of Directive 2014/34/EU of the European Parliament and of the Council, dated 26 February 2014, certifies that this component has been found to comply with the Essential Health and Safety Requirements relating to the design and construction of equipment intended for use in potentially explosive atmospheres given in Annex II to the Directive.		
7 The examination and test results are recorded in the confidential reports listed in Section 12.		
8 The 'U' suffix after the certificate number indicates that the component is subject to conditions of safe use (affecting correct installation or safe use). These are specified in Section 14.		
9 This EU Type Examination certificate relates only to the design and construction of the specified equipment or component. Further requirements of Directive 2014/34/EU Article 13 apply to the manufacture of the equipment or component and are separately certified.		
10 Compliance with the Essential Health and Safety Requirements, with the exception of those listed in the confidential report, has been demonstrated through compliance with the following documents:		
EN 60079-0:2012+A11:2013	EN 60079-1:2014	EN 60079-7:2015
10 The component shall be marked with the following:		
 II 2 G	 I M 2	
Ex db eb IIC Gb	Ex db eb I Mb	
Ta = -55°C to 40/50/60°C	Ta = -55°C to 40/50/60°C	
This certificate shall only be copied in its entirety and without change www.CMLEx.com		
1 of 4		
R C Marshall Certification Officer		

You can see the ex plug sockets used in the machine. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. The tightness of the unions should be checked from time to time. The cables must not move out or in when pulled.



You can see the ex plug sockets used in the machine. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. The tightness of the unions should be checked from time to time. The cables must not move out or in when pulled.



Documents for ex plug sockets used in the machine.

Han® Ex Sets

Connectors for explosion hazardous environments – standard sizes

Features

- Hoods and housings in the sizes 6, 10 B, 16 B, 24 B and 48 B
- Connectors especially for explosion hazardous applications
- Suitable for intrinsically safe circuits
- Inserts on basis of Han® E with 6 to 64 contacts
- Suitable for areas classed 1 and 2

WARNING! Industrial connectors of the Han® Ex series are designed exclusively for the use in pulverized safe electrical circuits of categories „I“, „II“ and „III“

The explosion group is defined by the corresponding intrinsically safe equipment.

- Temperature class according to DIN EN 60 079-11

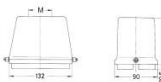
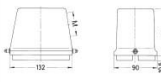
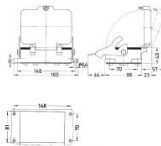
General description

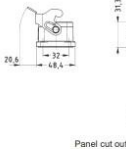
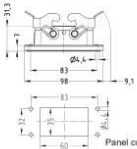
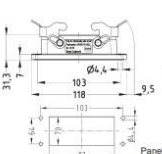
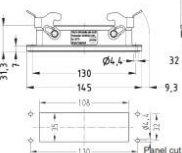
The connectors are designed to meet the intrinsic safety requirements for ignition protection class in explosion hazardous areas classed as 1 and 2. In intrinsically safe circuits, energy is limited in such a manner that even a potential spark cannot ignite an explosive environment.

The Han® Ex product portfolio offers complete connector systems consisting of housings and inserts, including housings made from an alloy that can be used in pulverized methane-coal dust atmospheres. They also offer ignition protection class 65 in the mated condition. The housing's blue colour indicates that an intrinsically safe circuit is present. The Han® sets provide a high number of pins and meet the standards of the ignition protection class even in the tightest of spaces.

Technical characteristics

Specifications	DIN EN 60 079-0, -11, -14 DIN EN 60 664-1 DIN EN 61 984
Hoods/ housings	
Material	zinc die cast
Colour	RAL 9015 (blue)
Surface	powder coated
Locking element	stainless steel
Lever type	metal lever
Seal	NBR
Ambient temperature	-20 °C ... +40 °C
Limit temperature for connectors	-20 °C ... +125 °C
Maximum surface temperature	T ₆ = 85 °C
Protection degree	according to DIN EN 60 529 in locked position
IP65	is achieved with cable gland
Inserts	
Number of contacts	6, 10, 16, 24
Rated current	16 A
Rated voltage	90 V
Insulation resistance	≥ 10 ¹⁰ Ω
Material	polycarbonate
Limiting temperature	-20 °C ... +40 °C
Mechan. working life - mating cycles	≥ 500
Contacts	
Material	copper alloy
Surface	
hard-silver plated	3 µm Ag
contact resistance	≤ 1 mΩ
Crimp termination	
AWG 20 ... 14	0.15 ... 2.5 mm ²
0.75 ... 2.5 mm ²	
Screw connection	
Tightening torque	0.5 Nm
Max. insulation diameter	3.0 mm

Han® Ex Sets				
Identification	Part number	Cable entry	Drawing	Dimensions in mm
Hood 48 B top entry 	19 36 048 0447 19 36 048 0448	M32 M40		
Hood 48 B side entry 	19 36 048 0547 19 36 048 0548	M32 M40		
Hood 48 B with cover 	09 36 048 0301			Panel cut out

Hoods and Housings – Sizes 6 B, 10 B, 16 B, 24 B, 48 B			
Identification	Part number	Drawing	Dimensions in mm
Housing, bulkhead mounted 6 B 	09 36 006 1301	 Panel cut out	31,3 7 32 48,4 85 70 70 1,0 45
Housing, bulkhead mounted 10 B 	09 36 010 1301	 Panel cut out	31,3 7 81 98 83 21 60 1,0 60 58 32 48,4
Housing, bulkhead mounted 16 B 	09 36 016 1301	 Panel cut out	31,3 7 103 118 103 64 82 58 32 48,4
Housing, bulkhead mounted 24 B 	09 36 024 1301	 Panel cut out	31,3 7 130 145 108 1,0 130 58 32 48,4

Documents for Ex plug sockets used in the machine.

CERTIFICATE **ZELM ex**

(1) Production Quality Assurance Notification

(2) Equipment and Protective Systems Intended for Use in Potentially Explosive Atmospheres - Directive 94/9/EC

(3) Notification Number: **ZELM 14 ATEX 1528**

(4) Product group(s): Development, production and distribution of RFID systems in the decisive type of protection "intrinsic safety"

A list of the EC-Type Examination Certificates covered by this notification is held by the notified body.

(5) Applicant: **HARTING Electric GmbH & Co. KG**
Marienwerder Strasse 3
D-32339 Espelkamp - Germany

(6) Actual manufacturer: **HARTING Electric GmbH & Co. KG**
Weher Strasse 151
D-32369 Rahden - Germany

(7) The Prüf- und Zertifizierungsstelle ZELM Ex, notified body No. 0820 for Annex IV in accordance with Article 9 of the Council Directive 94/9/EC of 23 March 1994, notifies to the applicant that the actual manufacturer has a production quality system which complies to the Annex IV of the Directive.

(8) This notification is based on the confidential audit report No. AB 1406004, issued 2014-06-27. This notification is valid until 2017-06-26 and can be withdrawn if the actual manufacturer no longer satisfies to the requirements of Annex IV.

Results of periodical reassessment of the quality system production within the period of validity are a part of this notification.

(9) According to Article 10 (1) of the Directive 94/9/EC the CE-Marking shall be followed by the identification number 0820 of ZELM Ex as the notified body which is involved in the production control stage.

Braunschweig 2014-07-04

ZELM ex
Zertifizierungs-
stelle

Zertifizierungsstelle ZELM Ex
Dipl.-Ing. Harald Zelm

ZELM ex
Prüf- und Zertifizierungsstelle
Strüßberger Str. 50-52, 38104 Braunschweig

Page 1 of 1

Notifications without signature and stamp are not valid. The notification may only be issued without alterations. Extracts or alterations are subject to approval by the Prüf- und Zertifizierungsstelle ZELM Ex.
This English version is based on the German text; in the case of dispute, the German text shall prevail.

Supplier's Declaration of Conformity
(in accordance with ISO/IEC 17050-1)
Konformitätserklärung eines Anbieters
(nach ISO/IEC 17050-1)

Issuer's name: HARTING Electric GmbH & Co. KG
Name des Ausstellers

Issuer's address: Wilhelm-Harting-Str. 1
32339 Espelkamp
Germany
Anschrift des Ausstellers

Product(s) Han Ex®
Produkt(e)

See Attachment
Siehe Anhang

The product(s) as described above are in conformity with the requirements of:
Die oben beschriebenen Produkte sind konform mit den folgenden Anforderungen:

Ex ia III T6

Applied standards:
Angewandte Normen:

Reference number Referenznummer	Edition Ausgabe
DIN EN 60079-0	2014
DIN EN 60079-11	2012
DIN EN 60079-14	2014

Espeikamp, 2016-11-25
Uwe Gräß
Managing Director

Espeikamp, 2016-11-25
Andre Benke
Director Product & Industry
Segment Management

The given information reflects our current knowledge. It is based on statements and information of our suppliers and third parties. This declaration does not include guarantee of quality or durability. Topicality, completeness and accuracy of the content are subject to change without notice. The safety instructions are to be respected. The user is self-responsible for the evaluation of the information and the concrete measures derived from it. Besides that, HARTING is liable only for culpable, intentional or grossly negligent acting as long as no mandatory legal liability for damages of life, body, health or Product Liability Act exists.
Die vorstehenden Informationen spiegeln unseren derzeitigen Kenntnisstand, dem insbesondere die Aussagen und Informationen unserer Lieferanten und Dritter zugrunde liegen, wider. Dies beinhaltet jedoch keine Beschaffenheits- oder Haltbarkeitsgarantie. Im Hinblick auf Aktualität, Vollständigkeit und Richtigkeit der Inhalte behalten wir uns Änderungen jederzeit vor. Die Sicherheitsanweisungen sind zu beachten, wobei der Anwender für die Bewertung der Informationen und die daraus abgeleiteten konkreten Vorkehrungen selbst verantwortlich ist. Im Übrigen haftet HARTING nur für schuldhaftes, vorsätzliches oder grob fahrlässiges Handeln, soweit nicht eine zwingende gesetzliche Haftung für Schäden an Leben, Körper oder Gesundheit oder nach dem Produkthaftungsgesetz besteht.

SDaC Han Ex DIN EN 60079.00.pdf 1/2 NOR 089301

A Attachment
Anhang

Part number Artikelnummer	Part number Artikelnummer	Part number Artikelnummer	Part number Artikelnummer
Hoods and Housings			
09360061301	09360241308	19360101521	19360241422
09360061305	09360241309	19360101540	19360241441
09360101301	09360480301	19360101541	19360241442
09360101305	19360081440	19360161421	19360241521
09360101307	19360081441	19360161422	19360241522
09360101308	19360081540	19360161441	19360241541
09360161301	19360081541	19360161442	19360241542
09360161305	19360101420	19360161521	19360480447
09360161307	19360101421	19360161522	19360480448
09360161308	19360101440	19360161541	19360480547
09360241301	19360101441	19360161542	19360480548
09360241307	19360101520	19360241421	
Inserts			
09360062601	09360102602	09360162616	09360242616
09360062602	09360102616	09360162701	09360242701
09360062616	09360102701	09360162702	09360242702
09360062701	09360102702	09360162716	09360242716
09360062702	09360102716	09360242601	09360483001
09360062716	09360162601	09360242602	09360483101
09360102601	09360162602		
Sets			
10360030001	10360040004	10360070001	10360080011
10360030002	10360040005	10360070002	10360080012
10360030004	10360040006	10360070003	10360080014
10360030005	10360040008	10360080006	10360120001
10360030008	10360040011	10360080007	10360120002
10360040001	10360040012	10360080008	10360120003
10360040002	10360050001	10360080009	10360120004
10360040003	10360050002	10360080010	

The given information reflects our current knowledge. It is based on statements and information of our suppliers and third parties. This declaration does not include guarantee of quality or durability. Topicality, completeness and accuracy of the content are subject to change without notice. The safety instructions are to be respected. The user is self-responsible for the evaluation of the information and the concrete measures derived from it. Besides that, HARTING is liable only for culpable, intentional or grossly negligent acting as long as no mandatory legal liability for damages of life, body, health or Product Liability Act exists.
Die vorstehenden Informationen spiegeln unseren derzeitigen Kenntnisstand, dem insbesondere die Aussagen und Informationen unserer Lieferanten und Dritter zugrunde liegen, wider. Dies beinhaltet jedoch keine Beschaffenheits- oder Haltbarkeitsgarantie. Im Hinblick auf Aktualität, Vollständigkeit und Richtigkeit der Inhalte behalten wir uns Änderungen jederzeit vor. Die Sicherheitsanweisungen sind zu beachten, wobei der Anwender für die Bewertung der Informationen und die daraus abgeleiteten konkreten Vorkehrungen selbst verantwortlich ist. Im Übrigen haftet HARTING nur für schuldhaftes, vorsätzliches oder grob fahrlässiges Handeln, soweit nicht eine zwingende gesetzliche Haftung für Schäden an Leben, Körper oder Gesundheit oder nach dem Produkthaftungsgesetz besteht.

SDaC Han Ex DIN EN 60079.00.pdf 2/2 NOR 089302

Grounding and braided wires at many points of the machine should not be removed. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. Ground and braided wires should be checked frequently and should be observed their suitability.



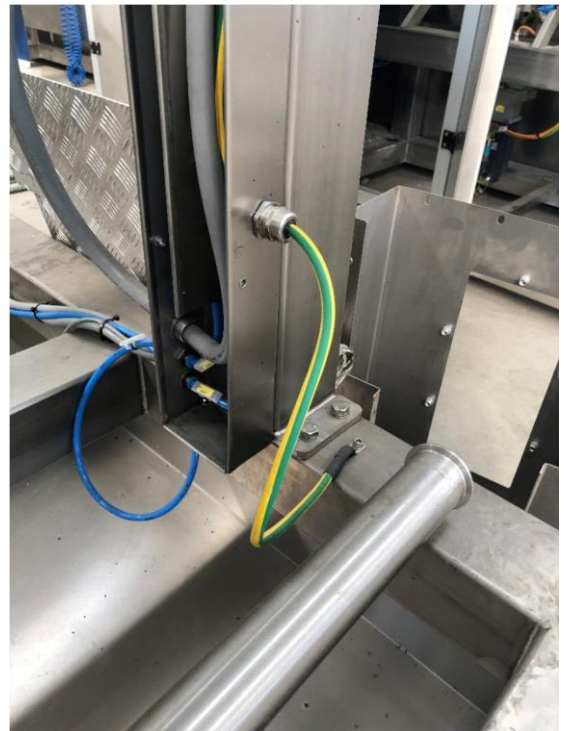
Grounding and braided wires at many points of the machine should not be removed. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. Ground and braided wires should be checked frequently and should be observed their suitability.



Grounding and braided wires at many points of the machine should not be removed. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. Ground and braided wires should be checked frequently and should be observed their suitability. Must be checked with a measuring device.



Grounding and braided wires at many points of the machine should not be removed. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. Ground and braided wires should be checked frequently and should be observed their suitability.



Grounding and braided wires at many points of the machine should not be removed. In case any change or intervention is required, the same product or equivalent products should be used and made suitable for atex. Ground and braided wires should be checked frequently and should be observed their suitability.



The exterior photos of the machine's panel are below.



3 COLOUR TOWER LIGHT
WITH SIREN

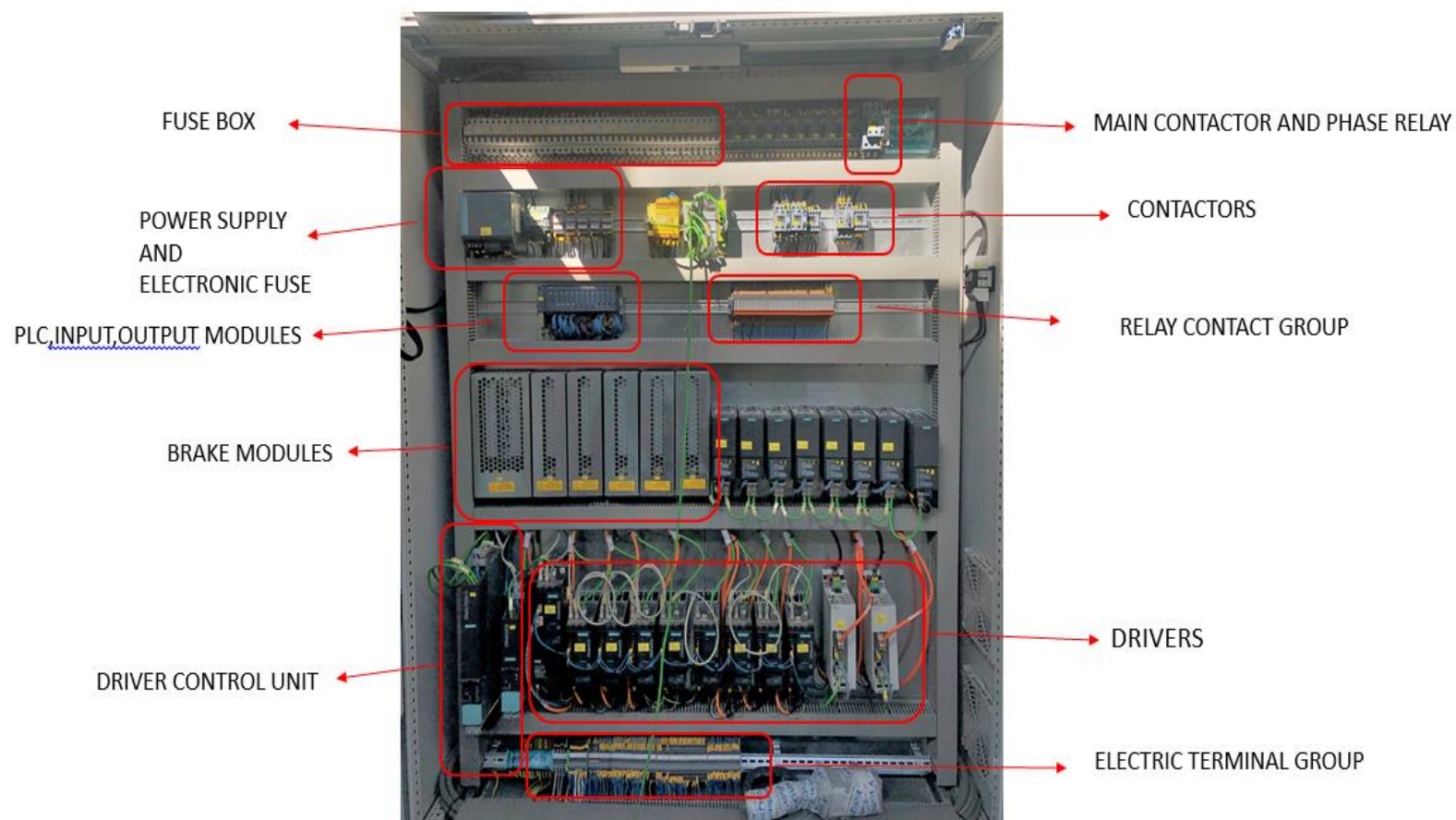
PANEL FAN



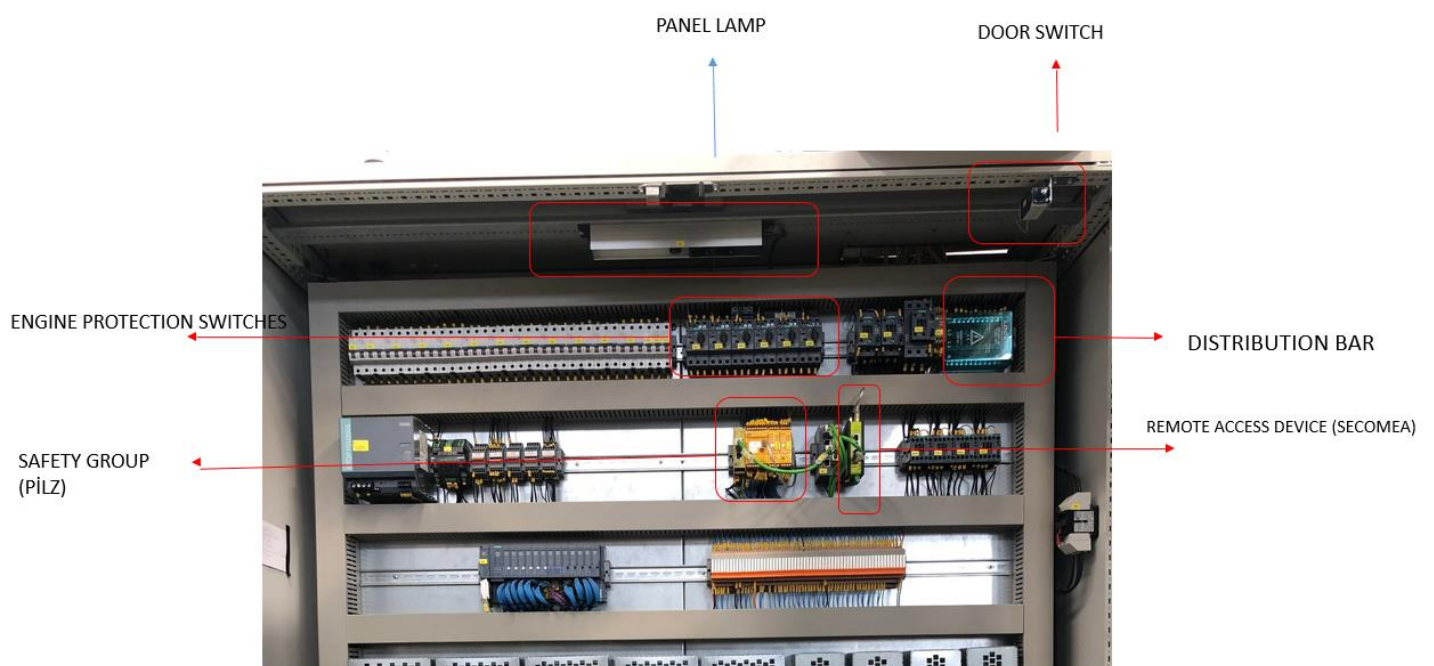
MACHINE ENERGY ON /
OFF (PAKO SWITCH)

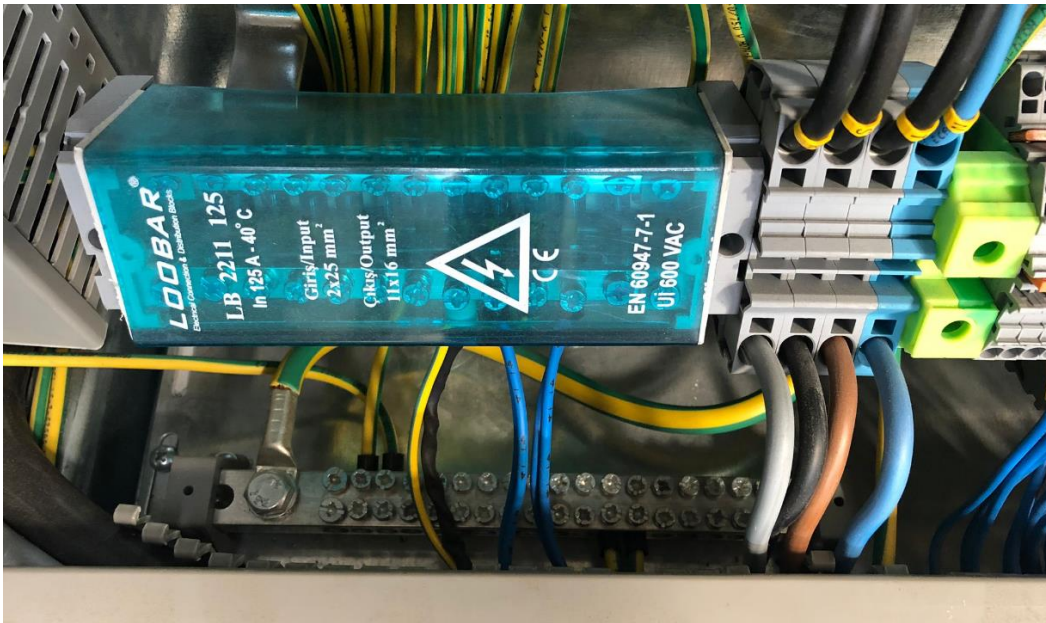
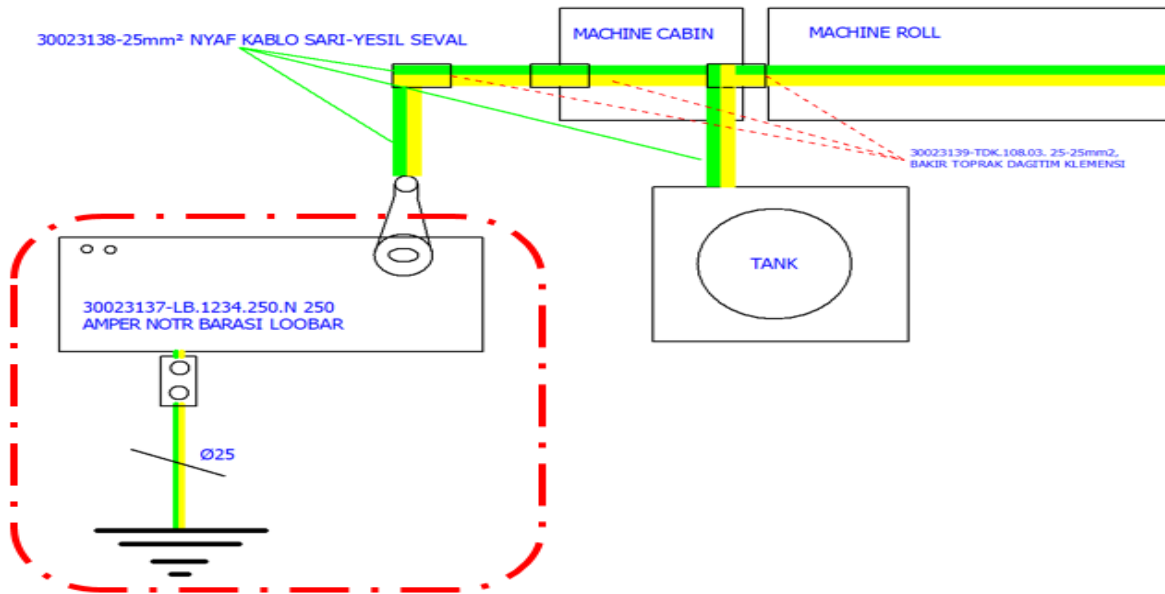
PANEL DOOR KEY

The panel view of the machine is as in the photo. In case any change or intervention is required, the same product or equivalent products should be used and made suitable.



The panel view of the machine is as in the photo. In case any change or intervention is required, the same product or equivalent products should be used and made suitable.





WIRE IT TO YOUR GROUND LINE BY USING 25mm² GROUND WIRE AS SHOWN IN THE ELECTRICAL DRAWING

Technical Features:

A. Capacity:

Cutting speed: 900 cuts/minute (design speed)*

800 cuts/minute (advisable speed with VAC)

Stacking speed: 150 stacks/minute

Linear speed: 150 meters/ minute*

Wipe amount: 2700wipes/minute (design speed)

2400 wipes/minute (advisable speed with VAC)

* The value reached before given numbers defines machine speed.

* Machine capacity may vary due to raw material and material quality.

B. Productivity:

Performance $\geq$ 85*

***Roll chance, cleaning, periodical maintenance etc. are taken into consideration while calculation machine operating conditions.*

Operation Continuity 24/7

C. Cutting System:

9 blades with rotary flexi system cuts with 1 helix angle. Therefore while the use time of blades extends, the cutting is quieter but more effective. With this system the impact is reduced to minimum and with the help of scissor type single point cutting, sensitive and effective cutting is possible with various raw materials with different compositions, especially materials with high polyester ratio. **Also, angled knife system enables easier blade adjustment.**

Maximum Wipe Count

500-600 cuts/minute (advisable speed with no VAC)

800 cuts/minute (advisable speed with VAC)

Cutting Range

120 mm – 230 mm (Optionally: 280 mm)

Blades

Specially hardened stainless steel materia

Bottom Anvil

Specially hardened stainless steel

Bottom and Upper Cylinders

Stainless steel

Designed to endure against high force and absorb vibration

Conveyor Belts

All conveyor belts are polyurethane and in compliance with FDA specifications

D.Unwinding Unit

Roll unwinder unit consists of a module for 1 bobbin. Roll finish sensor placed on the line gives alarm and stops the machine if roll finishes, tears or tangles.

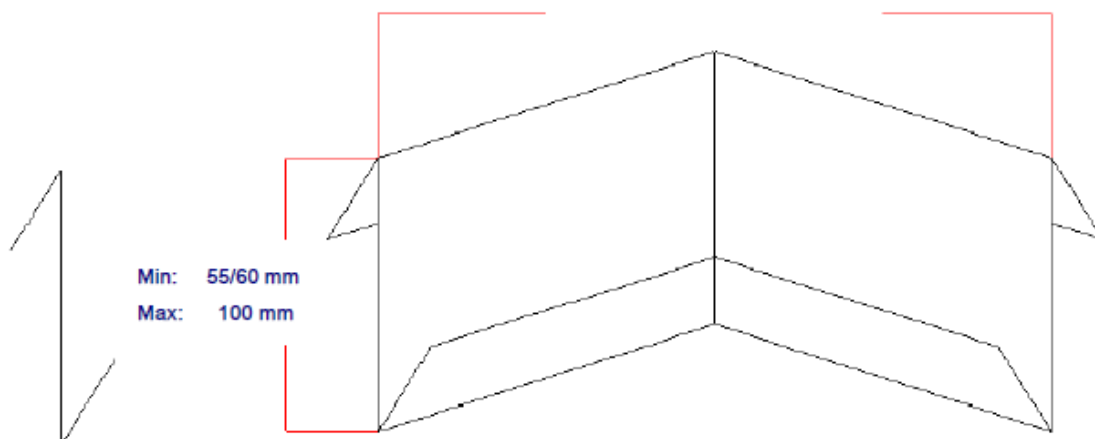
Optionally, automatic splice unit can be added to the bobbin unit.

Roll Specifications and Dimensions	<i>Roll width (for each line)</i>
	Max.: 200 mm*
	Min.: 120 mm
	<i>Max. Bobbin Diameter: 1200 mm</i>
	Core diameter: 76 mm
	<i>Max. Bobbin Weight: 50 kg</i>
	<i>Bobbin number: 1 for each module</i>
	<i>Bobbin change: Automatic non-stop flying splice</i>
	<i>Base nonwoven weight (gr/m²)</i>
	Min: 30 gr/m ²
	Max: 80 gr/m ²
	* It's possible to work with wider rolls for different folding types. Please contact and consult your sales representative for more information.

E. Folding Unit

There is a group of folding plates on the unit. All folding plates have a mechanism to enable easy adjustment on the plates. On this section the folding unit can make "Z" type folding.

Folding Unit Specifications	<i>Folded product width</i>
	Max : 100 mm (Unfolded 200 mm)
	Min. : 55-60 mm
	Unfolded : 120 mm
	<i>Folded product length</i>
	Max. : 115 mm (Unfolded 230 mm)
	Min. : 60 mm (Unfolded 120 mm)
	<i>Folding Type "Z" + Cross folded</i>



F. Cross Folding and Stacking Unit

This servo controlled unit realizes cross fold mechanically without VACUUM.

Hygiene is our number one priority on the folding system. Therefore this part of the machine is designed to give easy mechanic maintenance and intervene easily when needed.

Max. Stack Speed: 150 stack/min

Stackable Product Length (Folded)

Max. : 115 mm

Min. : 60 mm

Stackable Product Width (Folded)

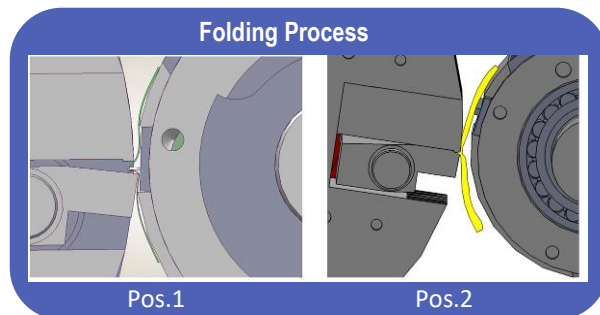
Max. : 100 mm

Min. : 55/60 mm

Minimum Wipe Count: 5

Maximum Stack Height *: 60 mm (depending on the base) / Optionally: 70-80 mm

* Product width must be equal to the value of max. product height



Stacking System Components and Features	
Transfer Unit	<i>Servo controlled. AISI 316 L stainless steel construction</i> <i>Controllable product transfer speed ramp</i>
Stacking Platform	<i>AISI 316 L stainless steel construction</i> <i>Independent servo control</i> <i>Electronic change option depending on stack height</i> <i>Torque control system to prevent mechanical jams</i>

G. Liquid Dosing System

Wetting system consists of below units:

High quality stainless steel 250 lt buffer tank;

Wetting nozzles on each line

Electronic pump has speed control which makes it work in sync with machine speed. It is possible to increase or decrease the amount of liquid dosing level with the help of this control mechanism. All welded joints are passivated against the possibility of rusting and blurring in pipes.

Wetting Amount

Minimum system flow 0,2 L/min. – 4 L/min

Maximum system flow: 0,8 L/min. – 12 L/min

Wetting type: By nozzles

Sensitivity: $\leq 3\%$ (On constant speed)

Liquid Dosing System Components	
Buffer Tank	Capacity: 250lt - AISI 316 L Stainless steel <i>Tri-Clamp Joints</i> <i>Automatic Feeding System:</i> System automatically checks the liquid level inside the tank. When liquid level probes sense low liquid amount, they send a signal to PLC which starts working liquid feeding pump. When the tank is full, PLC automatically stops the pump. <i>Feeding pump :</i> Alfa laval brand stainless steel centrifuge
Installation	<i>Filter :</i> AISI 316 L stainless steel <i>Piping maintenance :</i> Electronically controlled MANUAL EVACUATION mode <i>Easy cleaning nozzles</i> <i>All components AISI 316 L stainless steel</i> <i>Tri-Clamp Joints</i> <i>Wetting pump :</i> Ebara brand stainless steel centrifuge <i>Grundfosddi wetting pump</i>

Consumption

Electricity 380 VAC / 50 Hz / for each module 6Kw,

Air pressure: 6-8 bars

Components:

Electronic parts and PLC	: Omron
Electronic equipments	: Schneider, Telemecanique, Phoenix, Contact
Transmission	: Bonfiglioli, Tramec
Belts	: Optibelt, Megadyne, Isoflex, Gates
Pneumatic parts	: Pneumax, Festo, SMC, Mindman
Sensors	: Sick, Telemecanique, Turck, Leuze, FAG, Nachi, ORS
Bearings	: SKF, Asahi
Gear system	: SIT, KTR, RAN
Linear carriers	: INA, Hiwin, CPC
Stainless steel	: Finland, Sweden

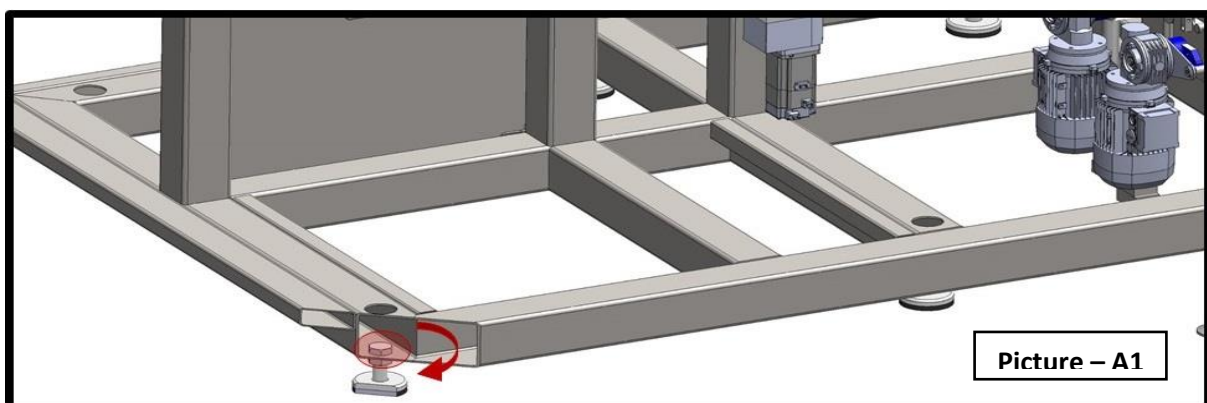
MACHINE LEVEL AND ELECTRICAL-PNEUMATIC CONNECTION

Frames of the machine is connected according to layout.

During this process, a forklift is needed. To check if the positioned modules of machine are on level, measuremental coordinates or a laser pointer is needed.

After placement, machine feet should be assembled with the help of a water gage to check the parallelism of machine with the ground. After this, the height of the machine should be between 30~60mm.

Tension of trigger and conveyor belts must be checked after levelling. After these controls, tension should be suitable to "1 cm stretch on 1 kg pressure" ratio.

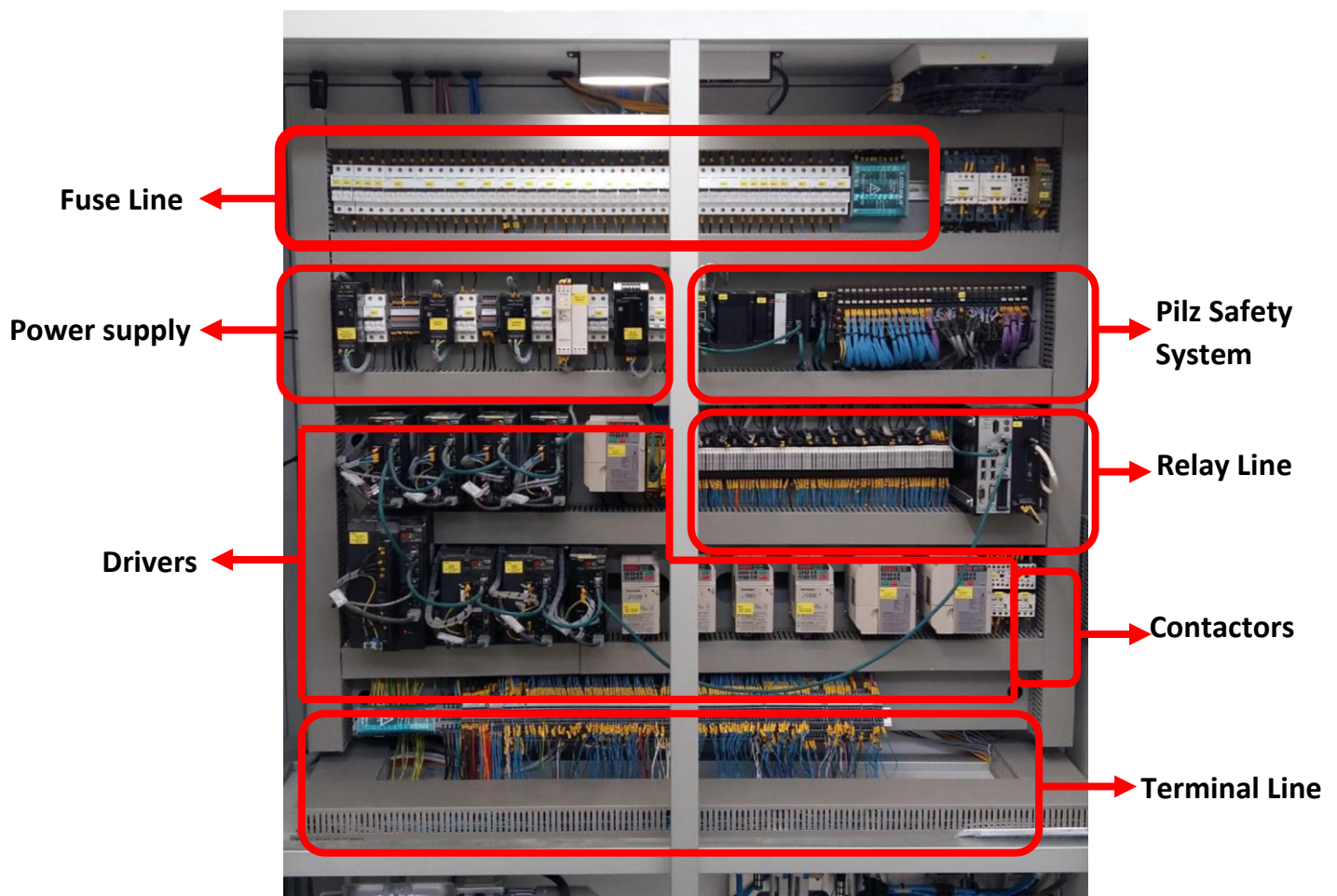


First thing to do on electrical and pneumatic connection is to make sure that main switch and pneumatic valve is off. Electrical (multi socket etc.) and pneumatic connections must be done properly according to the codes.

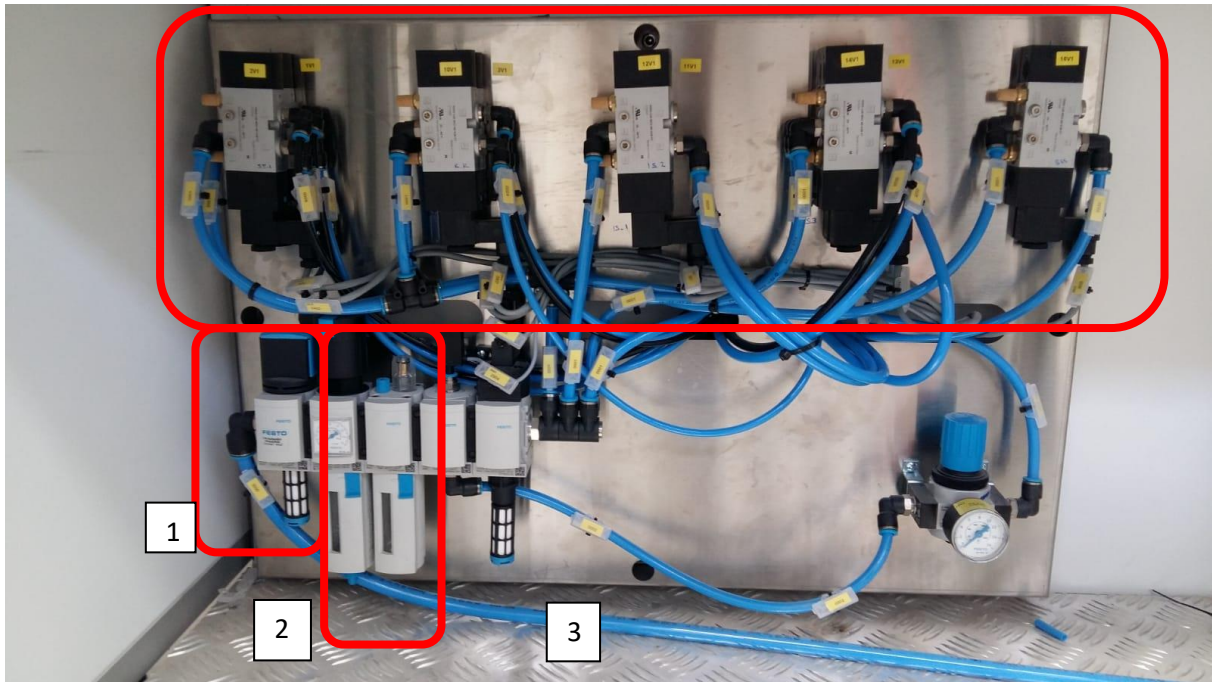
Electric; 3 PHASE – 1 NEUTRAL – 1 GROUND, between PHASE-NEUTRAL 220V, between PHASE-PHASE 380V, between GROUND – NEUTRAL should be 0V and must be tested.

Pneumatic main air pressure source must be 7-8 bars. Our machine's pneumatic working air pressure is 6 bars.

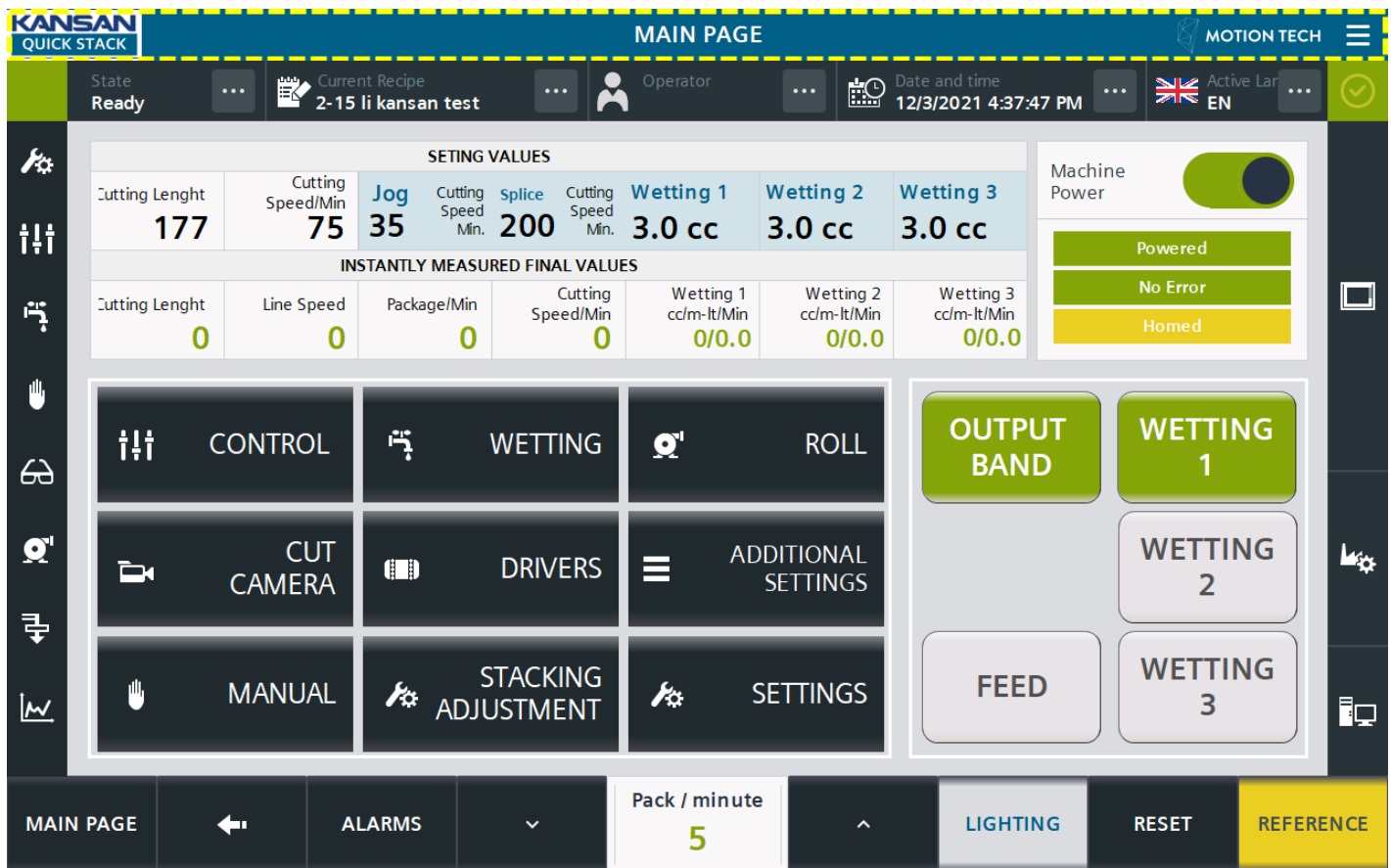
ELECTRICAL PANEL DETAILS



PNEUMATIC PANEL DETAILS



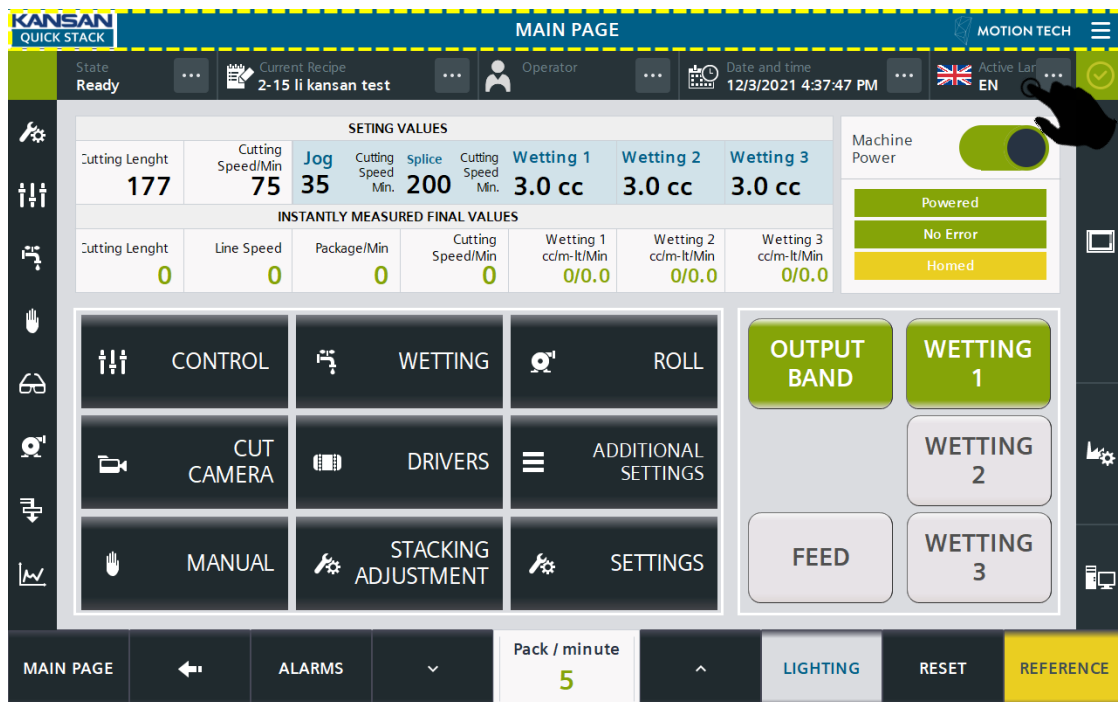
- 1. Main Air Input
- 2. Regulator
- 3. 5/2 – 1/8 Pneumatic Valve

HMI DISPLAY

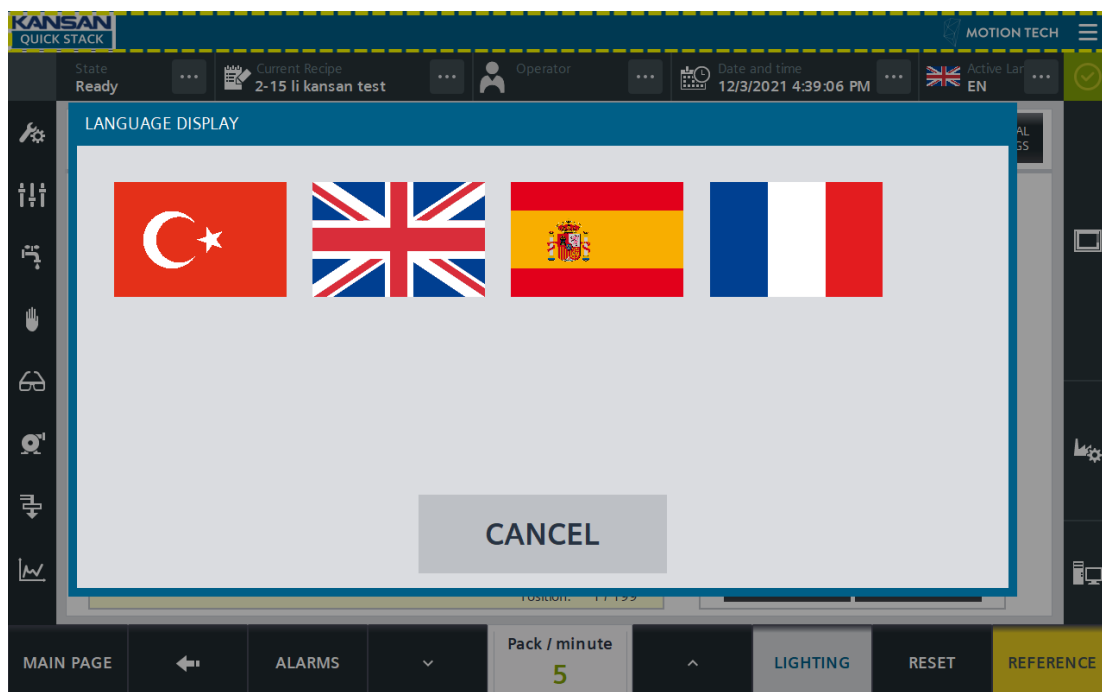
- Cutting Speed: Shows the speed of cutting.
- Pack Speed: Shows the speed of packs produced in one minute.
- Cutting speed setting and target-real values can be observed.

HMI DISPLAY

CHANGING LANGUAGE

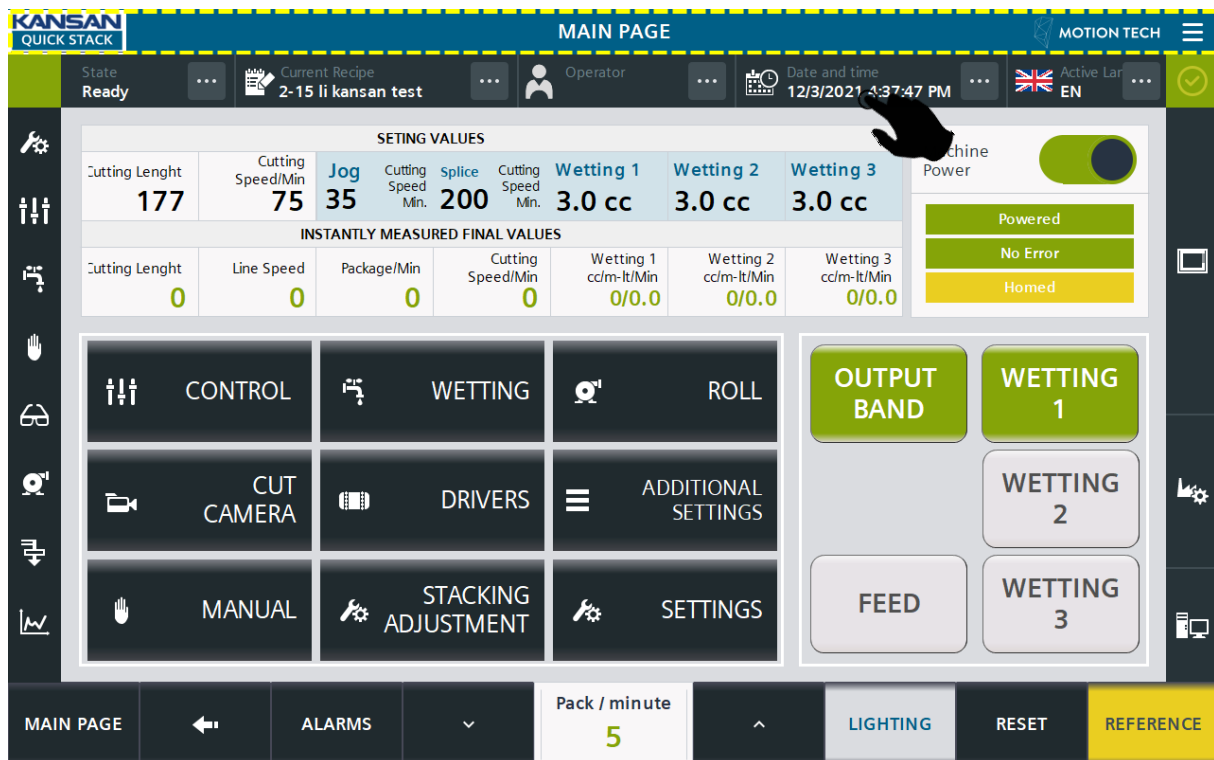


Touch the flag at the top right corner of the screen.

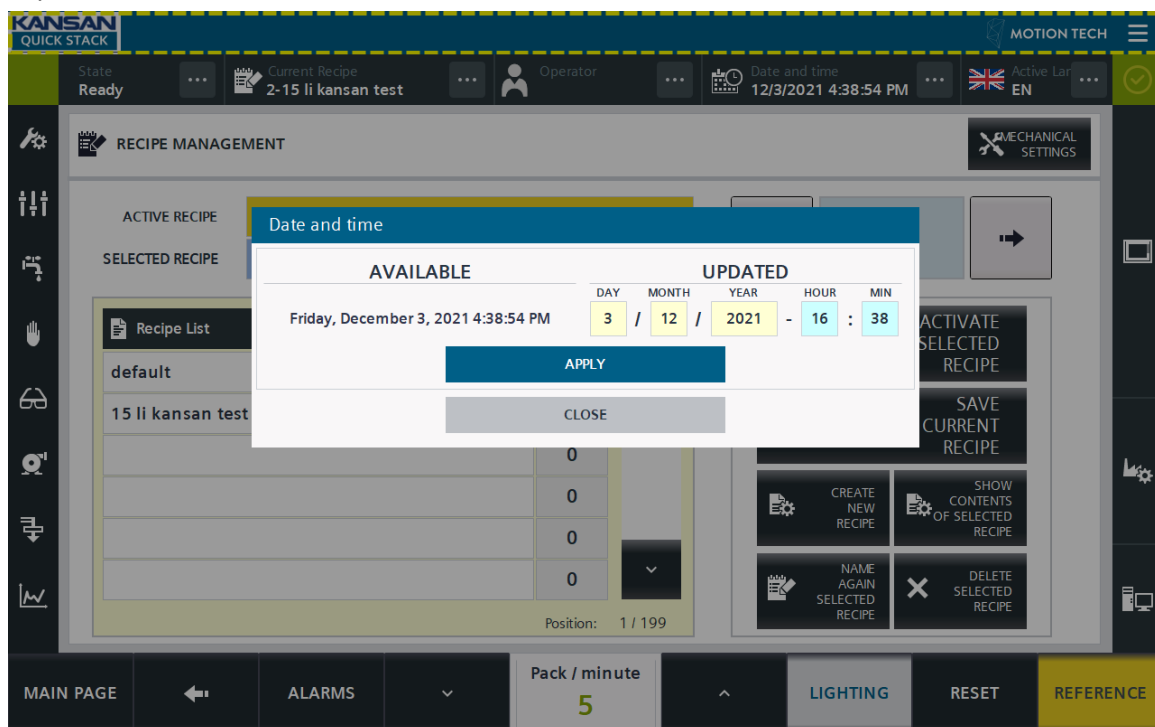


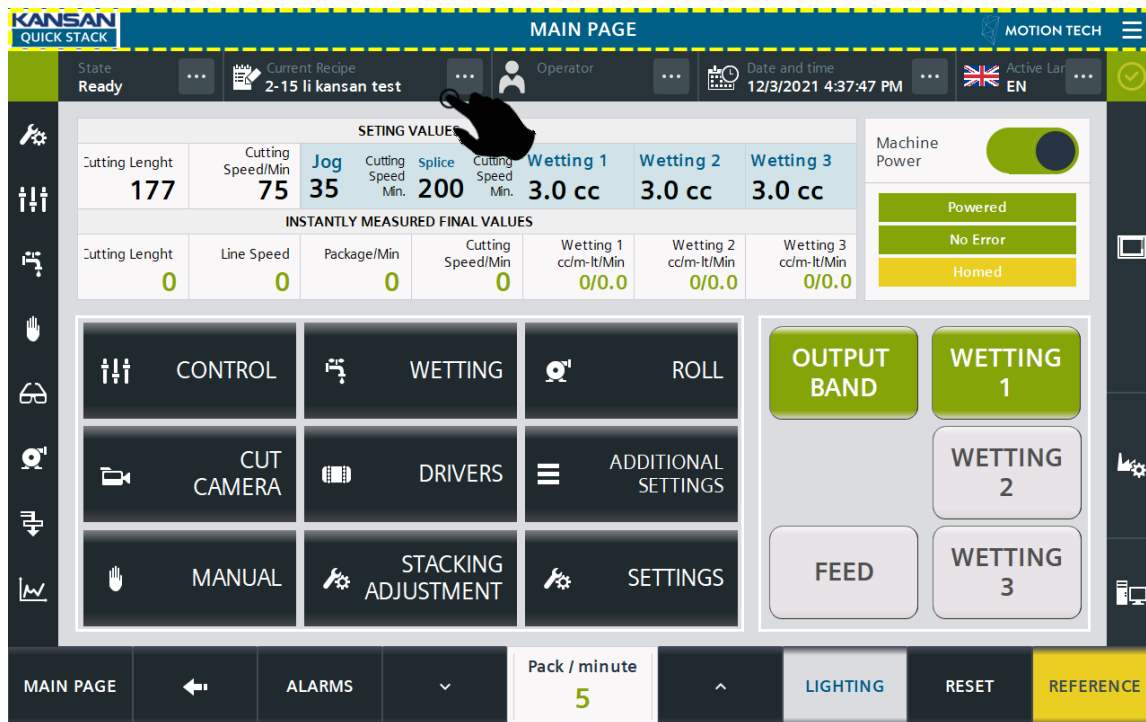
Choose the language you prefer, if you want to add more languages to the machine please ask from Kansan.

CHANGING DATE AND TIME

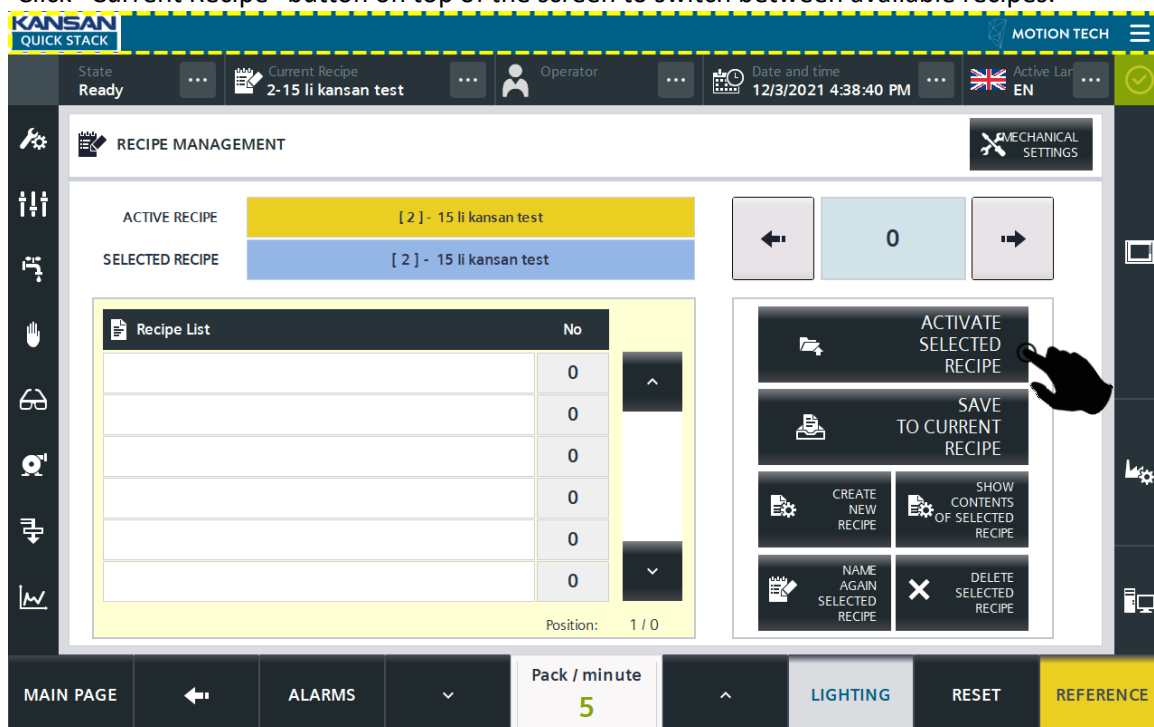


Click the date and time button on the top of the screen to change the date by using the pop-up keyboard.

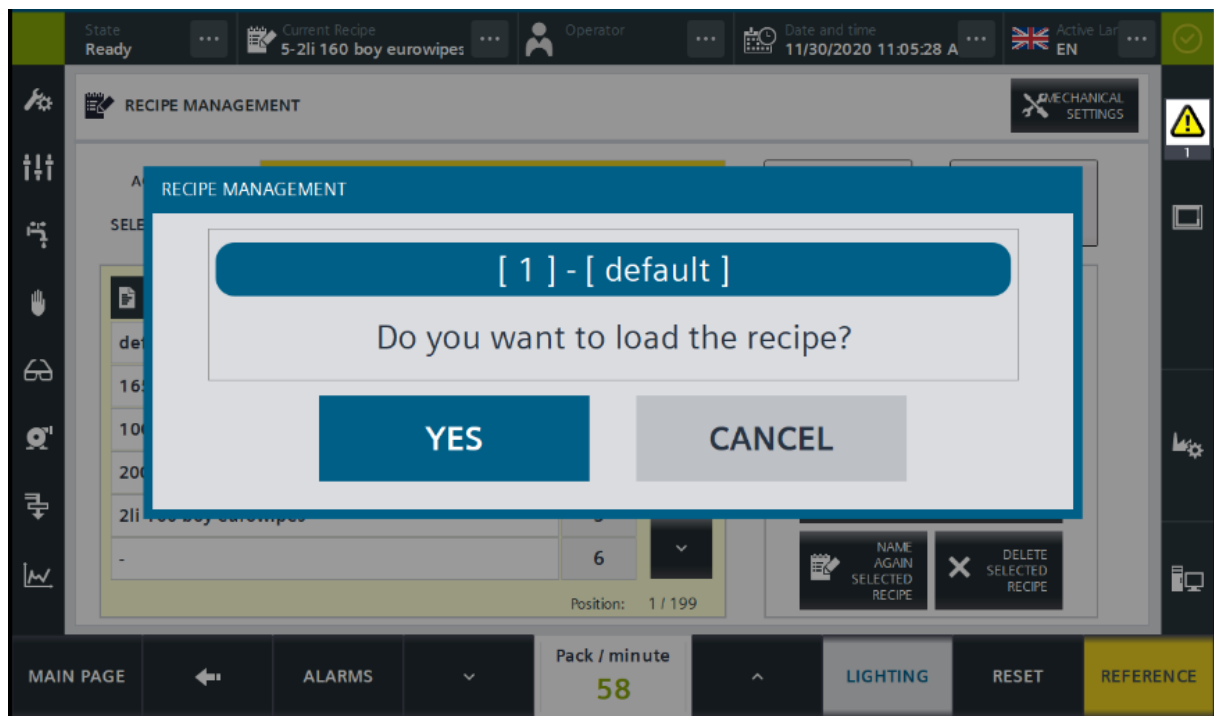


RECIPE PAGE

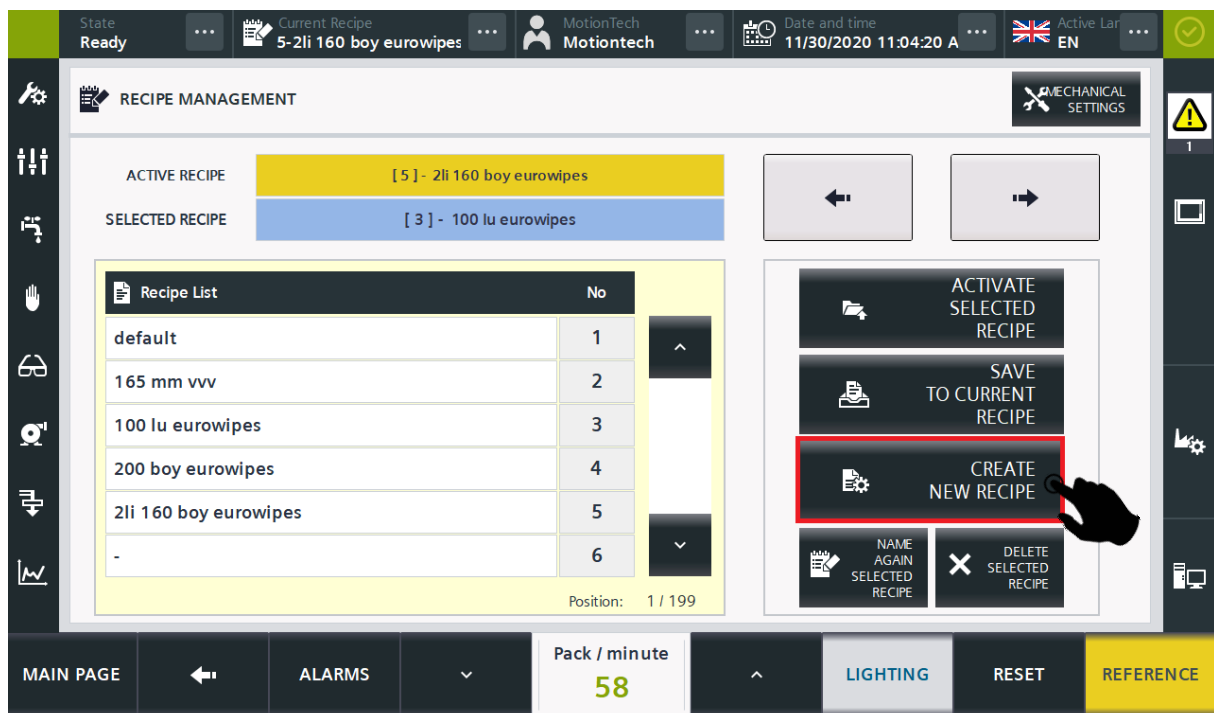
Click "Current Recipe" button on top of the screen to switch between available recipes.



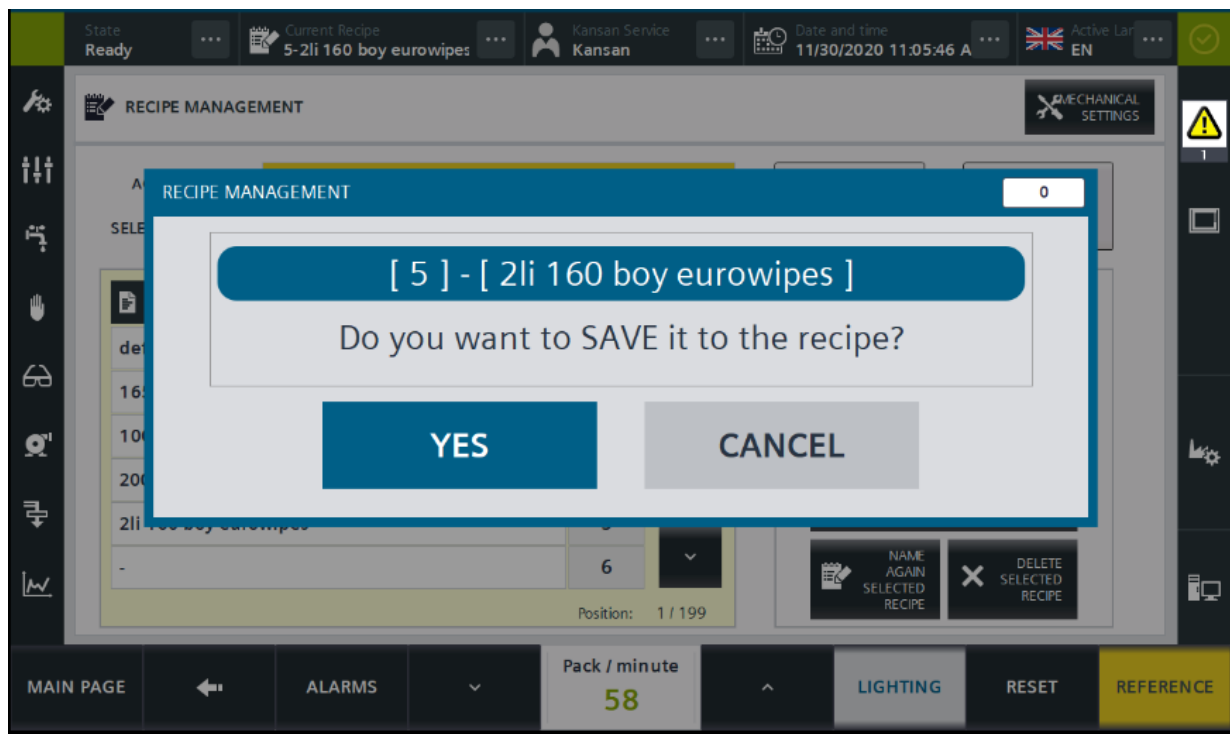
Choose desired recipe from Recipe List and click "Activate Selected Recipe" button.



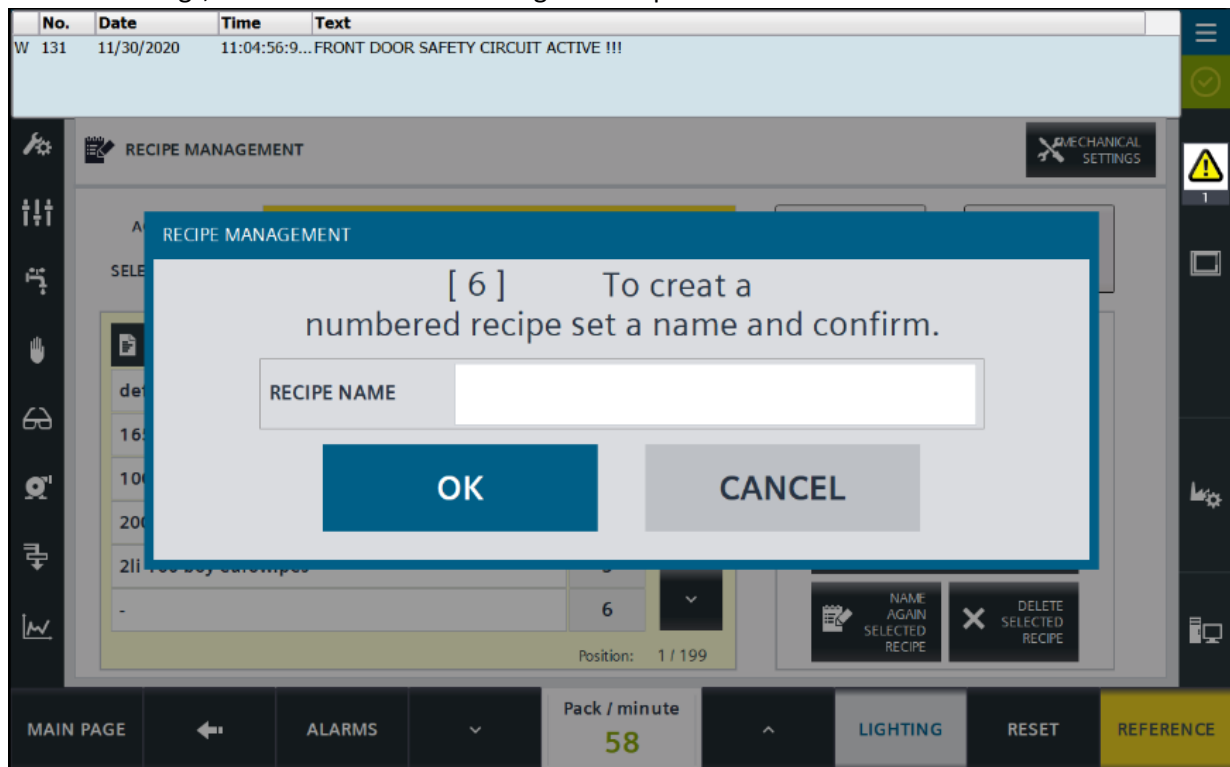
Click “Yes” to assign chosen recipe as active recipe.



You may create a new recipe by click on “Create New Recipe” button.



After all settings, click on “Yes” to save changes in recipe.



Then, write a recipe name by using pop-up keyboard and confirm to exit.

PRODUCTION PAGE

Production page can be used to track production record, actual & previous shift and total production numbers can be seen on this page.

Production		Active FAT		Previous -> 1 /	
START SHIFT	Last Start: 11/11/2021 3:39:42 PM Last Stop: 11/16/2021 3:35:37 PM	Start: 11/3/2021 11:37:08 AM		Start: 1/1/1992 12:00:00 AM Finish: 11/3/2021 11:37:08 AM	
Number of Cut Performed	: 414584 clips	81370 clips		333214 clips	
Number of Rejects	: 0 clips	0 clips		0 clips	
Number of Packs Ejected	: 0 packs	0 packs		0 packs	
Total Number of Stops	: 1663	274		1389	
Avg Speed Machine	: 208 clips/min - 14 packs/min	220 clips/min - 17 packs/min		199 clips/min - 12 packs/min	
Max Speed Reached	: 800 clips/min	600 clips/min		800 clips/min	
Total Duration Machine Running	13 hours 33 min 34 sec	6 hours 8 min		7 hours 25 min	
Total Duration Machine Stopped	43 hours 27 min 55 sec	19 hours 51 min		23 hours 36 min	
Total Duration of Shift	78 hours 40 min 31 sec	34 hours 42 min		63 hours 43 min	
Total Lotion Dosage Used	: 632.7 lt	292.4 lt		340.3 lt	
Total Wipes Used	: 1011316.6 m	282908.0 m		728408.6 m	

Use “Start Shift” button when the shift is over to change the shift.

SERVO SETTINGS

KANSAN QUICK STACK MOTION TECH

State: Ready | Current Recipe: 2-15 li kansan test | Operator: | Date and time: 12/3/2021 4:40:18 PM | Active Language: EN

Parameter	Value
Cutting Length	177 mm
Puller Speed %	103.4 %
Clips Rod Offset	60 mm
Stack-1 Horizontal Offset	-130 mm
Stack-2 Vertical Offset	-150 mm

MAIN PAGE | ALARMS | Pack / minute: 5 | LIGHTING | RESET | REFERENCE

WETTING PAGE

ANA SAYFA

Home page

Current Recipe: 5-2li 160 boy eurowipes

MotionTech Motiontech

Date and time: 11/30/2020 11:04:11 A

Active Language: EN

SETTING VALUES

Cutting Speed/Min	Jog	Cutting Speed Min.	Splice	Cutting Speed Min.	Wetting
348	30		348		158 cc / m

INSTANTLY MEASURED FINAL VALUES

Line Speed	Package/Min	Cutting Speed/Min	Wetting cc/m-lt/Min
0	0	0	0/0.0

Machine Power: 30 / 199

Powered

No Error

Homed

OUTPUT BAND

Wipe Counter: 0

Reset Counts

WETTING

FEED

MONITOR

BOBIN

STACKING ADJUSTMENT

Pack / minute: 58

LIGHTING

RESET

REFERENCE

Click on "Wetting" to see wetting settings.

KANSAN QUICK STACK

State: Ready

Current Recipe: 2-15 li kansan test

Operator

Date and time: 12/3/2021 4:40:24 PM

Active Language: EN

SETTING -1

LIQUID TANK EMPTY LEVEL CONTROL

Main Tank: 0.0%

Liquid Tank-1: 97.7%

WETTING ADJUST

	WETTING 1	WETTING 2	WETTING 3
Wetting Amount	3.0 cc	3.0 cc	3.0 cc
Wetting Alarm Lir	5.0 L	5.0 L	5.0 L
Target Wetting (l/min)	0.2 L	0.2 L	0.2 L
Measured Fluid Fl (l/min)	0.0 L	0.0 L	0.0 L
Measured Fluid Flow (cc/layer)	0.0 cc	0.0 cc	0.0 cc
Wetting Control C	0.0 %	0.0 %	0.0 %

MAIN PAGE

ALARMS

Pack / minute: 5

LIGHTING

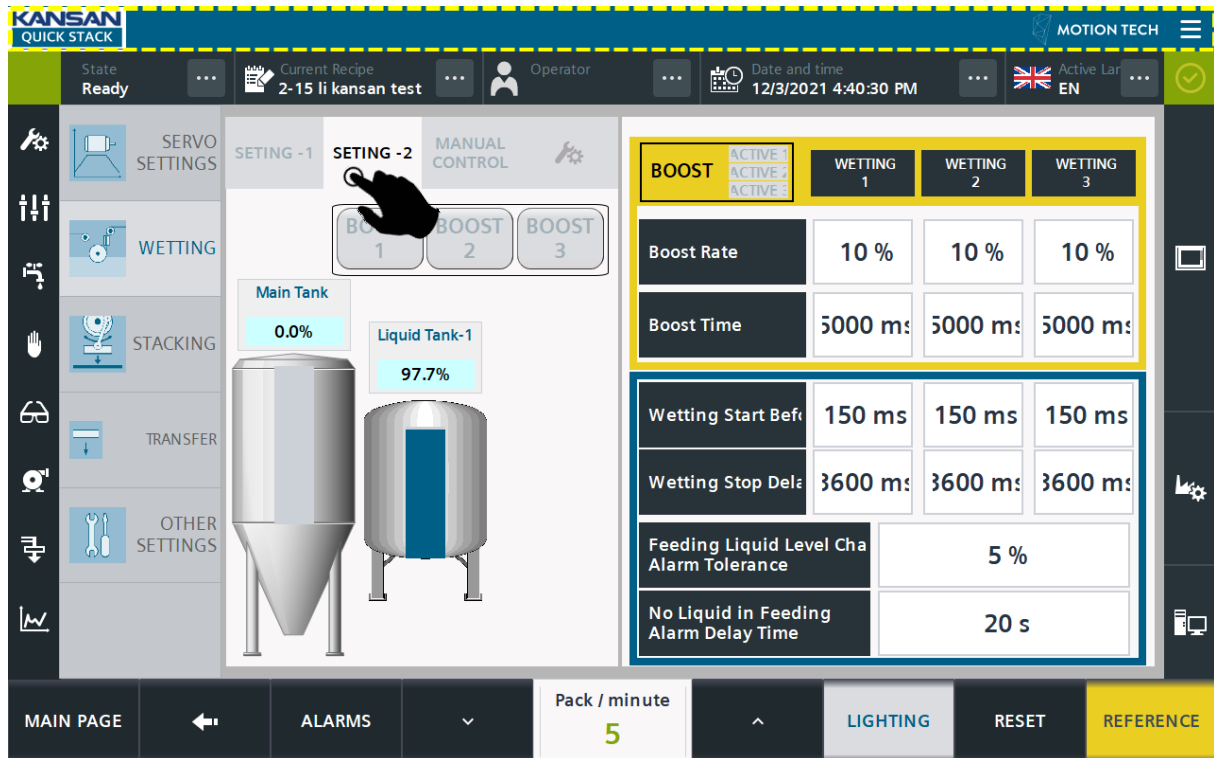
RESET

REFERENCE

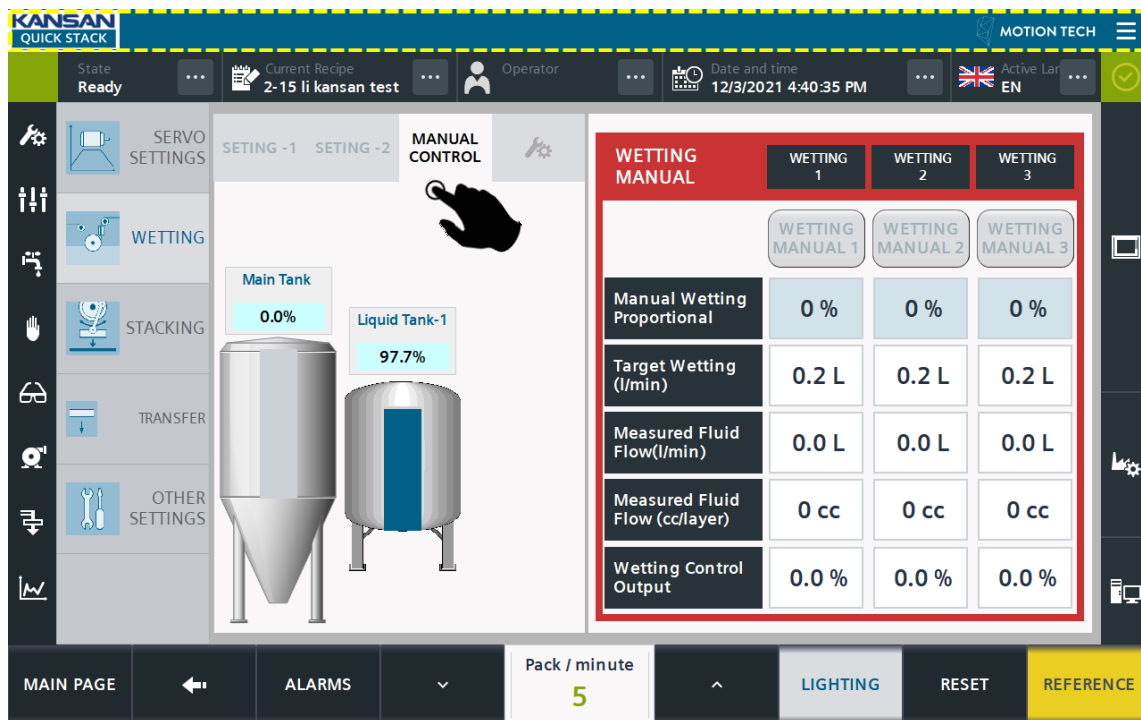
Wetting adjustment may be controlled in “Setting-1” interface.

Wetting Amount(cc/m): Enter the amount of wetting that needs to be applied.

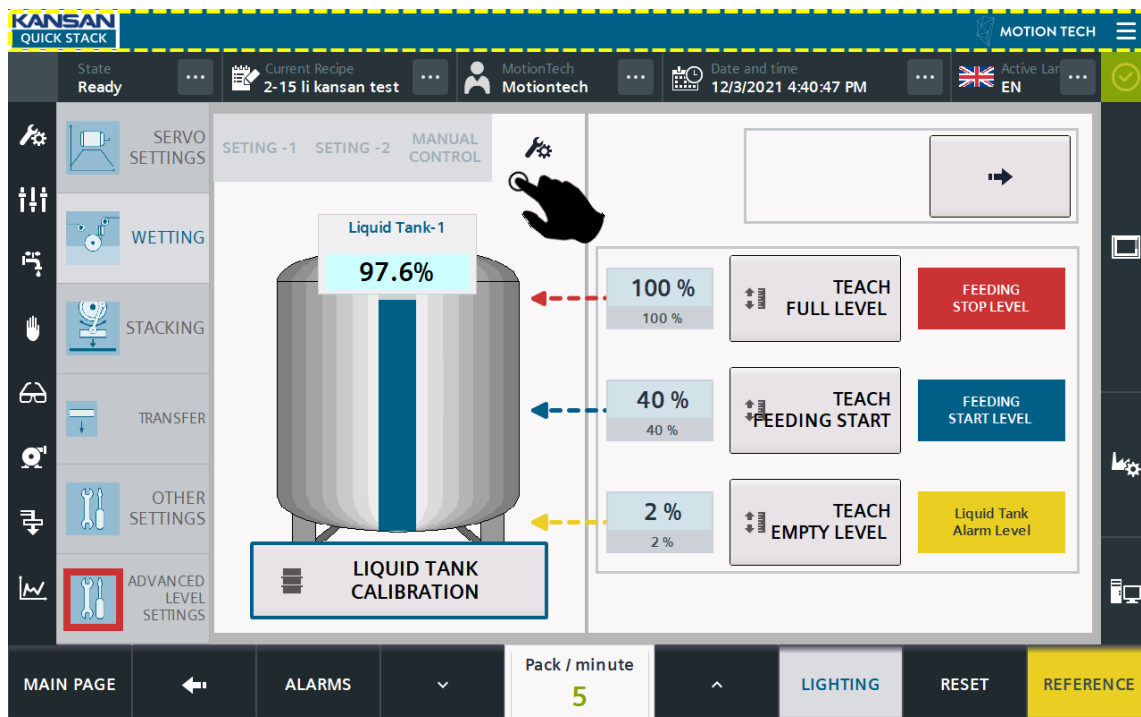
Target Wetting (L/min): The machine calculates/converts and displays the target wetting amount according to machine speed.



In “Setting-2”, “Boost Mode” can be activated and pneumatic-electronic delays are listed.

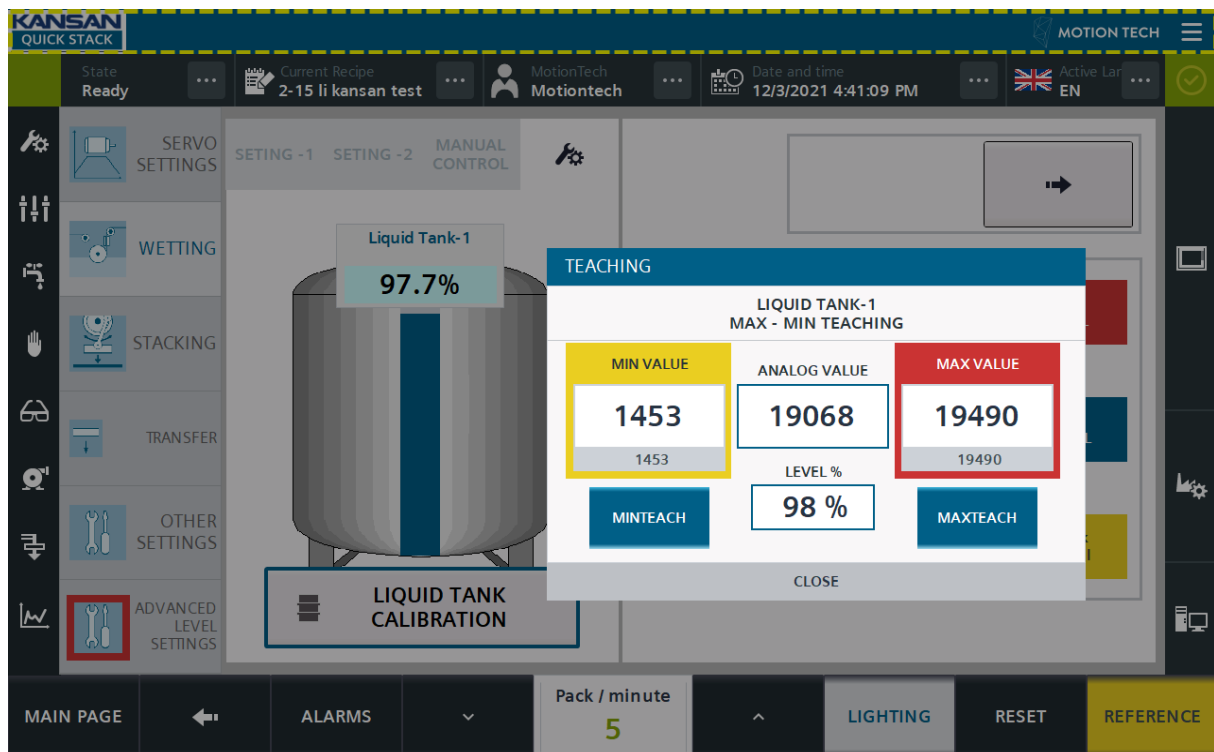


“Manual Control” tab, allows the user to run the wetting pump manually.



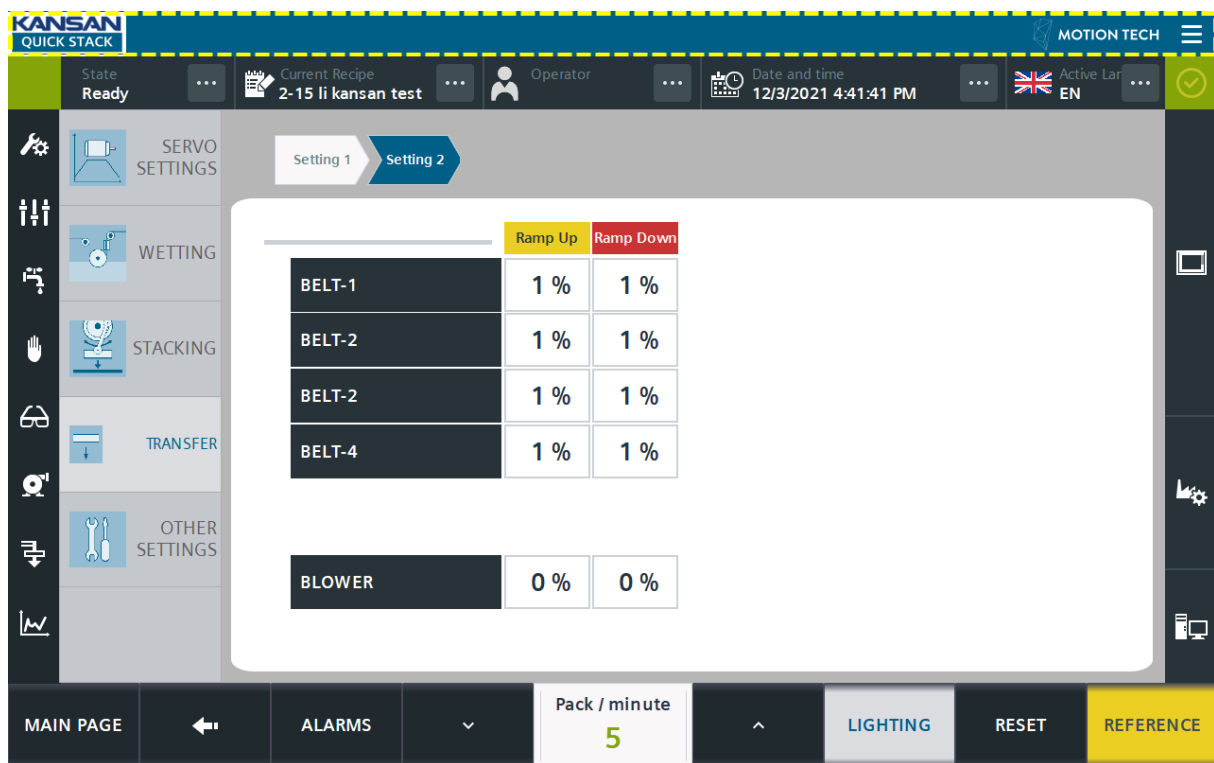
Last tab is used to control the levels of activation to make control system feed both main and liquid tanks.





Main tank level sensor calibration can be made from here. Click the MINTEACH button when the tank is empty, fill the tank fully and click the MAXTEACH button when the tank level is full to teach the sensor where the full level is.

TRANSFER SETTINGS



OTHER SETTINGS

KANSAN QUICK STACK MOTION TECH

State: Ready | Current Recipe: 2-15 li kansan test | Operator: | Date and time: 12/3/2021 4:41:52 PM | Active Language: EN

SERVO SETTINGS | **WETTING** | **STACKING** | **TRANSFER** | **OTHER SETTINGS**

Setting 1 | **Setting 2**

FAN MOTOR MANUEL

Air Flow Analog View

Roll Air Flow: ERROR | 0

AIR FLOW CONTROL DELAY: 500 ms | **AIR FLOW CONTROL UP LIMIT**: 18000

ANALOG VALUE: 0

AIR FLOW CONTROL DOWN LIMIT: 11000

MAIN PAGE | ← | ALARMS | ↓ | Pack / minute: 5 | ^ | LIGHTING | RESET | REFERENCE

KANSAN QUICK STACK MOTION TECH

State: Ready | Current Recipe: 2-15 li kansan test | Operator: | Date and time: 12/3/2021 4:41:47 PM | Active Language: EN

SERVO SETTINGS | **WETTING** | **STACKING** | **TRANSFER** | **OTHER SETTINGS**

Setting 1 | **Setting 2**

Other Settings

EDGE CONTROL PERIOD: 15 min | **EDGE CONTROL VALVE OUTPUT**: 5 s

EDGE CONTROL PERIOD: 5 s | **BLOWER SPEED**: 25 %

EDGE CONTROL VALVE OUTPUT: 1 s

MAIN PAGE | ← | ALARMS | ↓ | Pack / minute: 5 | ^ | LIGHTING | RESET | REFERENCE

ADVANCED LEVEL SETTINGS

KANSAN QUICK STACK MOTION TECH

State: Ready | Current Recipe: 2-15 li kansan test | MotionTech Motiotech | Date and time: 12/3/2021 4:41:26 PM | Active Language: EN

Parameter	Value
StackStep	1.7 mm
Pack Layer	15
Stacking-1 Layer Number	7
Stacking-1 Down Pos	80 mm
Stacking-2 Up Pos	117 mm

STOPPER -2 PISTON START POSITION: 20.0 mm | STOPPER -2 PISTON DELAY: 100 ms

MAIN PAGE | ALARMS | Pack / minute: 5 | LIGHTING | RESET | REFERENCE

CAMERA CONTROL

KANSAN QUICK STACK MAIN PAGE MOTION TECH

Camera Control

CAM TRIGGER POSITION (0°-360°)

Knife	Position	Trigger
KNIFE -1	0°	KNIFE -1 TRIGGER
KNIFE -2	0°	KNIFE -2 TRIGGER
KNIFE -3	0°	KNIFE -3 TRIGGER

EDGE CONTROL VALVE OUTPUT: 0 ms | OUT | CAMERA TRIGGER

CLOSE

MAIN PAGE | ALARMS | Pack / minute: 5 | LIGHTING | RESET | REFERENCE

BLADE CHANGE

KANSAN QUICK STACK

State: Ready | Current Recipe: 2-15 li kansan test | Operator: | Date and time: 12/3/2021 4:39:20 PM | Active Language: EN

RECIPE MANAGEMENT

ACTIVE RECIPE: [2] - 15 li kansan test

SELECTED RECIPE: [2] - 15 li kansan test

Recipe List	No
default	1
15 li kansan test	2
	0
	0
	0
	0

Position: 1 / 199

Production Pages:

- PRODUCTION
- BLADE CHANGE

Documents:

- PDF Viewer
- Videos

MAIN PAGE | ALARMS | Pack / minute: 5 | LIGHTING | RESET | REFERENCE

Click “Blade Change” button to see both the knife condition and alteration order.

KANSAN QUICK STACK

State: Ready | Current Recipe: 5-2li 160 boy eurowipes | Kansan Service: Kansan | Date and time: 11/30/2020 11:16:01 A | Active Language: EN

Current Knife 1 Cut: 589230 | BLADE 1 CHANGE | Current Knife 2 Cut: 839435 | BLADE 2 CHANGE

BLADE 1		
	Changing Time	Total Cut
1	11/20/2020 12:45:07 PM	0
2	11/20/2020 12:44:46 PM	250464
3	10/8/2020 9:33:26 PM	1
4	10/8/2020 9:33:21 PM	1669
5	10/4/2020 10:49:26 AM	43678
6	9/16/2020 11:49:45 AM	30
7	9/16/2020 11:49:11 AM	13881
8	9/5/2020 4:51:58 PM	164
9	9/5/2020 4:04:14 PM	0
10	9/5/2020 4:03:42 PM	0

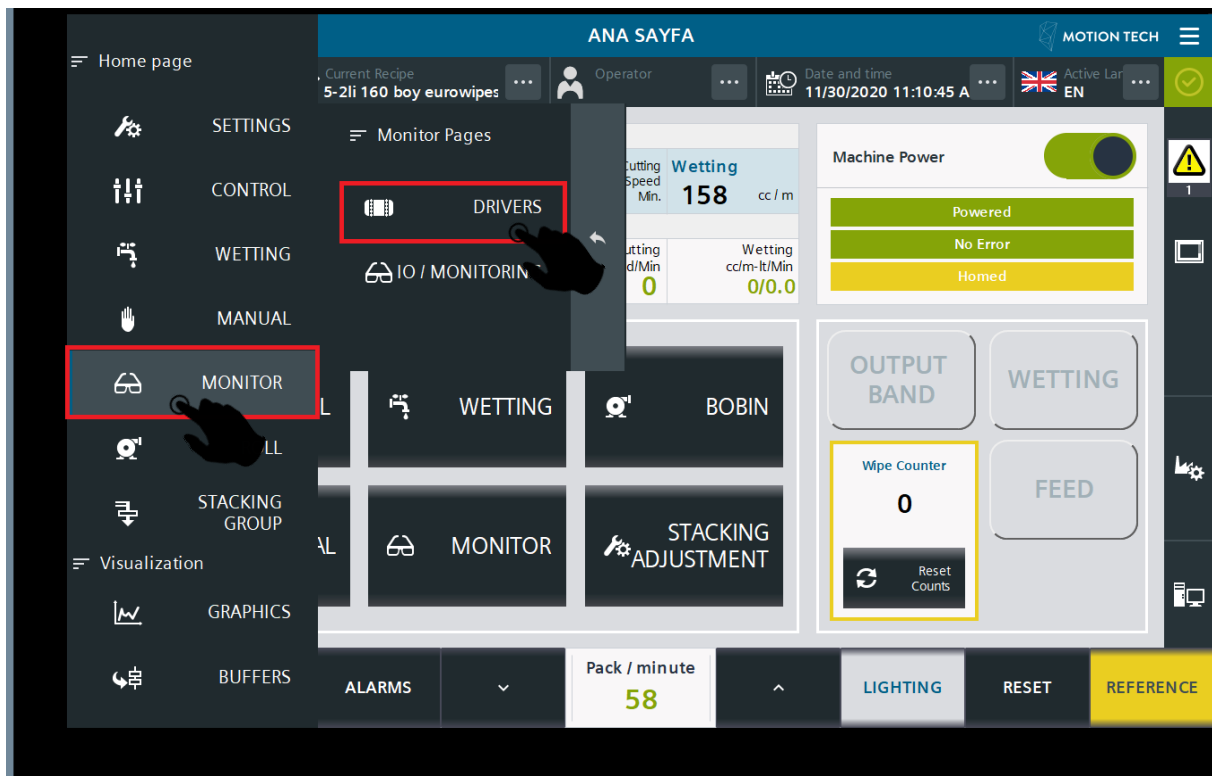
KNIFE 1 RECORDS COMPLETE RESET

BLADE 2		
	Changing Time	Total Cut
	10/8/2020 9:33:24 PM	145
	10/4/2020 10:49:48 AM	43641
	9/16/2020 1:04:44 PM	0
	9/16/2020 1:04:40 PM	0
	9/16/2020 1:04:38 PM	22
	9/16/2020 11:49:44 AM	29
	9/16/2020 11:49:15 AM	9868
	9/5/2020 4:52:03 PM	162
	9/5/2020 4:04:36 PM	0
	9/5/2020 4:04:18 PM	47

KNIFE 2 RECORDS COMPLETE RESET

MAIN PAGE | ALARMS | Pack / minute: 58 | LIGHTING | RESET | REFERENCE

This page can be used to track knife lifetimes, click “Blade 1 Change” button after you inserted a new knife to the machine. This will reset the counter data to start recording for the new.

MONITORING DRIVER PARAMETERS

This instruction above guide user to observe actual status of the machine drivers and their current parameters.

KANSAN
QUICK STACK

STATUS GENERAL

MOTION TECH

State
Ready

...

Current Recipe
2-15 li kansan test

...

Operator

...

Date and time
12/3/2021 4:42:33 PM

...

Active Language
EN

...

All Axis Status

Servo Axis Status

Async Status

Drive Topology

Roll

Folding

Puller

Clips Rode

Pusher-1

Pusher-2

Stack -1

Stack-2

Pump - 1

Pump - 2

Pump - 3

Belt-1

Belt-2

Belt-3

Belt-4

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ACTIVE

ERROR

ERROR

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ERROR

ERROR

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

Error ID: 0

-0.5 [rpm]

0.0 [rpm]

0.5 [rpm]

0.5 [rpm]

0.0 [rpm]

0.0 [rpm]

0.0 [rpm]

0.0 [rpm]

+0.0 [rpm]

+0.0 [rpm]

+0.0 [rpm]

+0.0 [rpm]

+0.0 [rpm]

+0.0 [rpm]

+0.0 [rpm]

+0.0 [rpm]

+0.1 [Nm]

+0.5 [Nm]

+0.3 [Nm]

-0.3 [Nm]

+0.0 [Nm]

+0.0 [Nm]

+0.6 [Nm]

+0.7 [Nm]

+0.0 [Nm]

+0.0 [Nm]

+0.0 [Nm]

+0.0 [Nm]

+0.0 [Nm]

+0.0 [Nm]

+0.0 [Nm]

+0.0 [Nm]

0 [%]

1 [%]

1 [%]

2 [%]

0 [%]

0 [%]

6 [%]

6 [%]

MAIN PAGE

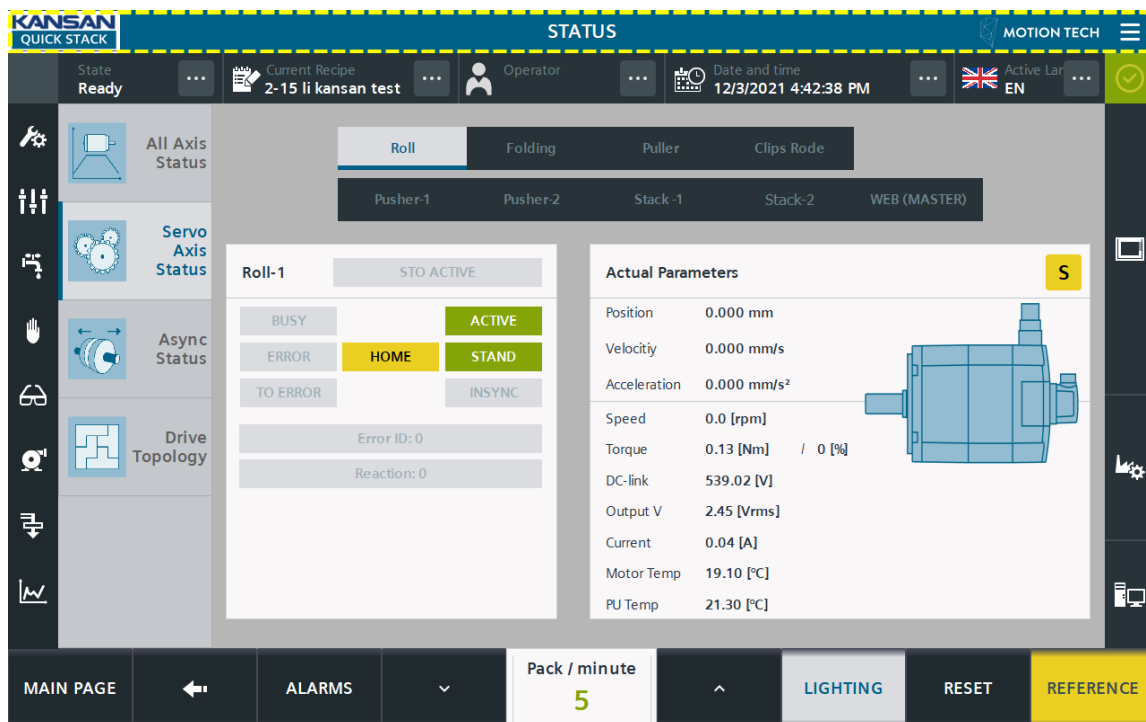
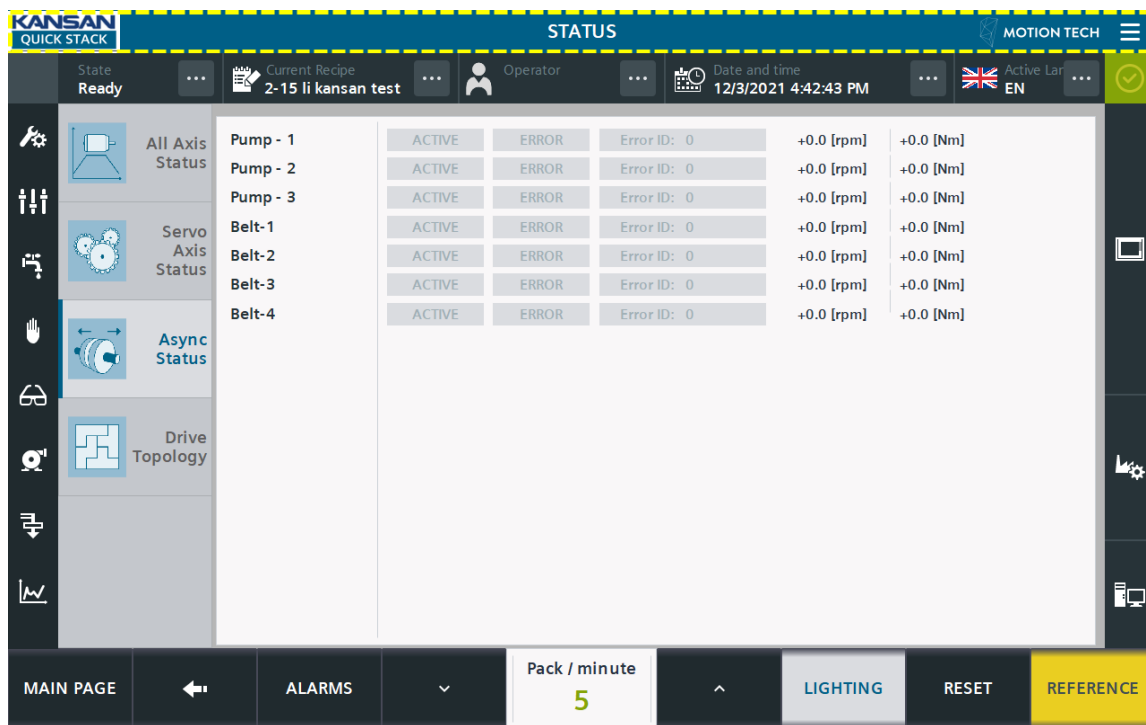
ALARMS

Pack / minute
5

LIGHTING

RESET

REFERENCE



ROLL PAGE

KANSAN QUICK STACK MOTION TECH

State: Ready | Current Recipe: 2-15 li kansan test | Operator: | Date and time: 12/3/2021 4:42:07 PM | Active Language: EN

ROLL PAGE

ROLL | ROLL SPLICE | ROLL SPLICE

ROLL 1 ADDING | ROLL 2 ADDING | ROLL 3 ADDING

ROLL BROKEN SENSOR

1	2	3
---	---	---

ROLL ADDING PISTON OUTLET

1	2	3
---	---	---

ROLL ADDING READY BUTTON

1	2	3
---	---	---

ROLL ADDING SENSOR

1A	2A	3A
1B	2B	3B

ROLL ADDING SENSOR

1A	2A	3A
1B	2B	3B

ACCUMULATOR ABOVE SENSOR

1A	2A	3A
1B	2B	3B

SPLICE DELAY DISTANCE 100 mm

MAIN PAGE | ALARMS | Pack / minute 5 | LIGHTING | RESET | REFERENCE

KANSAN QUICK STACK MOTION TECH

State: Ready | Current Recipe: 2-15 li kansan test | Operator: | Date and time: 12/3/2021 4:42:00 PM | Active Language: EN

ROLL PAGE

ROLL | ROLL SPLICE | ROLL SPLICE

ROLL 1 | ROLL 2 | ROLL 3

ROLL BROKEN SENSOR

1	2	3
---	---	---

ROLL ADDING PISTON OUTLET

1	2	3
---	---	---

ROLL ADDING READY BUTTON

1	2	3
---	---	---

ROLL ADDING SENSOR

1A	2A	3A
1B	2B	3B

ROLL ADDING SENSOR

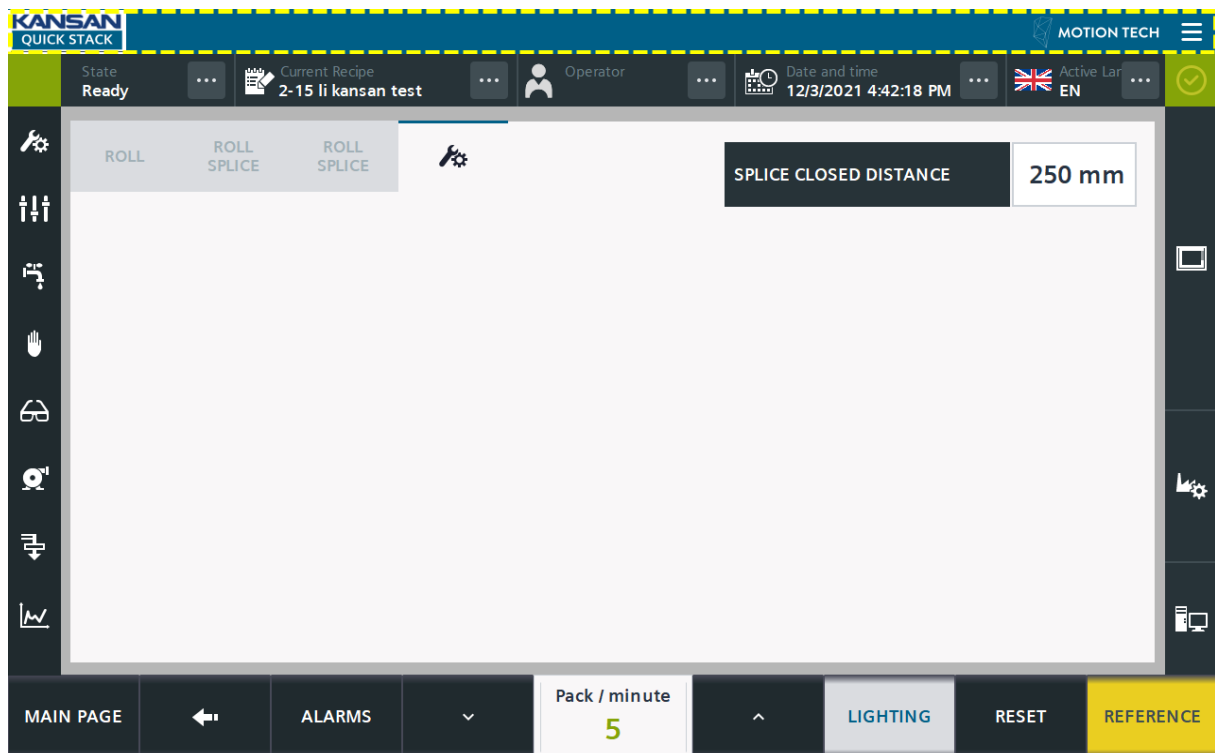
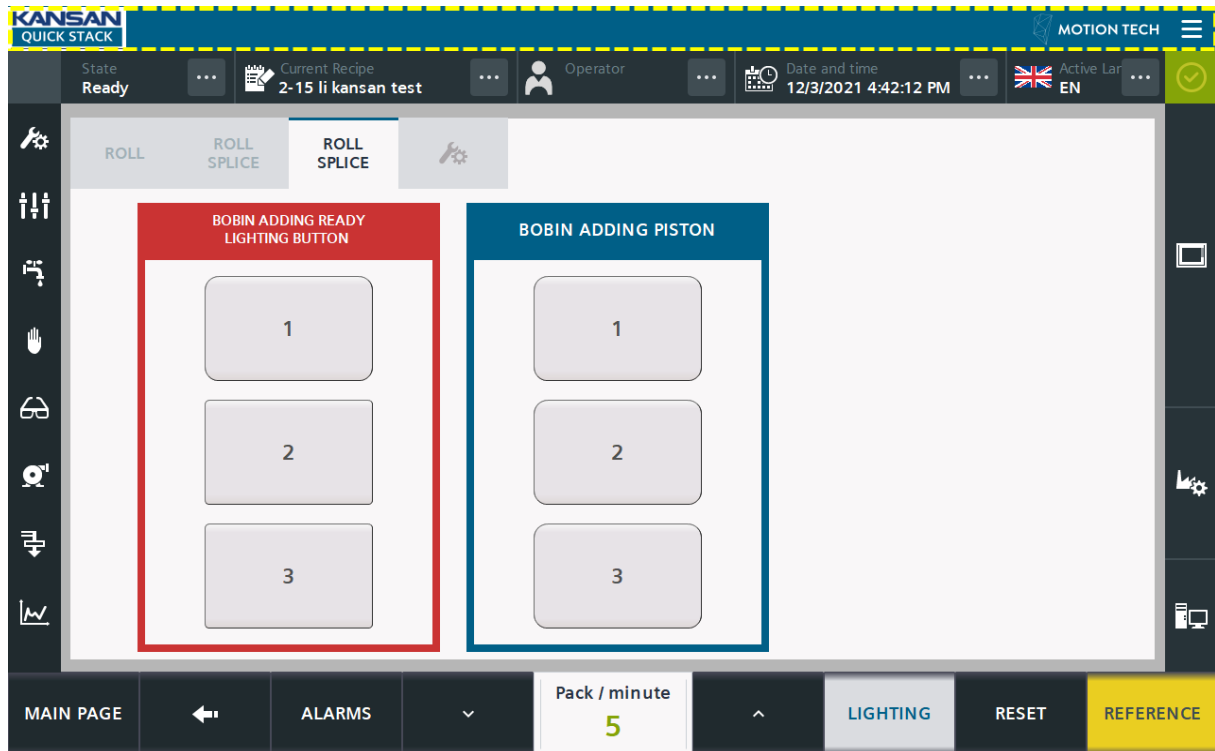
1A	2A	3A
1B	2B	3B

ACCUMULATOR ABOVE SENSOR

1A	2A	3A
1B	2B	3B

SPLICE DELAY DISTANCE 100 mm

MAIN PAGE | ALARMS | Pack / minute 5 | LIGHTING | RESET | REFERENCE



SETTINGS

EXAMPLE RECIPE CREATION

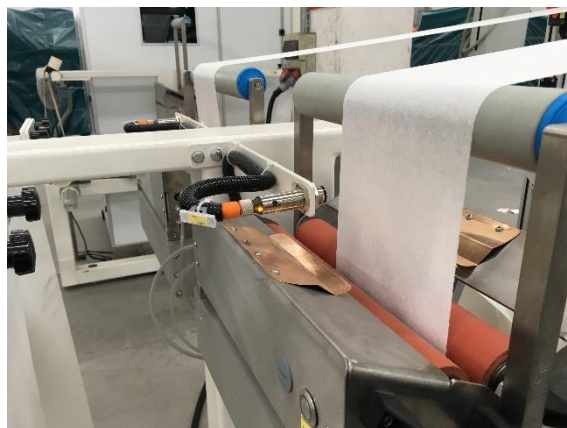
Example Product Description: 200*180 "Z" type folding. Total product weight is 240 gr and wipe count is 25. Recipe name will be saved as KANSAN01. Total product width is 90mm.

1. Cut length is decided according to 180mm or 200mm roll width. In our example we'll use 180mm roll width. Therefore cut length will be 200mm.
2. Roll with 180mm width (Pictures_R1) centered on pneumatic shafts. (Pictures_R2) With air gun, pneumatic shafts are pumped with air. Roll's unwinding direction should be suitable according to the way it will go through splice unit. (see scheme)



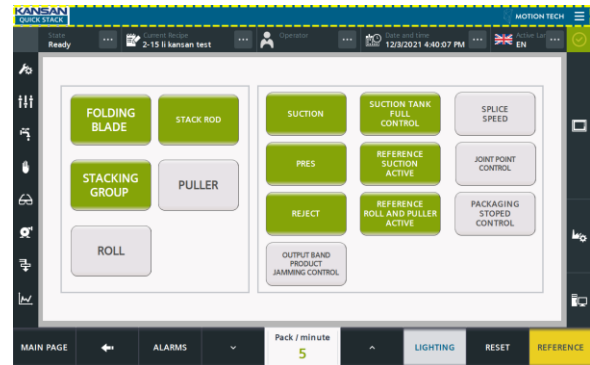
3. After roll is passed through splice system properly, brake weights should be positioned on the roll. (see picture)
Raw material is passed through unwinder rolls and brought until 90° cross shafts. (see pictures)





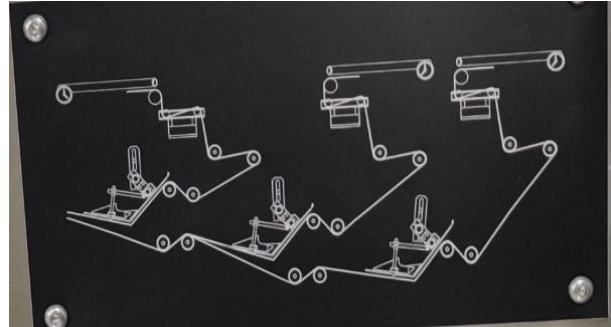
4. After this, roll servos and puller servos are activated on HMI through control page on the main page and selected modules are set into motion. (see screenshot)

*** TIP: By activating SETTING on the main page, roll and puller servo group can be activated.



5. While passing the raw material according to flow scheme, it should be centered on folding plates and center position should be chosen on edge control and reached to the point of the puller with the help of machine start button.

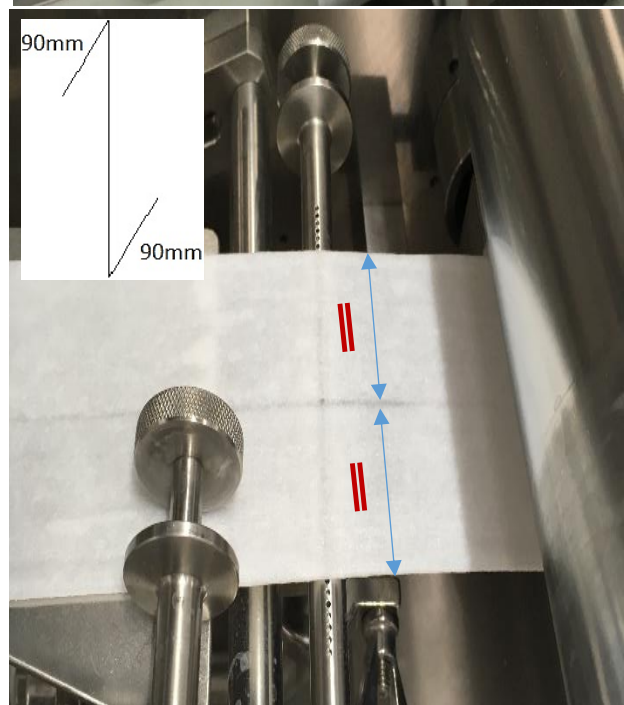
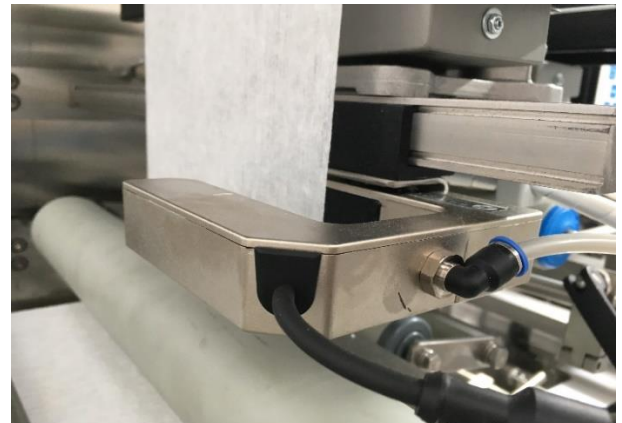
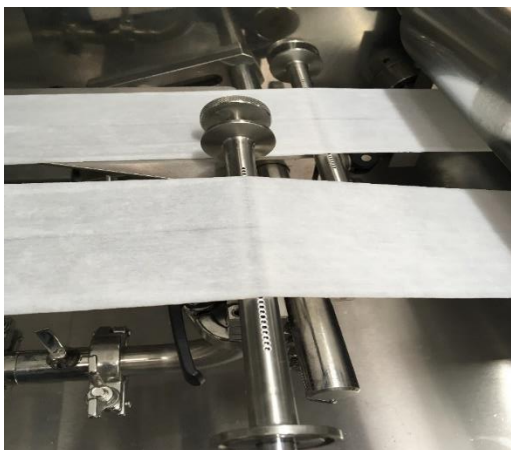
After this, puller speed is entered as 101 from the servo setting page. Wipe length is entered as 200. During and after material flow, in case 101 value is too high, flowing material will have tension. This tension will reduce 180 mm original roll width down to 160-170 mm. Under these circumstances, puller roll speed must be decreased. If puller roll speed is decreased too much, the material will be loose and folding plates will not be able to make desired folding. 100 should be regarded as starting point and ideal value must be adjusted by increasing or decreasing the value according to the tension or looseness of the flowing material. When ideal value setting is done, enter point and exit point of folding plates should be measured and roll width must be 180 mm as a result. Material should be on a stable flow and be prevented from sliding left-right.



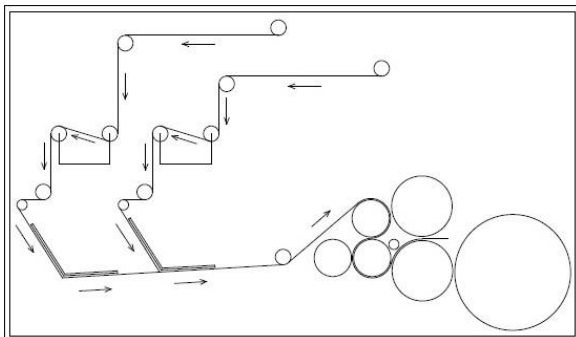
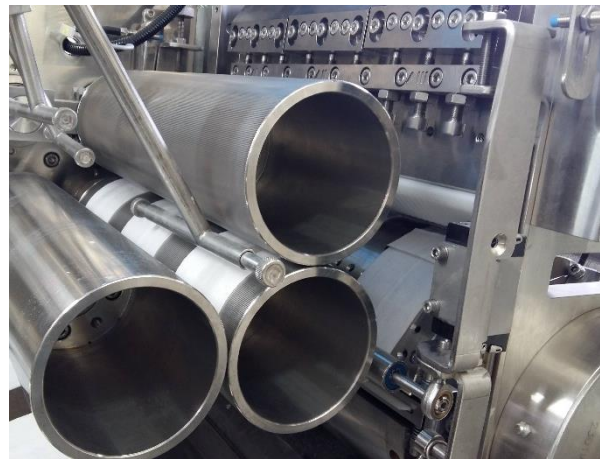
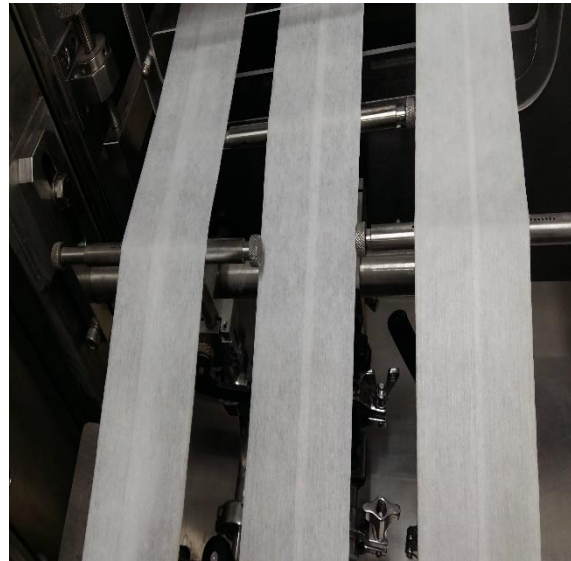
6. After puller roll, material width 180 mm must be folded into 90 mm, which is the recipe value, by going through the folding plates. To adjust this fold dimension, the button (in the picture) is turned to left or right which moves the folding plate and therefore desired folded width can be obtained.
7. The material should not touch against the wide surface of the folding plate. (see picture)
8. It must be checked that product touching rolls are turning and turning easily (see picture)
9. Wetting nozzle must be adjusted to impregnate the material going between two rolls around 5-10 mm (see picture)
10. Wetting nozzles must be positioned vertically against the surface of raw material. (see picture) Material's top surface should be impregnated. The surface that touches the folding plate must not be wet. (see picture) If nozzle tube pistons are not adjusted according to product width, liquid will overflow outside. (see picture) Tube pistons must be adjusted in a way not to overflow the liquid outside but enough to wet all the surface so no part of the product stays dry.



11. After material flow is complete and folded product is obtained from the puller, cross shaft is moved to left or right to adjust the folding axis of the product right into the middle of the product. (see picture) Since folding type is "Z", edges folded from the sides to top and bottom should be in line and meet in the middle of the product creating 90 mm width.
12. After this, edge control sensor centering settings are done (see Fife user guide) and automatic mode is selected. So if looseness or axis shift occurs, control system will tolerate it to some point.
13. After product width is adjusted and dimensions are obtained according to the recipe values, last wetting tube is positioned under the product with nozzles facing up (see picture) and product is widened 1 cm. If nozzle tube pistons are not adjusted according to product width, liquid will overflow outside. (see picture) Tube pistons must be adjusted in a way not to overflow the liquid outside but enough to wet all the surface so no part of the product stays dry.



14. After product's exit from puller roll with correct dimensions and homogeneous wetting, the material goes through the cutting roll. (see picture)
15. Make sure the blade is positioned linear to the cap.
After leaning the blade onto the cap, tighten the bolts on the cap enough to let set screws move the blade.
While the cutting edge of the blade is on tangent point to the bottom anvil, move the blade away from the set screws evenly and get it closer to the anvil enough to be 0-0,05 mm parallel to it.
Knife cutting is adjusted while machine runs, adjusting is repeated from the set screws until both edges cut evenly.
After cutting is adjusted, cap bolts are tightened completely.
See below detailed drawings.
16. There must be 6-6,5 mm between top point of plastic sheet apparatus and edge of plastic sheet material. (see picture)
You need to pay attention that while the plastic sheet enters between jaws, it should enter from the center without touching top or bottom jaw. **See below detailed drawings.**



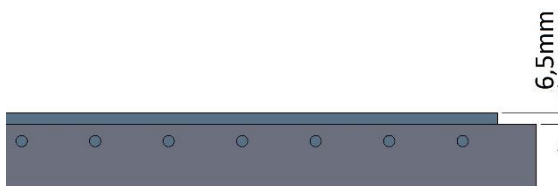
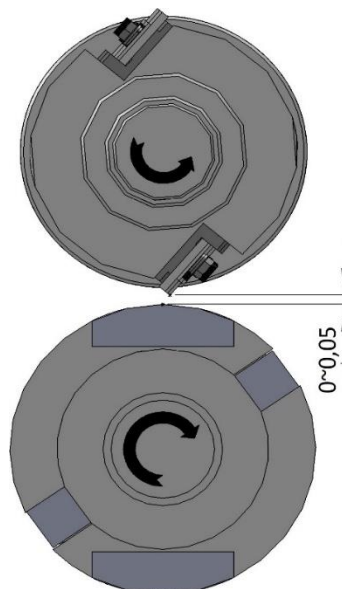
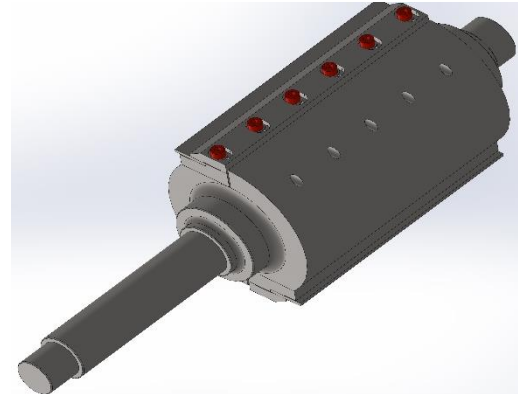
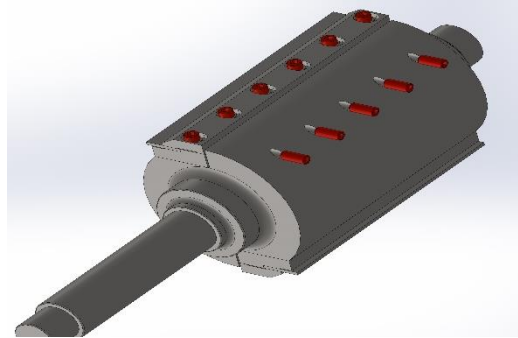
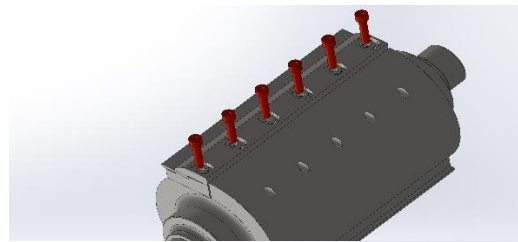
For cutting adjustment, first untighten stabilizing bolts of the blade very gently.

Tighten the set screws until blade position is proper.

After bringing the blade to the desired point, tighten stabilizing bolts and fix the blade.

When blade is first put on the roll, instructions on the top should be followed respectively before running the machine and blade should be adjusted to leave 0-0,05 mm distance between the anvil and itself. This adjustment must be done while edge of the blade is on the top tangent point of bottom anvil. After adjusting the space between, put a sheet of paper for testing, run the machine on jog speed and repeat the instructions above until optimum cutting is obtained. During this process, set screws must be tightened equally.

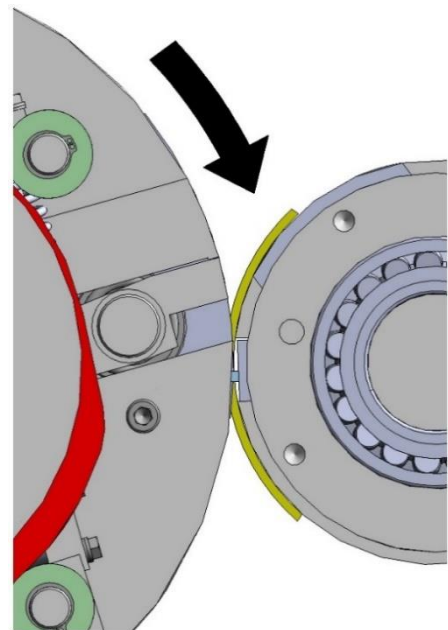
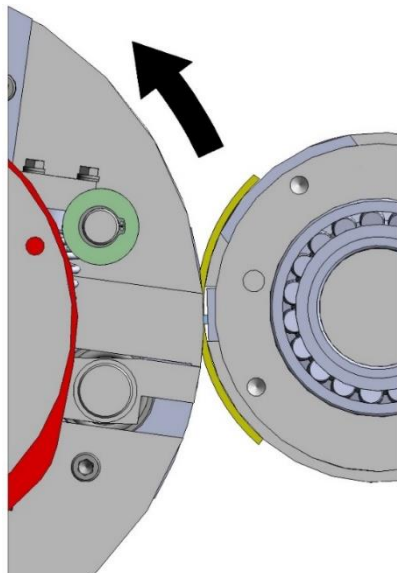
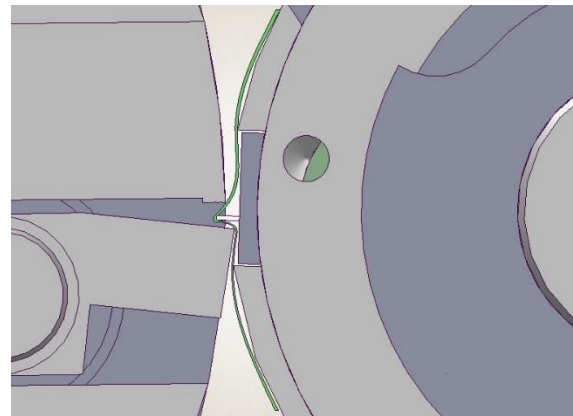
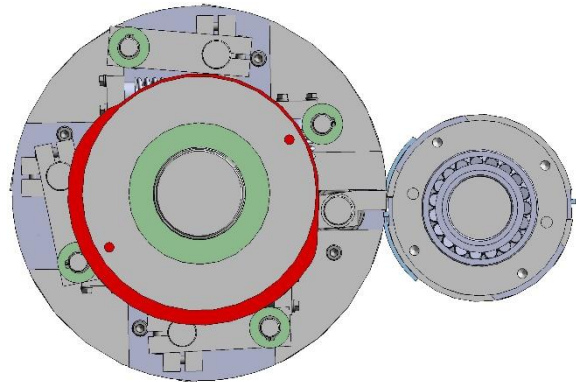
NOTE: While adjusting the cutting, speed should be %10-20 (jog speed)



When plastic sheet is held to the holder, it should be 5 mm higher.

While adjusting the cross folding roll, the plastic sheet on bottom cutting roll are called to reference. Plastic sheet should be around 2 mm inside of stable holder clip. The plastic sheet on bottom cutting roll and moving jaw on the cross folding roll should be on the same axis.

If moving jaw is more forward than the plastic sheet's axis, turn the cross folding roll counter clockwise and make it on the same axis with the plastic sheet. If moving jaw is more backward than the plastic sheet axis, turn the cross folding roll clockwise and make it on the same axis with the plastic sheet.



17. When stacker fingers are lowered, the distance between the bottom of the fingers which touch the product and the outer diameter of the cross folding roll should be 10 mm (see picture)

Working time of stacker fingers (time of going up and down) should be checked like this: when the plastic sheet comes to jaw center, bottom part of stacker fingers must go down from the outer diameter of the cross folding drum. This timing is adjusted by turning the running gear of stacker fingers

Interval measure of stacker finger's stopper and knife gusset is 300mm. (see picture)

Product stopper place and stacker fingers stopper area must be parallel. Jaw should open when stacker fingers are 35 mm away. (see picture)

Opening time of jaw is adjusted by changing the position of cam (see picture)

Distance between folding drum and upper position of co-stacker should be around 20 mm. On page STACK SETUP, STACK STEP SET value must be set as 2-2.5 mm. This value may vary due to material type and thickness. For example: When step set is little, product will be jammed and every wipe will be put on the stack earlier than the previous one which will create a stair type look.

To make main stacker and co-stacker work and transfer on the right time, STACK 1 OFFSET AND OFFSET 2 settings must be done from the SERVO SETUP page.

STACK 1 OFFSET: Co-stacker's timing of moving forward. If wrong value is set, co-stacker will hit against the wipes on the top. This means timing of co-stacker is too early.

If co-stacker is too late, then it will not catch the first wipes of the pack. In this case first wipes of the packs will tend to pop out. Since move forward time of co-stackers is independent from the machine speed, when optimum speed is found, it is not suggest to change it unless there is a material change.

STACK 2 OFFSET: Main stack platform's timing of moving down. It can be also referred as "pack wipe count complete, transfer start". If wrong value entered, below situations may occur:

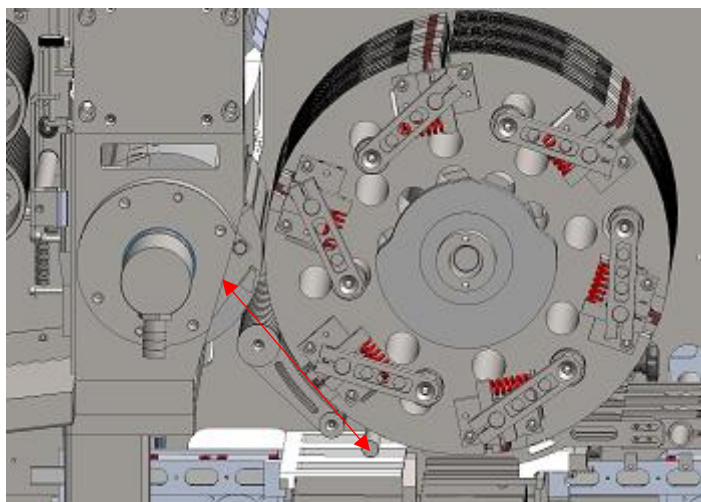
If stack platform goes down too early, because the last wipe won't be on correct position, the first wipe of pack will be outer. If stack platform goes down too late, co-stacker will hit against the wipes on top.

STACK 1 OFFSET – STACK 2 OFFSET settings are interconnected. Therefore first STACK 2 OFFSET (MAIN STACK) settings are done.

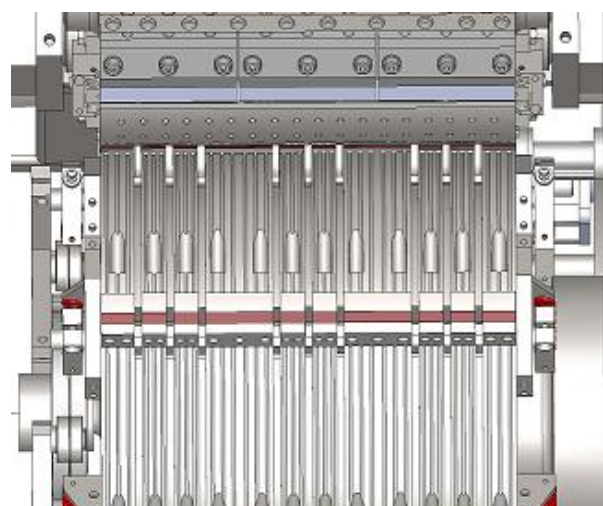
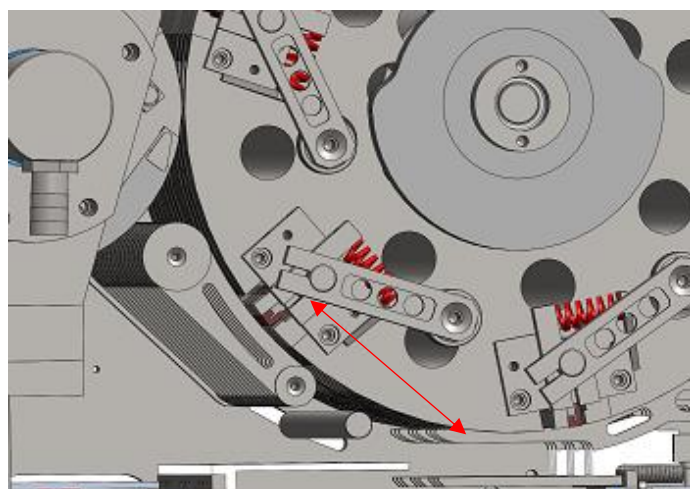
STACK 2 OFFSET value is increased, platform's early down movement and top wipe's wrong positioning is observed, little by little the value is decreased until top wipe finds the correct position and transfer is realized.

After optimum transfer time is set, STACK 1 OFFSET setting is done too.

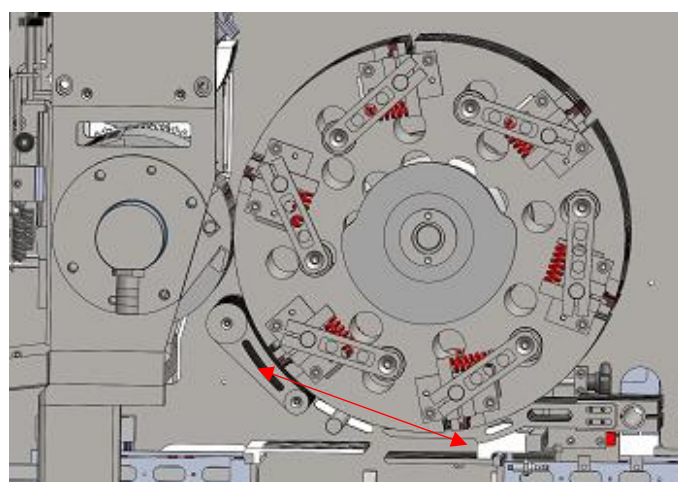
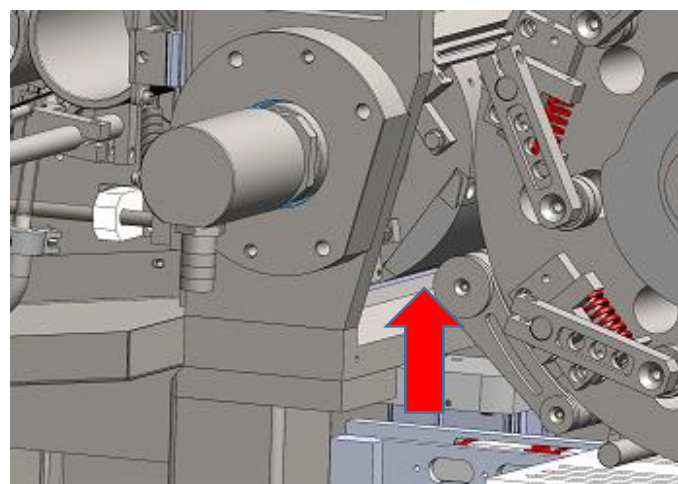
STACK 2 OFFSET value is reduced until co-stacker hits against the top wipe. This means platform goes down too early and the value is increased until there is no more hitting and optimum value is found like this.

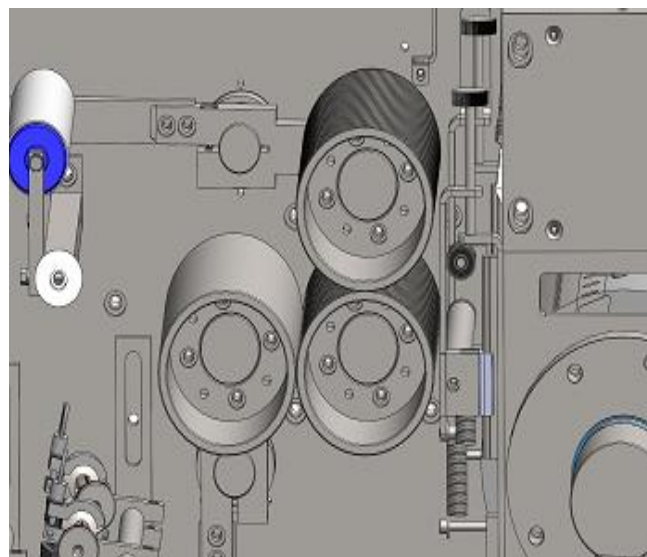
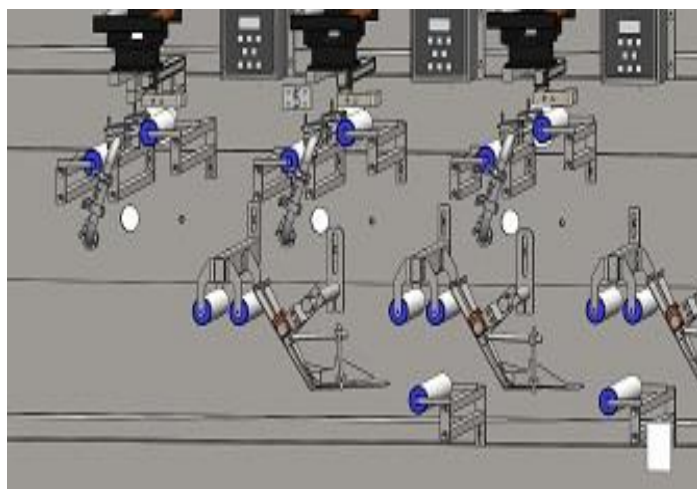
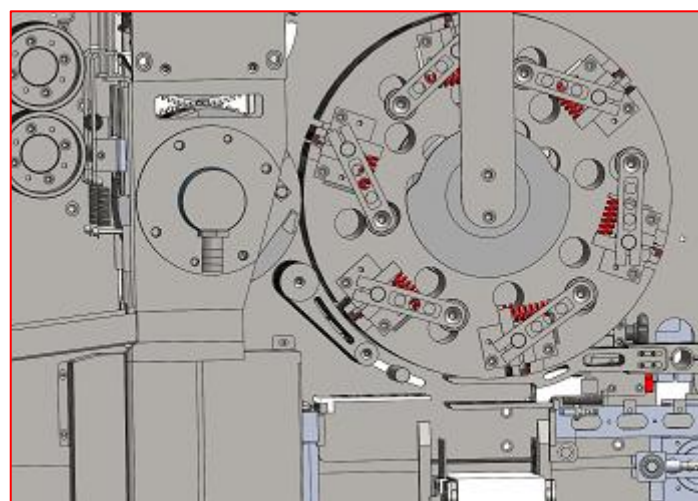
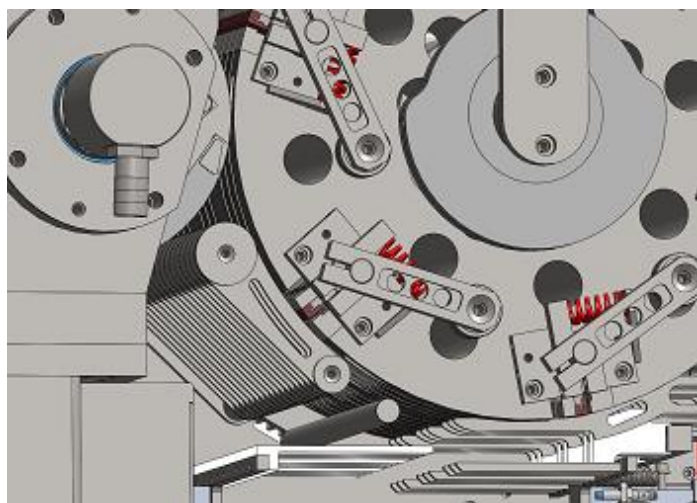


DISTANCE MUST BE 154 MM



DISTANCE MUST BE 2 MM





WARNING AND ALARM SYMBOLS

WARNING LABEL MEANINGS

- Stable panel should not be opened by anyone except electrician or authorized person!
- See relevant documents before intervening with electrical panel.



Contact us in case of servo driver problems.



- High voltage warning.
- In case of static reaction, human health hazard and malfunction in electronic cards occur.



Do not interfere before reading relevant documents.



- Network & high voltage line warning



- Mobile, boat or plugs are active

WARNING AND ALARM SYMBOLS

WARNING LABEL MEANINGS

- Read the instructions for air filter change and cleaning.



- Warning that shows trigger belt systems exist.
- Do not interfere if machine is running!
- Make sure machine is stopped or off during belt change!
- On areas protected with cover plate, make sure cover plate is assembled.



- Do not interfere except the operator!
- People that are not trained for the machine may cause work accidents!



- Warning label showing the areas with high risk

WARNING AND ALARM SYMBOLS

WARNING LABEL MEANINGS

- Do not interfere with the sensor
- Designates sensor reference point. If setting is not correct then it won't bring reference and will keep the machine from working correctly.



- Air hose connection label

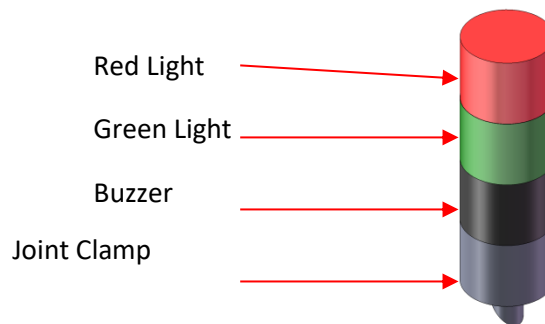


- Liquid hose connection label



- Moving parts and gears label

WARNING LABEL MEANINGS



RED LIGHT: This light turns on when:

- If there is a part not working during machine operation
- If there is any electrical malfunction
- If there is any alarm occurs

GREEN LIGHT: This light turns on when:

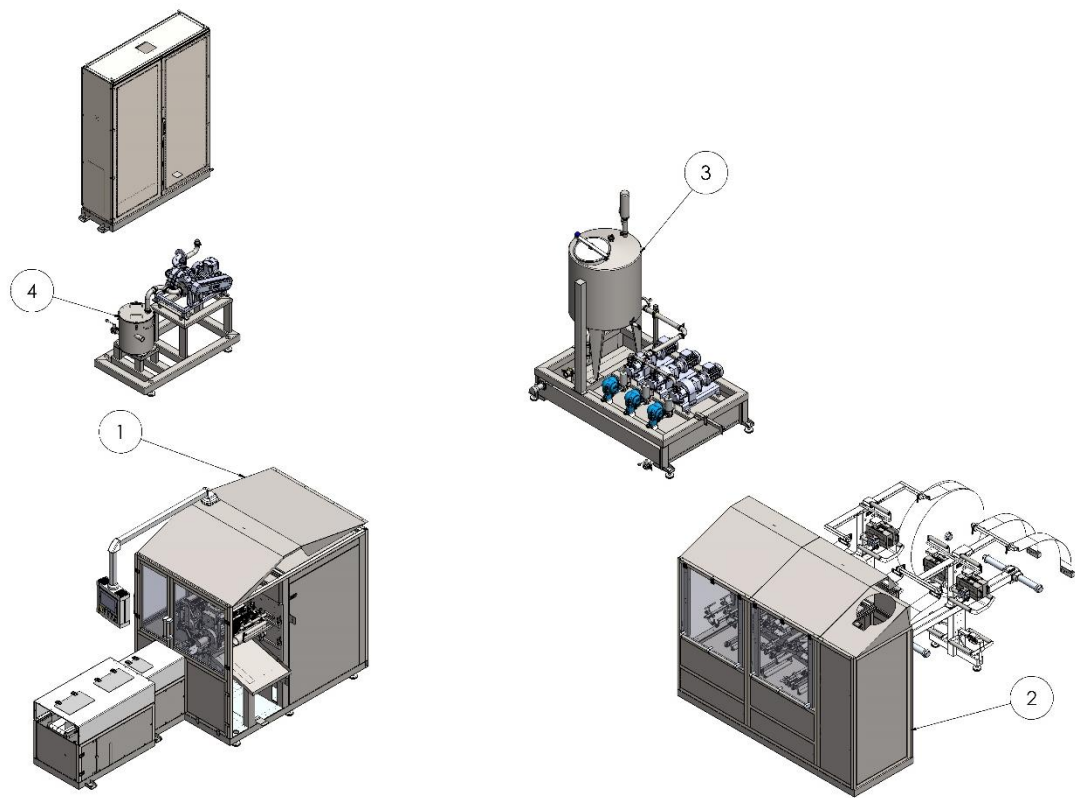
When all functions are normal and machine is on starting position.

BUZZER: When there is a situation that requires the red light to turn on, buzzer is activated with the red light to make audial stimulation.

Also it works like an alert just before the start-up of machine.

RED – GREEN LIGHT: When a warning or alarm occurs that does not require machine to stop; red light, green light and buzzer is turned on together. When the operator resets the alarm from HMI, buzzer is disabled and red and green warning lights keep flashing.

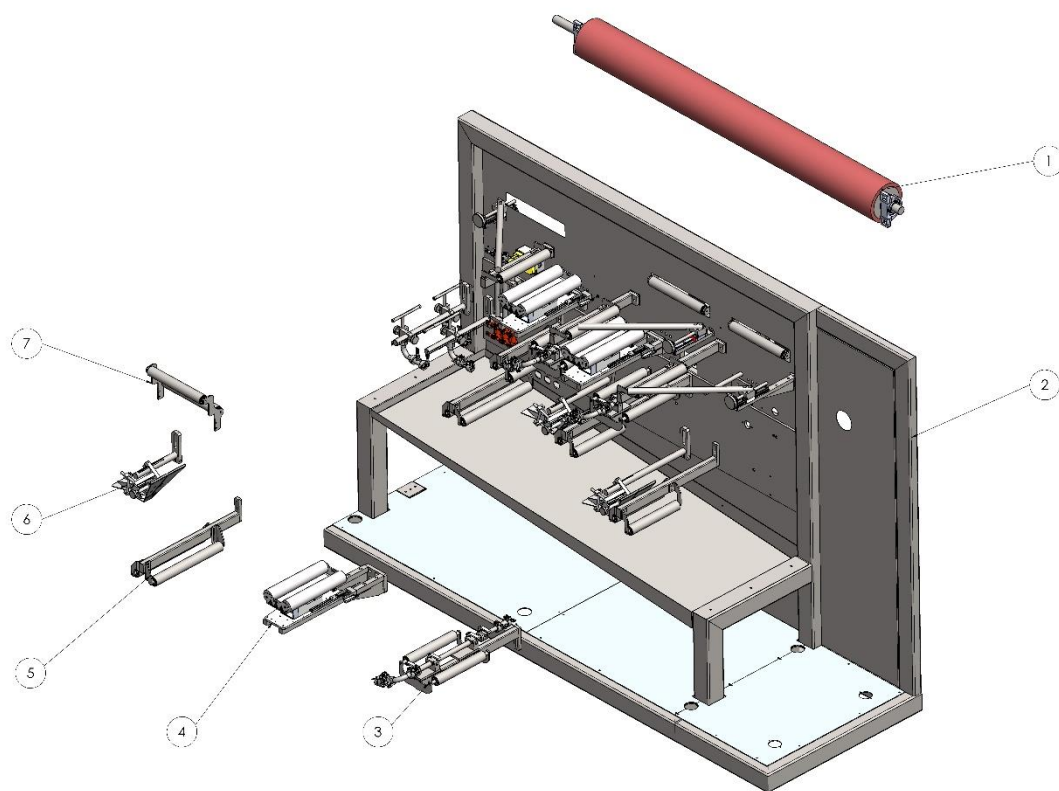
GENERAL MACHINE AND MODULES INTRODUCTION



ATEX FLEXIWORK 3 consists of 4 main parts as

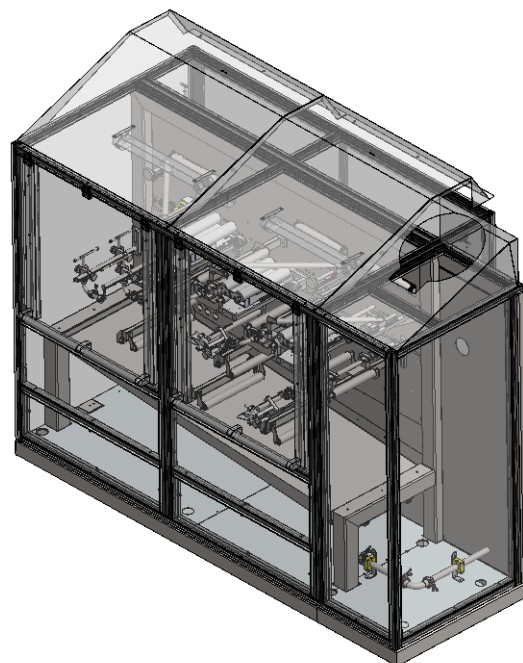
- 1- CABIN UNIT
- 2- BOBBIN UNIT
- 3- WETTING INSTALLATION UNIT
- 4- BLOWER STAND

BOBBIN UNIT INTRODUCTION

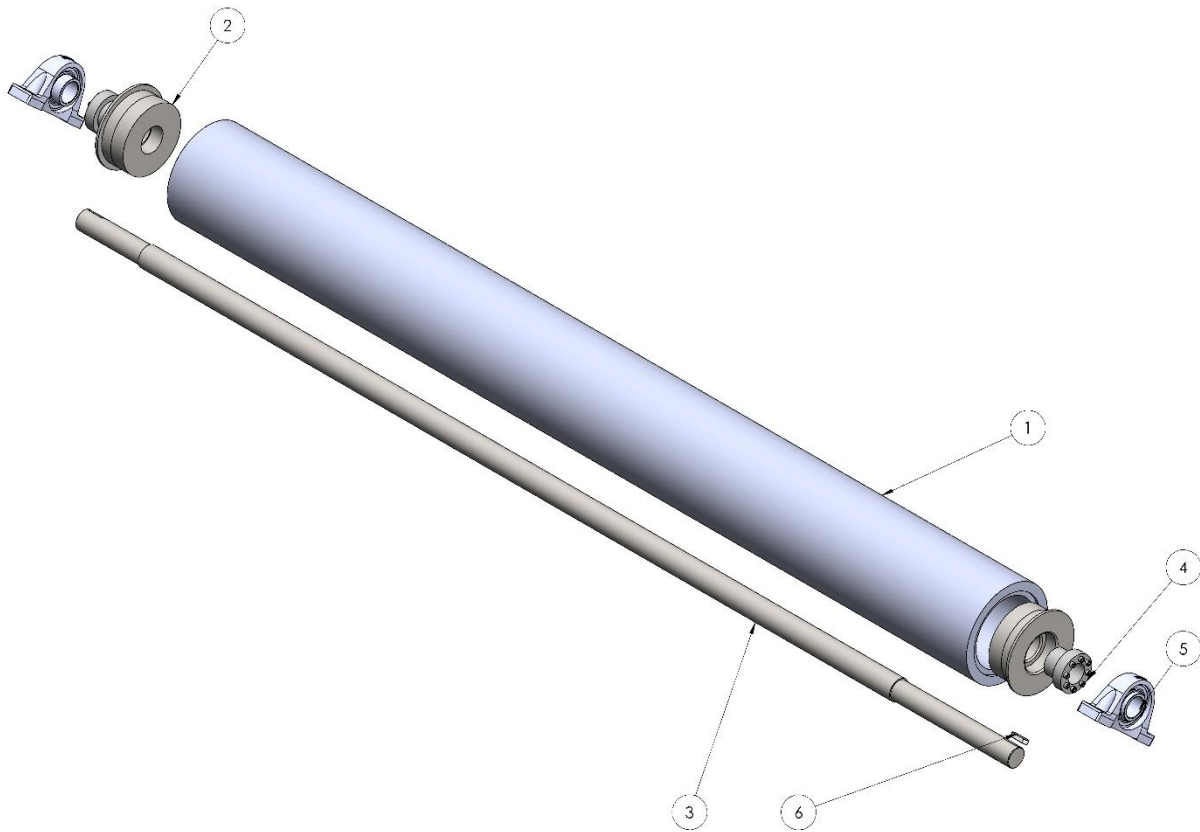


BOBBIN UNIT CONSISTS OF TWO PARTS:

- 1- UNWINDER MODULE
- 2- BOBBIN CHASE
- 3- PVC IDLE UNIT-1
- 4- EDGE CONTROL UNIT
- 5- PVC IDLE UNIT-2
- 6- FOLDING UNIT
- 7- UNDER UNWINDER IDLE UNIT

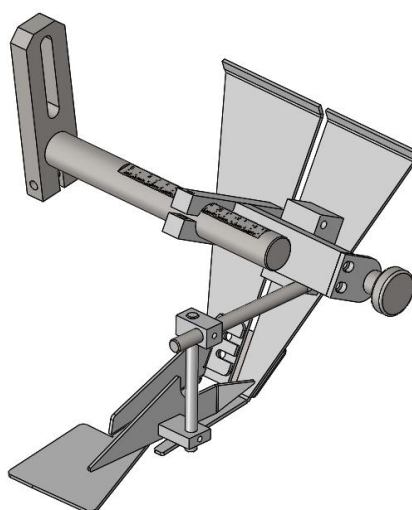
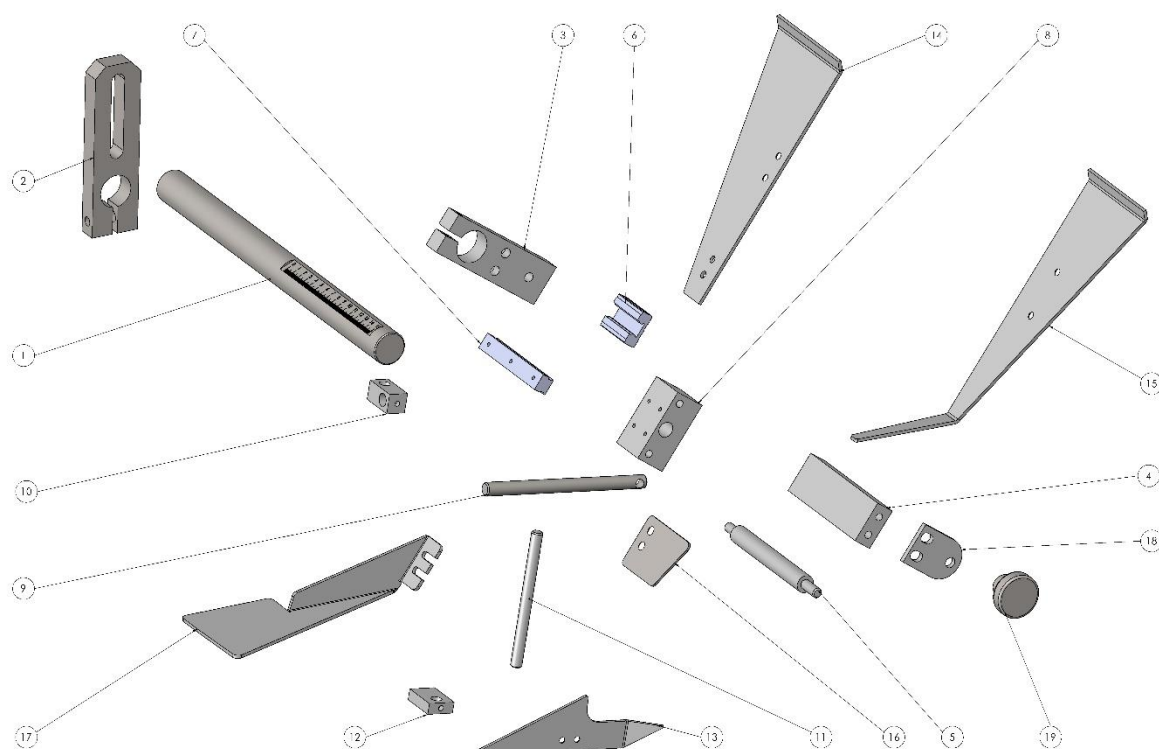


UNWINDER MODULE INTRODUCTION



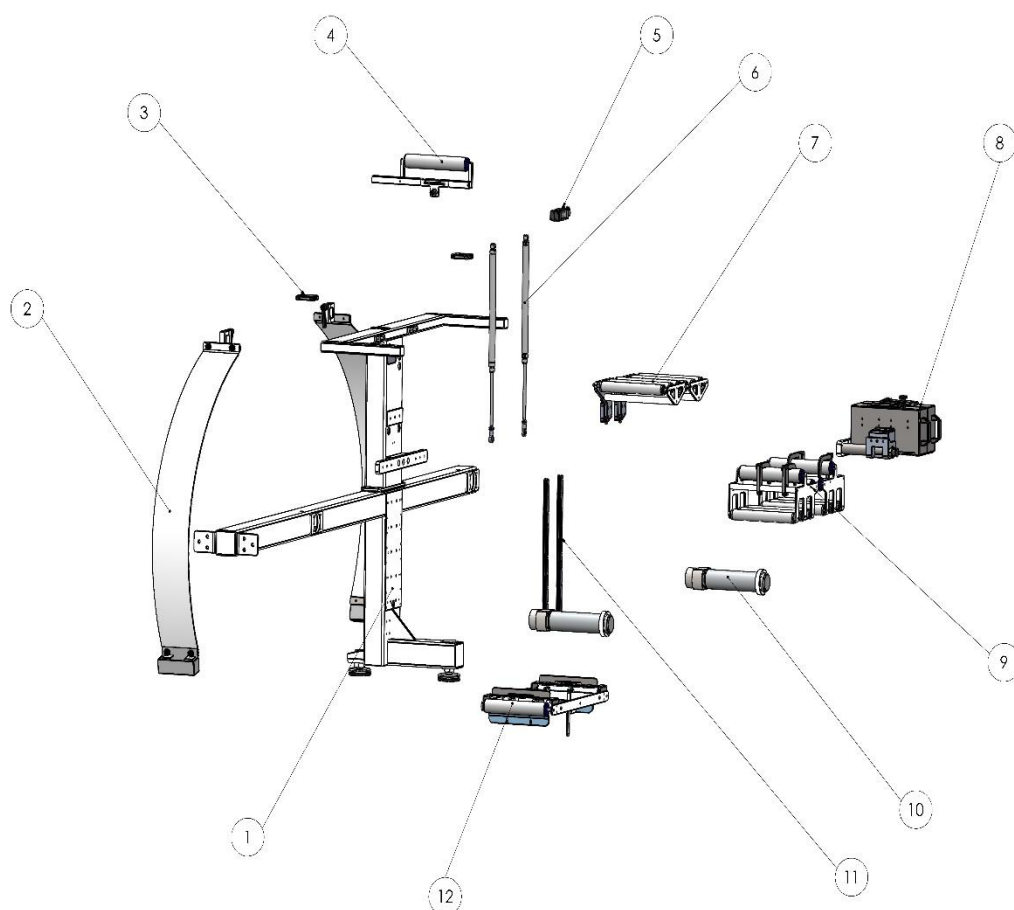
COMP NO.	CODE	EXPLANATION	QTY.
1	22012471	POLYAMID PULLER CYLINDER	1
2	22006222	PULLER CYLINDER PLUG	2
3	22015537	SHAFT OF PULLER CYLINDER	1
4	30008787	CTL 100-40 CONICAL CLAMP	2
5	30005244	UCP 208 BEARING HOUSE	2
6	---	12 x 8 x 40 PARALEL KEY	1

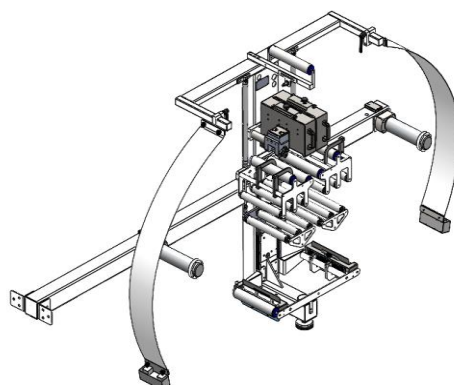
FOLDING PLATE INTRODUCTION



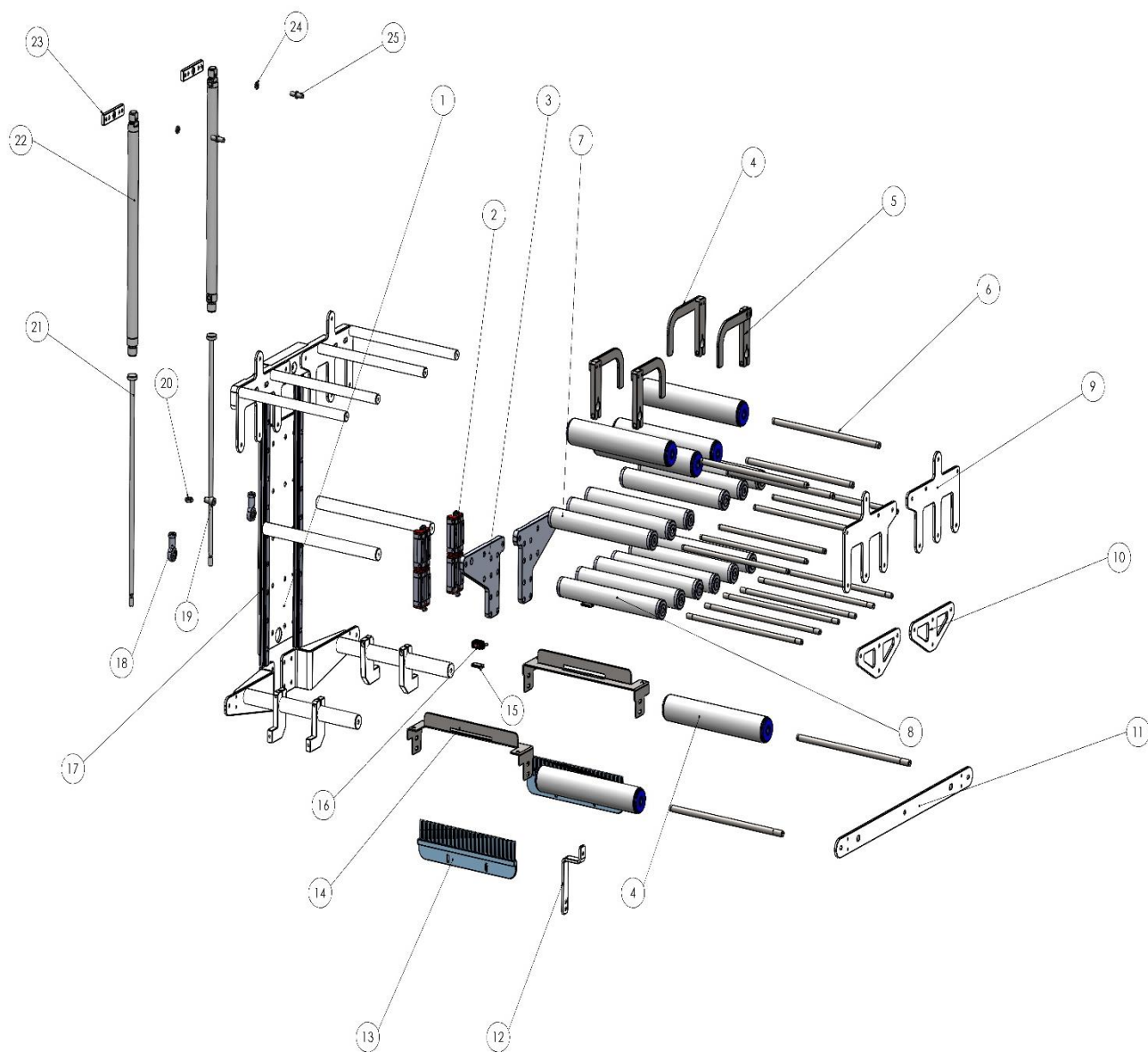
COMP. NO	CODE	EXPLANATION	QTY
1	22000733	FOLDING PLATE_2 RIGHT	1
2	22000734	FOLDING PLATE _1 RIGHT	1
3	22006169	FOLDING MOUNT	1
4	22006168	FOLDING CONNECTOR CLAMP	1
5	22002852	FOLDING ADJUST CONNECTION POINT	1
6	22000738	FOLDING APPARATUS SLEDGE CONNECTION	1
7	22000739	FOLDING APPARATUS ADJUST CONNECTOR	1
8	23017630	MR 15X100mm (CPC) LINEAR SLEDGE (MINIATURE STAINLESS) HOLE DIMENSION FROM EDGES 10 MM	1
9	22000740	MR 15 MN (CPC) LINEAR CARRIER	1
10	24102089	MECHANIC NUMERATOR OB14-D-1-S (LINPOTEK)	1
11	23005953	CE 85851 COUNT METER BUTTON	1
12	24102091	FOLDING APPARATUS SIDE CONNECTION SHAFT	1
13	30013235	FOLDING APPARATUS SHAFT CONNECTION MOUNT	1
14	23001003	FOLDING APPARATUS SIDE CONNECTION SHAFT	1
15	30008490	FOLDING APPARATUS SHAFT JOINT POINT	1
16	23008687	FOLDING PLATE CONNECTION SHAFT _1	1
17	30001326	2. CONNECTION APPARATUS SHAFT	1
18	30010925	FOLDING PLATE _3 RIGHT	1
19	22019234	MIDDLE FOLDING PLATE	1
20	22019235	FOLDING PLATE SIDE PLATE	1

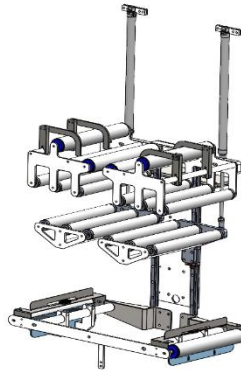
BOBBIN STAND INTRODUCTION





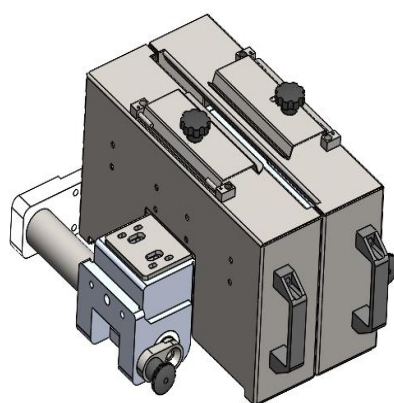
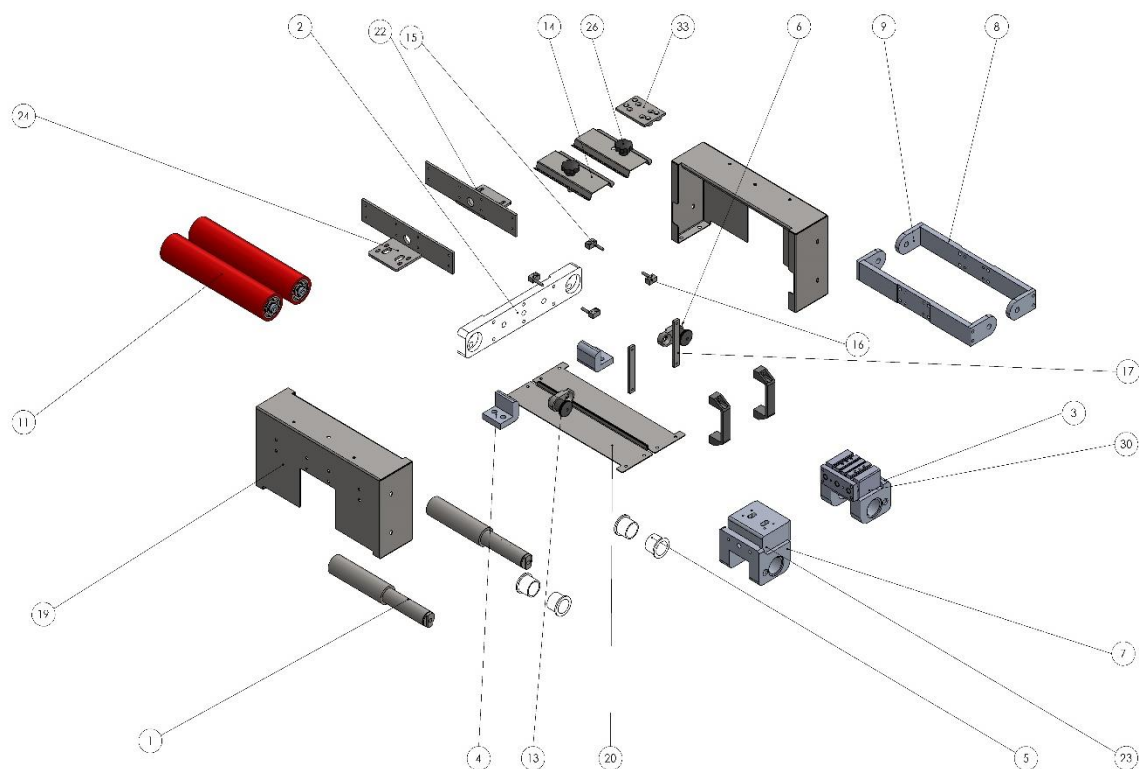
COMPONENT NO.	CODE	EXPLANATION	QUANTITY
1	-----	BOBBIN CHASE	1
2	30018826	CONVEYOR BELT	2
3	23013911	SHEET OF SENSOR	2
4	-----	UPPER IDLE GROUP	1
5	-----	FESTO VALVE	4
6	30017476	DSNU-25-400-PPV-A PISTON (35193) FESTO	2
7	-----	MOVABLE IDLE GROUP	2
8	-----	SPLICE GROUP	1
9	-----	STABLE IDLE GROUP	2
10	20005883	BOBBIN SHAFT	2
11	30004129	HGR 20 LINEAR RAIL	2
12	-----	LOWER IDLE GROUP	2

ACCUMULATOR DETAIL



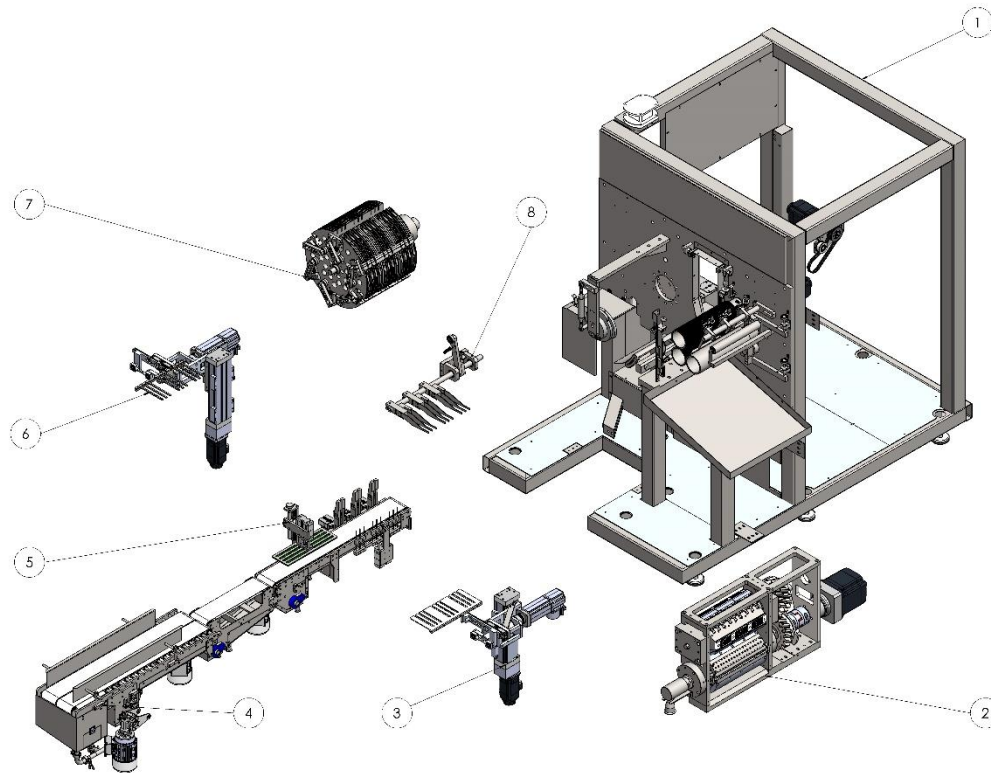
NO	CODE	EXPLANATION	QTY
1	30017701	UNDER RAIL CHASE CONNECTION SHEET	1
2	30000905	LINEAR CARRIAGE	1
3	22013679	PVC PIPE	1
4	22013679	PVC PIPE	1
5	22014327	ADJUSTMENT PLATE	1
6	22013680	ROD OF IDLE	1
7	22013683	PVC PIPE-2	1
8	22013683	PVC PIPE-2	1
9	21061189	IDLE CONNECTION SHEET	1
10	21061192	MOVABLE IDLE CONNECTION SHEET	1
11	21190936	LOWER IDLE CONNECTION SHEET	1
12	21190850	UNDER SPLICE IDLE SUPPORT	1
13	22014540	BRUSH	1
14	23014133	SENSOR SHEET	1
15	23013968	SENSOR SHEET-2	1
16	30004389	SENSOR	1
17	30004129	LINEAR RAIL	1
18	30010813	SGS M10*1,25 ROD EYE FESTO	1
19	22013807	PISTON CONNECTION PIN	1
20	30016426	M10 NUT	1
21	30017476	DSNU 25-400-PPV-A PISTON FESTO	2
23	21190812	PISTON WELDED LOWER CHOCK	2
24	22014584	PISTON CONNECTION WASHER	2
25	22014002	PISTON CONNECTION PIN	2

SPLICE GROUP DETAIL



NO	CODE	EXPLANATION	QTY
1	22013665	SPLICE GROUP CONNECTION SHAFT	2
2	21190804	CHASE CONNECTION PLATE	1
3	30017796	HOUSED PISTON	1
4	22013663	SPLICE STOPPER CHOCK	2
5	22013666	MOBILE CHOCK HOUSE RING	4
6	22013664	SPLICE LOCK CONNECTION PLATE	2
7	21151305	MOVABLE CONNECTION CHOCK	1
8	22013966	SPLICE CYLINDER CONNECTION PLATE	2
9	22013661	CYLINDER SIDE CONNECTION PLATE	4
11	21040048	SPLICE POLYAMID CYLINDER	2
12	23012584	WIPE CLAMP SHEET	2
13	22013670	PIN OF WIPE CLAMP SHEET	4
14	22013669	CHOCK OF PIN OF WIPE CLAMP SHEET RIGHT	2
15	22013668	CHOCK OF PIN OF WIPE CLAMP SHEET RIGHT	2
16	22013668	CHOCK OF PIN OF WIPE CLAMP SHEET LEFT	2
17	23012810	SPLICE GROUP ENCLOSURE SHEET	2
18	23012809	SPLICE GROUP UNDER ENCLOSURE SHEET	2
19	20201146	LID SHEET SUPPORTER PLATE	1
20	21151320	SPLICE GROUP CYLINDER CONNECTION CHOCK	1
21	20201345	LID SHEET SUPPORTER PLATE-2	1
22	30009918	HANDLE	2
26	30017846	BAKALITE HOLDER	

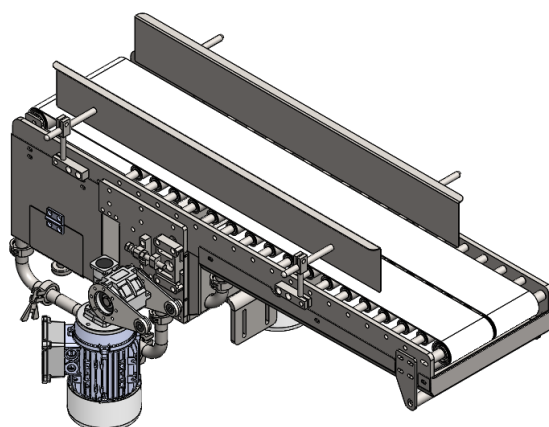
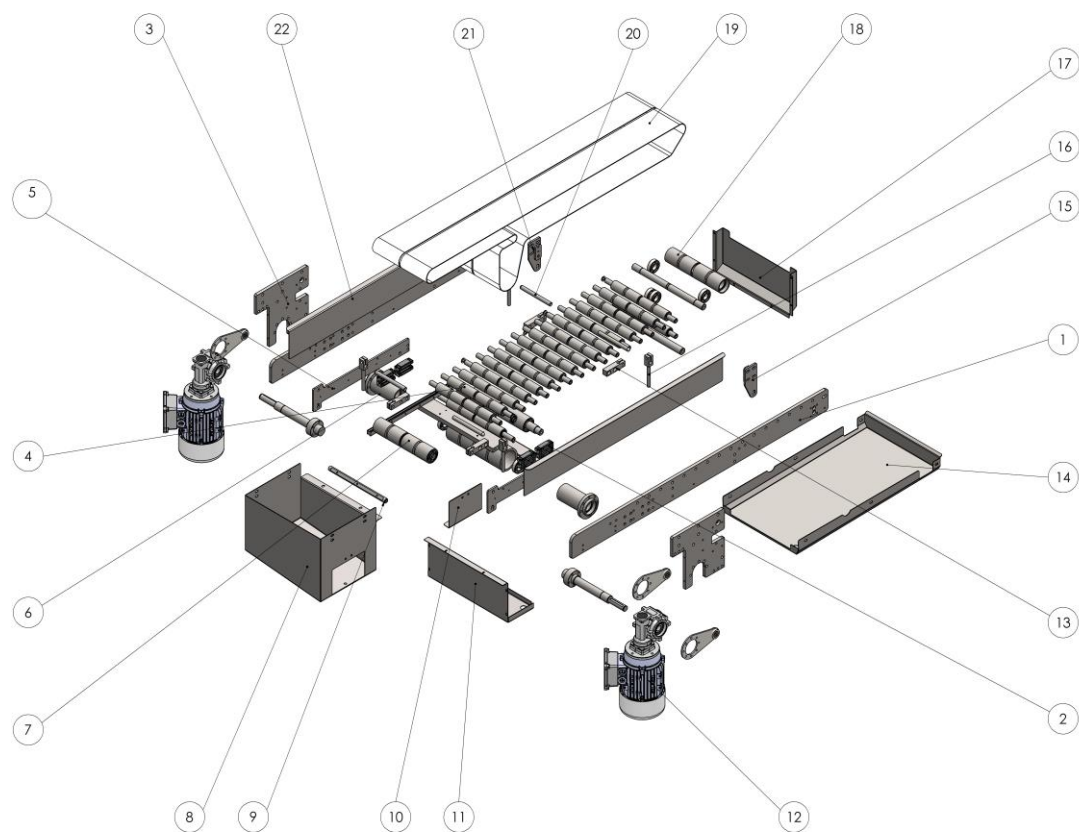
CABIN UNIT INTRODUCTION



CABIN UNIT CONSISTS OF 9 MAIN PARTS:

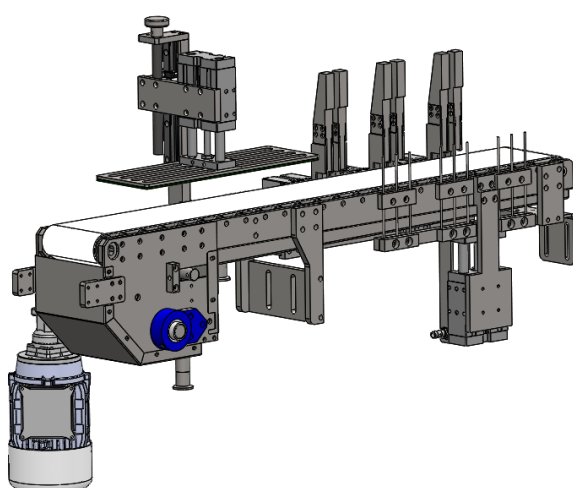
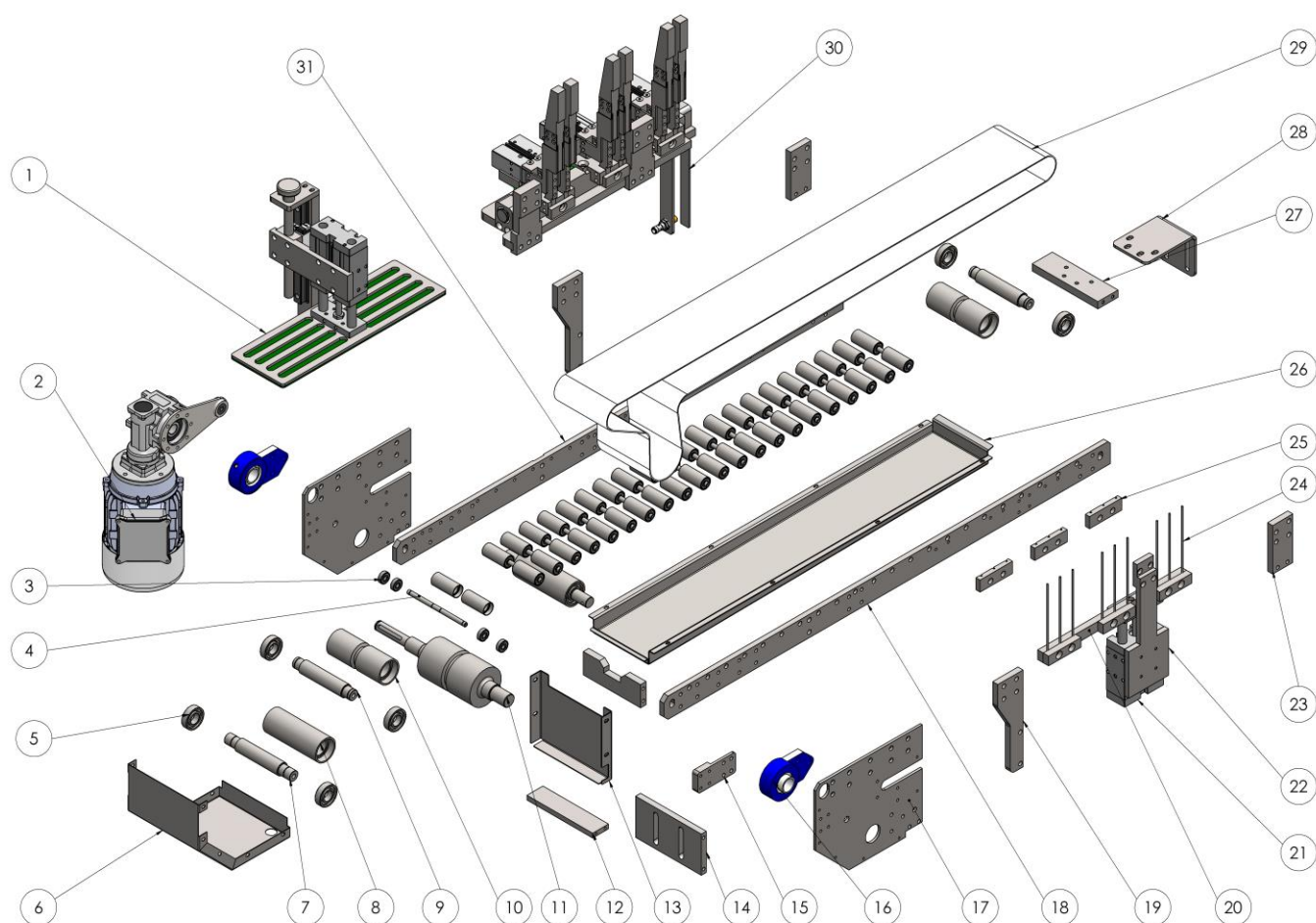
- 1- CABIN CHASE
- 2- KNIFE MODULE
- 3- STACKING MODULE-1
- 4- REJECT CONVEYOR MODULE
- 5- EXIT CONVEYOR MODULE
- 6- STACKING MODULE-2
- 7- FOLDING DRUM MODULE
- 8- MOON BAR MODULE

REJECT CONVEYOR MODULE



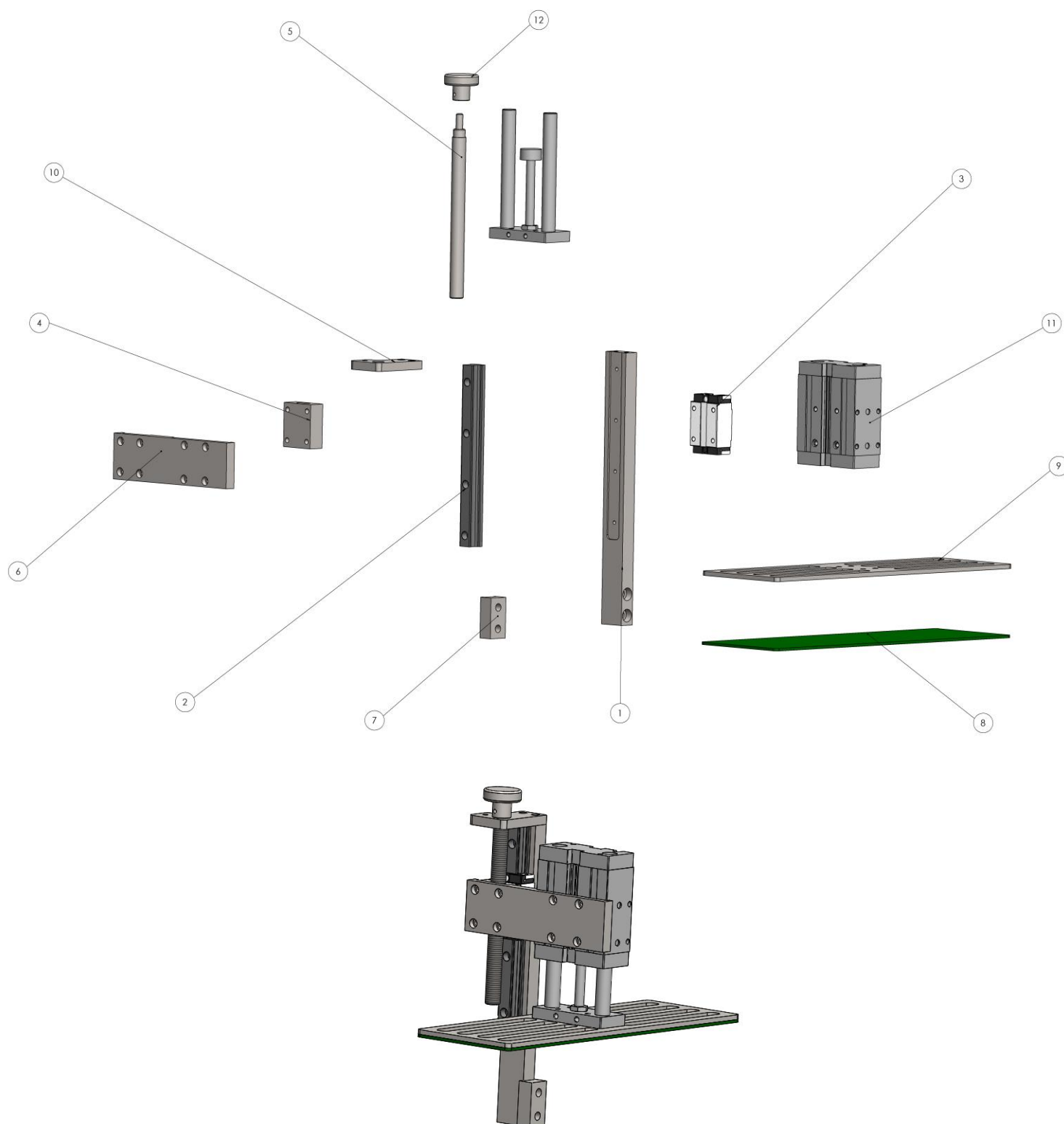
COMP NO.	CODE	EXPLANATION	QTY.
1	23016799	CONVEYOR LOWER PLATE	2
2	30004605	KWVE 15 BS G3 V1 LINEER CARRIAGE	4
3	22018540	MOTOR CONNECTION SHEET	2
4	22015114	IDLE PIPE	16
5	23016798	RAIL CONNECTION PLATE	2
6	22015109	DRIVER IDLE ASSEMBLY	2
7	22015112	IDLE ASSEMBLY	1
8	23016794	REJECT DRAIN LID	1
9	22014772	BRUSH CONNECTION ROD	1
10	23016797	REJECT DRAIN LID	1
11	23016796	REJECT DRAIN LID	1
12	30022787- 30022357	MOTOR VE GEARBOX	1
13	22009374	SIDE PLATE CONNECTION CHOCK	4
14	23016803	LOWER PAN	1
15	23013325	MID CONNECTION SHEET	2
16	22015242	SIDE PLATE CONNECTION CHOCK	4
17	23016795	LOWER PAN 2	1
18	-----	IDLE ASSEMBLY	1
19	30023084	CONVEYOR BAND	2
20	22015118	CONVEYOR ROD	4
21	23013331	IDLE CONNECTION PLATE	2
22	23016802	SIDE CONNECTION SHEET	2

EXIT CONVEYOR MODULE



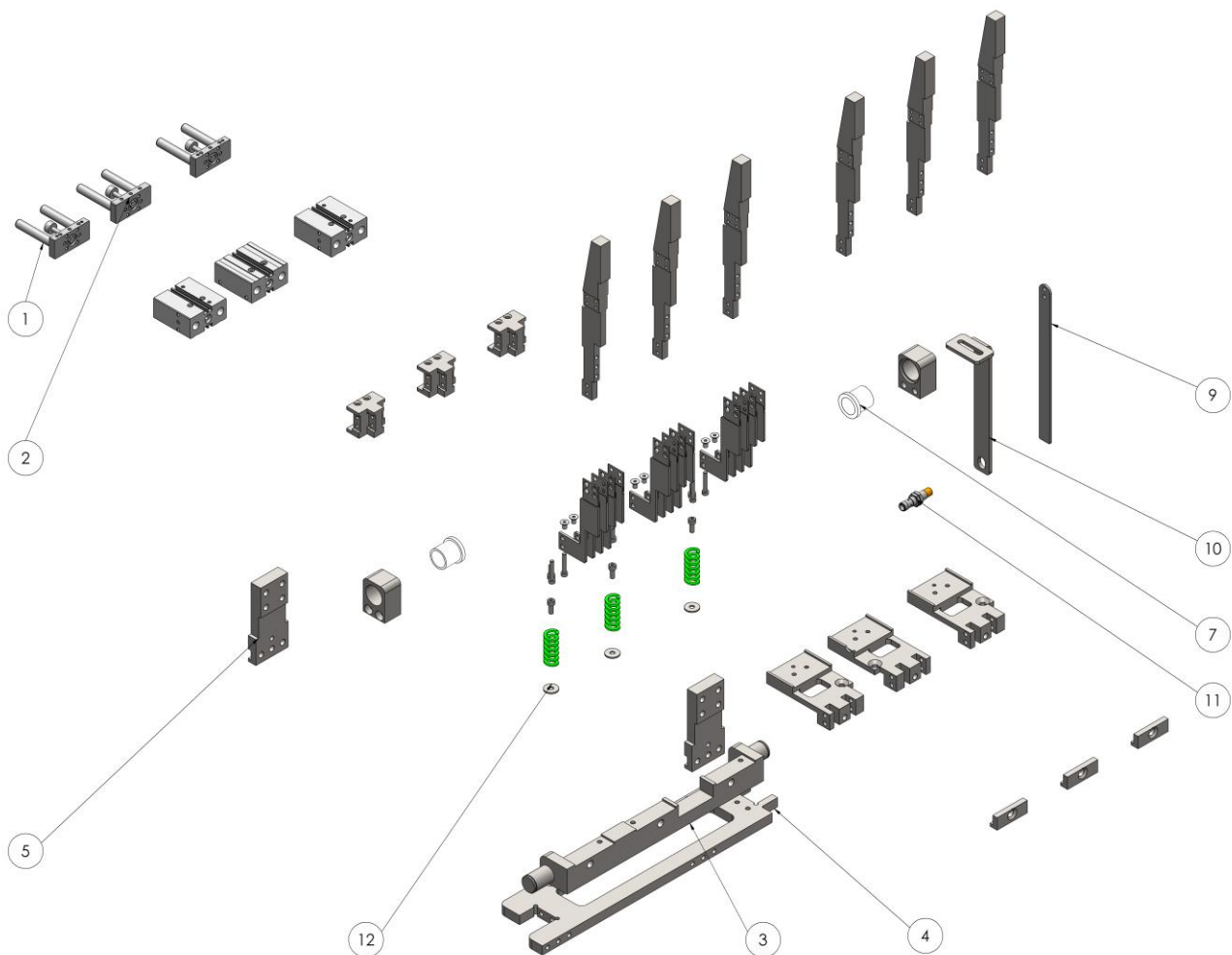
COMP NO.	CODE	EXPLANATION	QTY.
1	20401209	PRESS GROUP	1
2	30022787	GEARBOX	1
3	30011655	S608 2RS BEARING	88
4	22018526	IDLE ROD	22
5	30004835	SS6004 ZZ BEARING	8
6	23016787	CONVEYOR PAN	1
7	22018525	IDLE ROD	1
8	22015503	IDLE PIPE	2
9	22018525	IDLE ROD	2
10	22015495	IDLE PIPE	2
11	22015110	DRIVER CYLINDER	1
12	22018537	SUPPORT PLATE	1
13	23016785	CONVEYOR PAN	1
14	22018527	CONNECTION PLATE	1
15	22018532	CONNECTION SHEET	2
16	30001245	UCFK 205 BEARING	2
17	23016789	DRIVER CONNECTION SHEET	2
18	22018521	SIDE PLATE	1
19	22015508	MID CONNECTION PLATE	2
20	22012526	ROD CONNECTION-1	1
21	30023073	532317 FESTO MODULE	2
22	22018528	PISTON CONNECTION PLATE	1
23	22018527	CONNECTION PLATE	2
24	22018530	STOPPER ROD	9
25	22018531	ROD ALIGNMENT CHOCK	6
26	23016786	CONVEYOR PAN	1
27	22018534	MID PLATE	1
28	23016788	CONVEYOR HOLDER SHEET	1
29	30023082	BELT 125x2430	1
30	20401211	STOPPER GROUP	1
31	22018520	SIDE PLATE	1

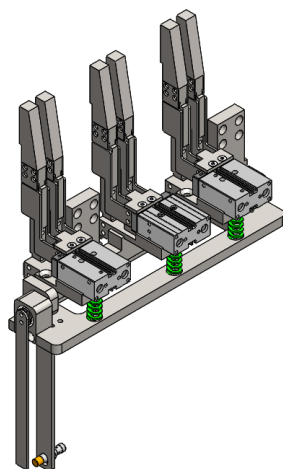
PRESS GROUP



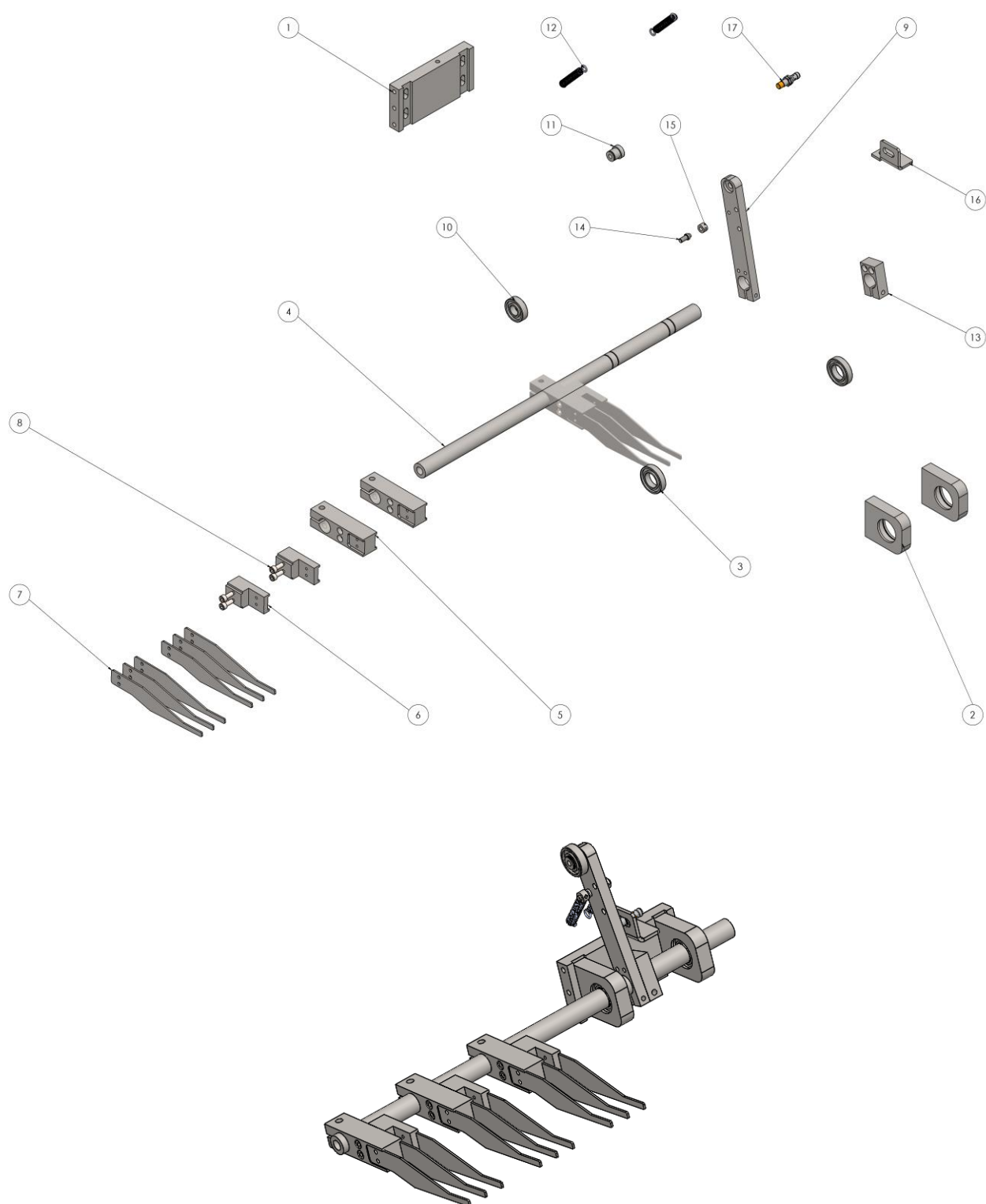
COMP NO.	CODE	EXPLANATION	QTY.
1	22015517	CONVEYOR CONNECTION PLATE	1
2	30004129	LINEER RAIL CPC	1
3	30007661	LINEER CARRIAGE CPC	1
4	22011356	CARRIER CONNECTION PLATE	1
5	22018507	ADJUSTMENT ROD	1
6	22012536	CONNECTION PLATE	1
7	22018506	CONNECTION PLATE	1
8	30023074	PRESSING BELT	1
9	23011465	PRESS SHEET	1
10	22017187	CAFE HOT CONNECTION PLATE	1
11	30023073	532317_DFM-25-50-B-PPV-A-GF-EX3 FESTO ATEX	1
12	22018300	PULLER ADJUSTMENT HEAD	1

HINGE STOPPER GROUP

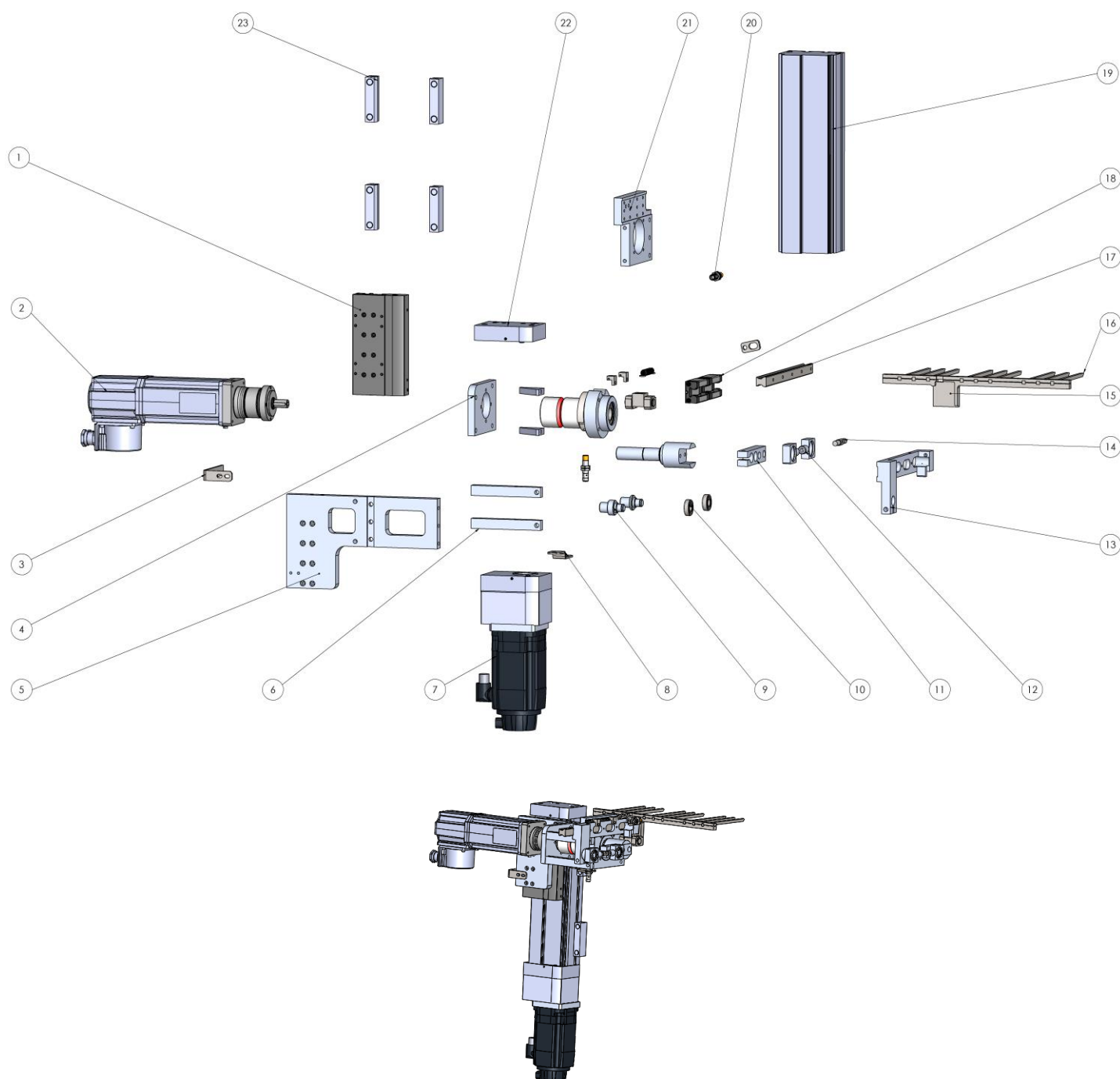




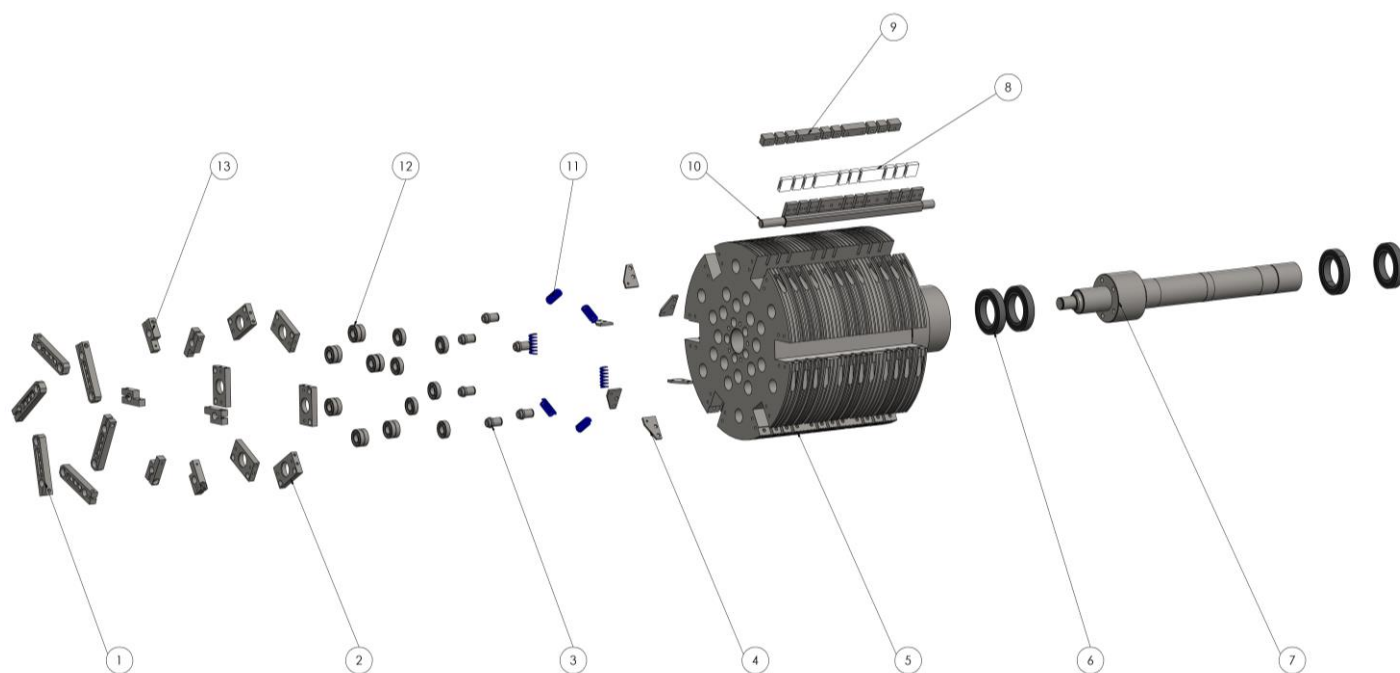
COMP NO.	CODE	EXPLANATION	QTY.
1	30023079	529120 FESTO MODULE	3
2	30023079	529120 FESTO MODULE	3
3	22014800	HINGE CHOCK	1
4	22014802	LOWER PLATE	1
5	22018513	CONNECTION PLATE	2
6	22014804	HINGE CHOCK HOUSE	2
7	22014805	SPACER OF CHOCK HOUSE	2
8	-----	M6x15 BOLT	3
9	23013345	SENSOR SHEET	1
10	23016778	SENSOR CONNECTION SHEET	1
11	-----	SENSOR	1
12	23013451	SPACER OF SPRING	3

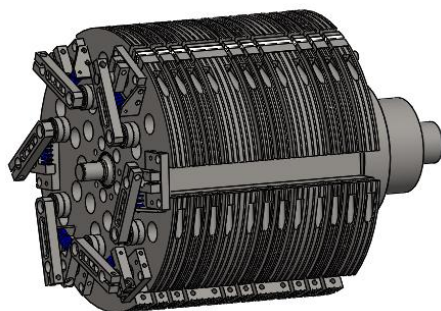
MOON BAR MODULE

COMP NO.	CODE	EXPLANATION	QTY.
1	22014427	MAIN CONNECTION CHOCK	1
2	22014428	SIDE CONNECTION CHOCK	2
3	30019258	SS6006 ZZ BEARING	2
4	22015576	CONNECTION PIPE	1
5	22014384	CONNECTION CHOCK	3
6	22014385	CONNECTION CHOCK	3
7	23012975	MOON BAR	9
8	-----	M8 x 20 HEXOGANE SOCKET HEAD STAINLESS	6
9	22017029	MOON BAR ARM	1
10	30023140	6204 RADIAL CONTACT BALL BEARING	1
11	22014429	ARM BEARING PIN	1
12	30021297	SPRING	2
13	22014633	CLAMP ARM	1
14	22015562	SPRING SPACER	2
15	22015561	SPRING SPACER	2
16	23016779	SENSOR CONNECTION SHEET	1
17		SENSOR	1

STACK MODULE

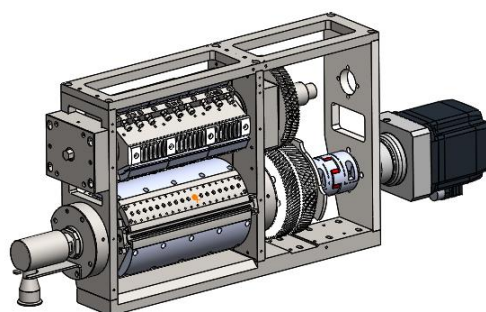
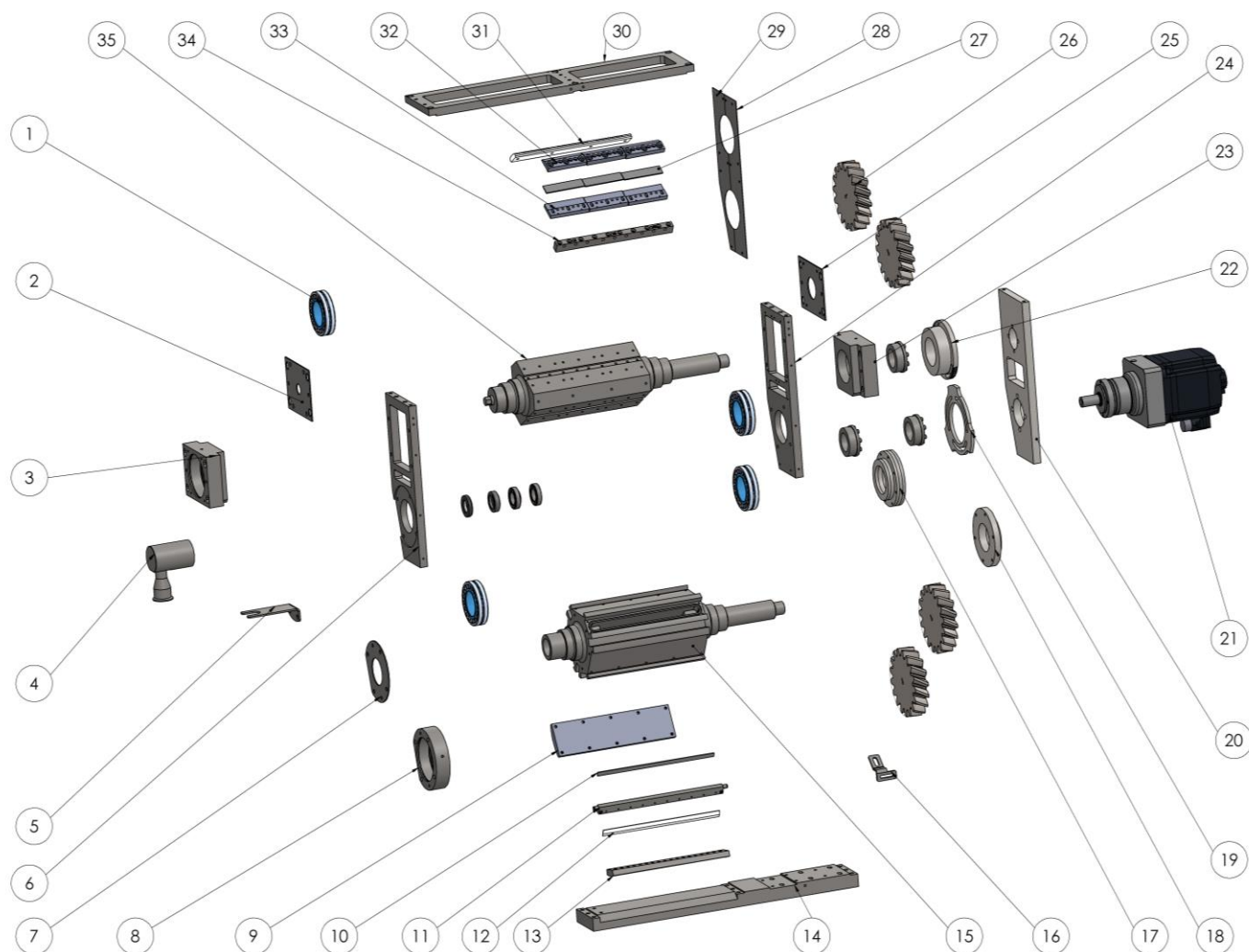
COMP. NO.	CODE	EXPLANATION	QTY.
1	30023343	(556809) EGC-120-150-BS-25P-KF-0H-ML-GK - PART 3	1
2	30023342	MKE047B-XXX-XGX-BEXX GEARBOX	1
3	23014033	SENSOR SHEET	1
4	21151565	GEARBOX CONNECTION PLATE	1
5	21151821	MAIN CONNECTION PLATE	1
6	21150339	GEARBOX CONNECTOR CHOCK	2
7	30023343	(556809) EGC-120-150-BS-25P-KF-0H-ML-GK - PART 1	1
8	23017051	SENSOR SHEET	1
9	21150335	CONNECTION PIN	1
10	30001243	6002 ZZ RADIAL CONTACT BALL BEARING	2
11	21150333	CONNECTION CLAMP	1
12	21120338	BEARING HOUSE	1
13	21150462	HINGED MOTION ARM	1
14	22011383	ADJUSTMENT GIGON	1
15	22017381	CONNECTION PLATE	1
16	22018696	STACKING ROD	9
17	30013417	BOSCH REXROTH-RAIL	1
18	30013416	BOSCH REXROTH-CARRIAGE R201189404	2
19	30023343	(556809) EGC-120-150-BS-25P-KF-0H-ML-GK- PART 4	1
20	-----	SENSOR	2
21	21151266	CARRIAGE CONNECTION PLATE	1
22	30023343	(556809) EGC-120-150-BS-25P-KF-0H-ML-GK- PART 5	1
23	30023343	(556809) EGC-120-150-BS-25P-KF-0H-ML-GK- PART 6	4

FOLDING DRUM MODULE

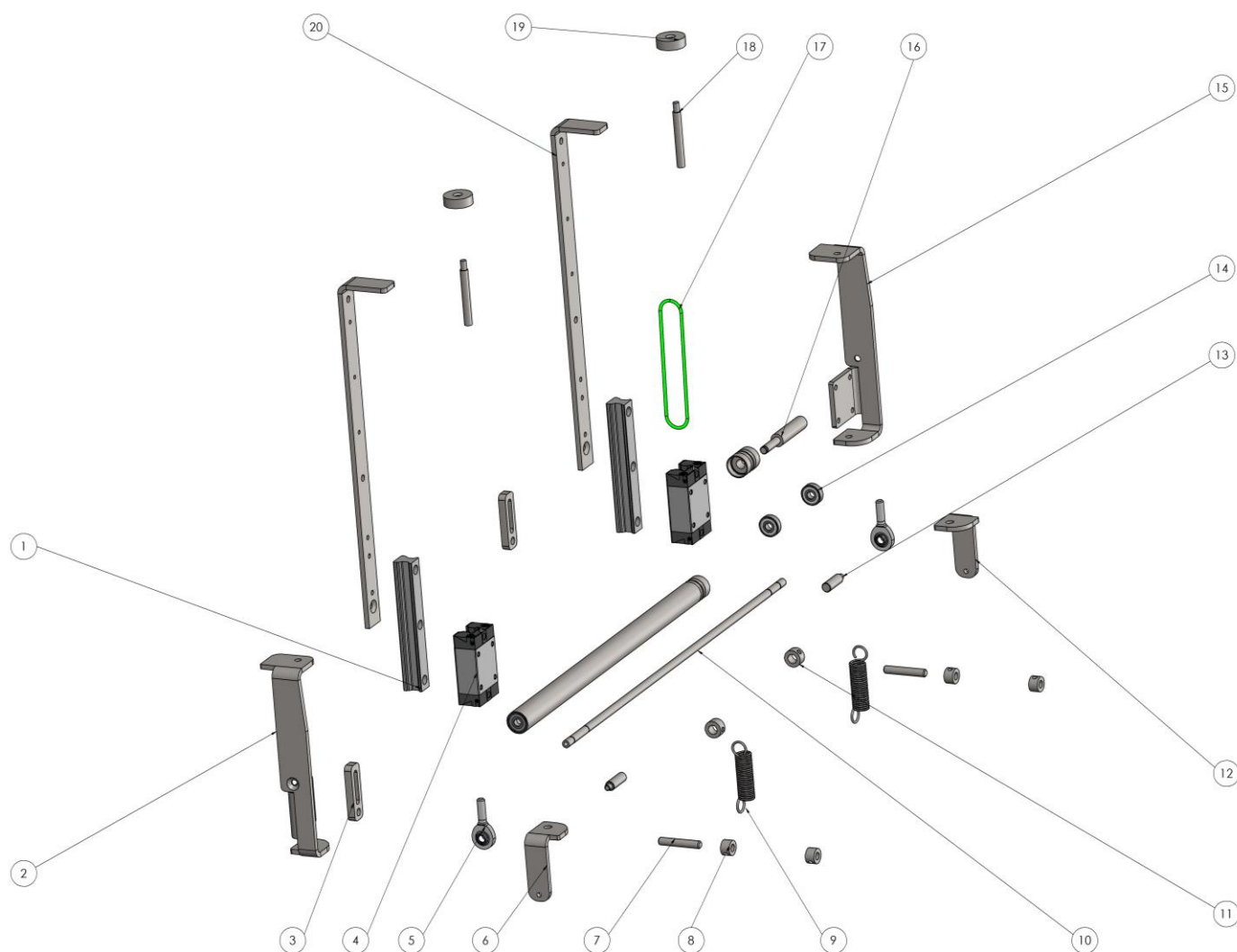


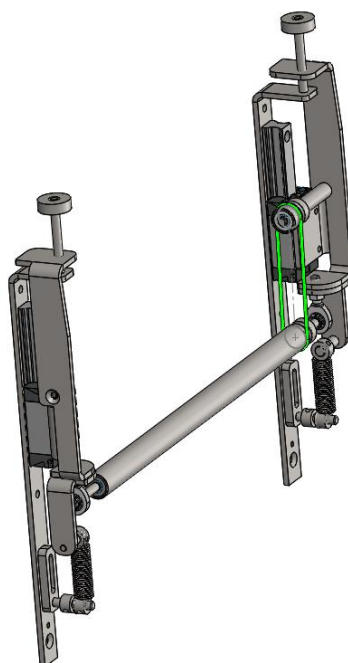
COMP NO.	CODE	EXPLANATION	QTY.
1	22012388	CAM ARM	6
2	22011304	BEARING HOLDER	12
3	22012389	CAM ARM PIN	6
4	23010615	SPRING CONNECTION SHEET	12
5	22013176	FOLDING DRUM SIX CLAMP	1
6	30023076	SS6014 ZZ BEARING	5
7	22012378	SHAFT OF FOLDING DRUM	1
8	-----	MOVABLE CLAMP ASSEMBLY	12
9	21090504	STABLE CLAMP	12
10	21090514	CONNECTION PLATE	6
11	30014636	SPRING	12
12	30004835	SS6004 ZZ BEARING	24
13	22011301	SPRING CONNECTION PLATE	12

KNIFE MODULE



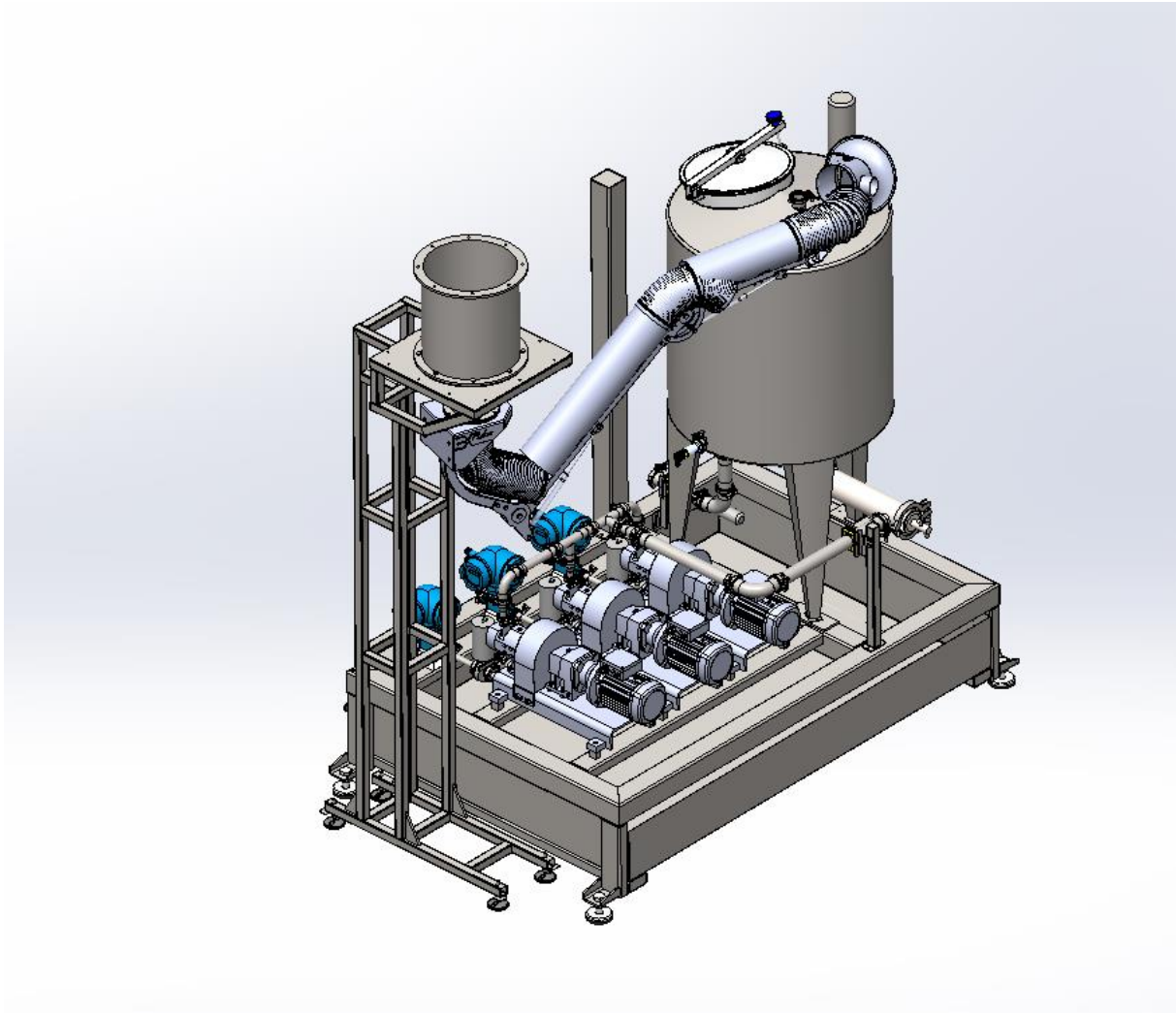
PART NO.	CODE	EXPLANATION	QTY
1	30017123	22216 E BEARING	4
2	23011415	LID OF BEARING HOUSE	1
3	22012402	BEARING HOUSE	2
4	22018695	VACUUM CONNECTOR PART	1
5	23017050	VACUUM CONNECTOR HOLDER PLATE	1
6	22015567	REAR CONNECTION PLATE	1
7	23011417	LID OF BEARING HOUSE	1
8	22012403	BEARING HOUSE	2
9	21090627	LOWER CUTTER ANVIL CHOCK	3
10	23011419	NAIL HOLDER SHEET	3
11	22012411	NAIL HOLDER PLATE	3
12	22013406	NAIL MATERIAL (2MM)	3
13	21090590	CHOCK OF CYLINDER	3
14	22018516	LOWER CONNECTION PLATE	1
15	21090653	LOWER CUTTER	1
16	23016783	LOWER CUTTER SENSOR SHEET	1
17	21016414	LOWER CUTTER DELRIN GEAR HUB	2
18	21016275	COUNTER FLANGE HUB	1
19	21016279	COUNTER FLANGE	1
20	21016272	REAR CONNECTION PLATE	1
21	30023081	GEARBOX PEII 120-005 / SIEMENS 1FK7101-2AC71-1QA0-Z M03 APEX	1
22	21082792	UPPER CUTTER GEAR HUB	1
23	22012402	BEARING HOUSE	2
24	21016272	CONNECTION PLATE	1
25	23011416	BEARING HOUSE	1
26	21016325	UPPER CUTTER GEAR	1
27	21090652	FLEXI KNIFE	9
28	23016782	ENCLOSURE SHEET	1
29	23016781	ENCLOSURE SHEET-1	1
30	22018515	CONNECTION PLATE	1
31	22016330	MID CONNECTION PLATE	3
32	22017162	HRC42 KNIFE UPPER HOLDER	9
33	22017161	HRC42 KNIFE LOWER HOLDER	9
34	22016935	BOLT ADJUSTMENT PLATE	1
35	21090629	UPPER CUTTER	1

WRAP PREVENTION MODULE



COMP NO.	CODE	EXPLANATION	QTY
1	30013417	BOSCH REXROTH-RAIL	2
2	23016772	CARRIAGE CONNECTION SHEET	1
3	22015546	SPRING PLATE	2
4	30013416	BOSCH REXROTH-CARRIAGE	2
5	30018507	BULLEYE	2
6	23013871	SPRING SHEET	1
7	22015545	SPRING PIN	2
8	22015561	SPRING SPACER	4
9	30019300	REVERSE PRESSING SPRING	1
10	22015584	IDLE ROD	1
11	22017324	SPACER	2
12	23013872	SPRING SHEET	1
13	22015547	TENSIONER ROD	2
14	30011655	S608 2RS BEARING	4
15	23013874	CARRIAGE CONNECTION SHEET	1
16	22015583	IDLE ROD	1
17	30004432	BELT	1
18	20201317	ADJUSTMENT ROD	2
19	22013191	ADJUSTMENT BAKELITE	2
20	23013875	CONNECTION SHEET	2

WETTING INSTALLATION INTRODUCTION



WETTING INSTALLATION UNIT CONSISTS OF:

- 1- LOBE PUMP-3 PCS
- 2- PNEUMATIC VALVE-3 PCS
- 3- FLOWMETER-3 PCS
- 4- WETTING TANK-500 LT
- 5- ACROBAT ARM
- 6- FAN

PERIODIC MAINTENANCE INSTRUCTIONS

- Do general cleaning of the machine.
- Clean the tank and filter.
- When necessary, clean the installation but pushing CLEANING button.



Liquid is pumped to whole installation and nozzles and it is cleaned as long as the time value entered on the page of CLEANING TIME SETUP

- Check the tightness of bolts on stacking and cutting systems.
- While cleaning electric panel, use vacuum cleaner.
- Clean the ventilation fan on top of the electric panel.
- Check the tension on all belts and do their maintenance. If there are worn out belts, change them.



During the periodical maintenance of the belts on the machine, first blow air to get rid of dust or any small particules. Then wipe them with a moist cloth.

- Check all derlin gears. Change if there are any damaged ones.
- Check the body heat of redactors.
- Empty the drain water out of the conditioner.

PERIODIC MAINTENANCE INSTRUCTIONS

- Check the tightness of all tie bolts on sight.
- Take out all the belts on the machine and check the gears, pulleys and bearings. Turn the mechanisms by hand, check if there is any anomaly. Change the bearings of mechanisms that don't turn.
- Clean the optic sensors without taking out from their places.
- Check the lids of ventilation filters every month.
- Check the tension of conveyor belts and give maintenance.



Clean the conveyor belt with a cloth and acetone during periodical maintenance.

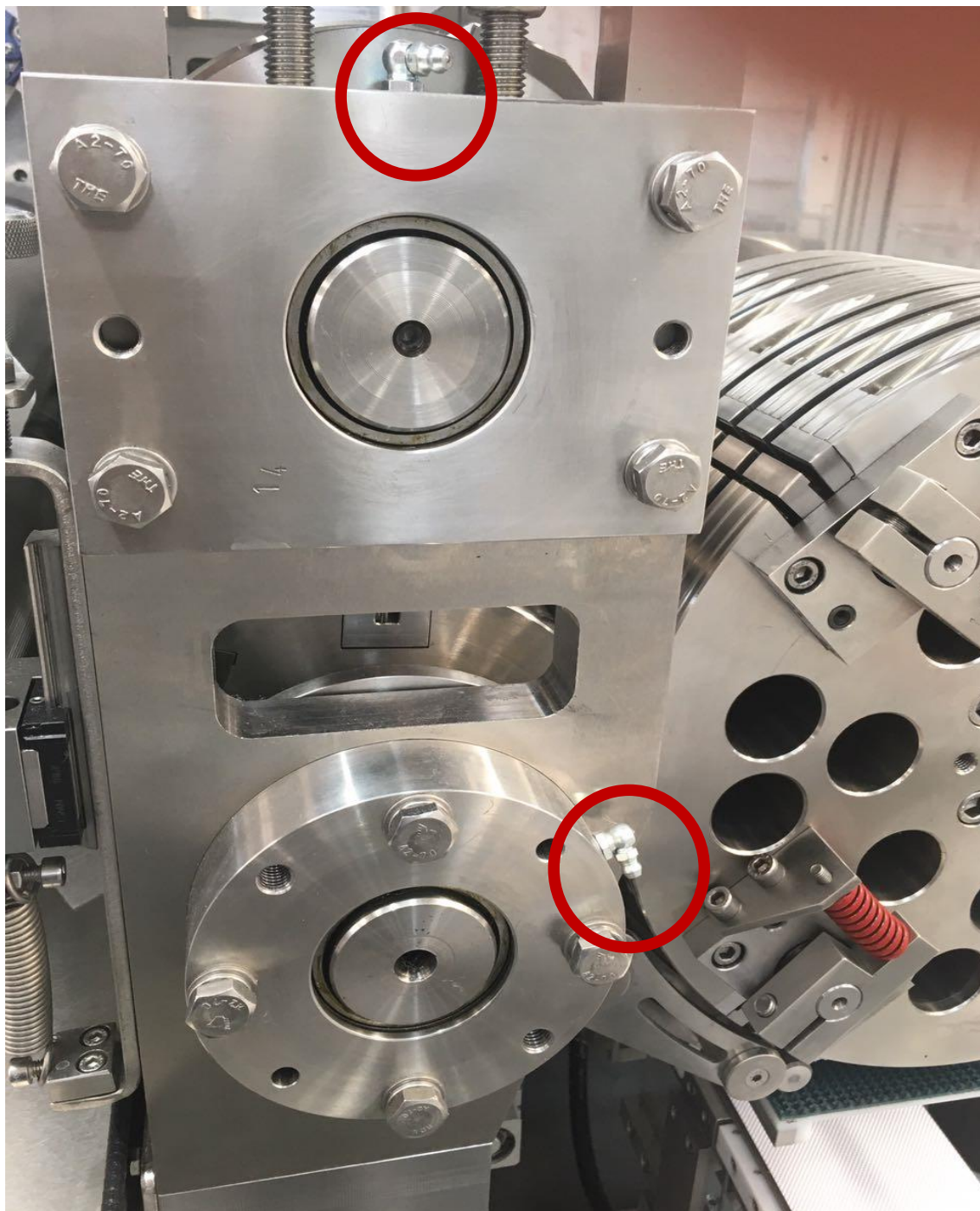
- Clean the rubber rolls with water and after they are well dried, apply powder.
- Check the oil of the conditioner.

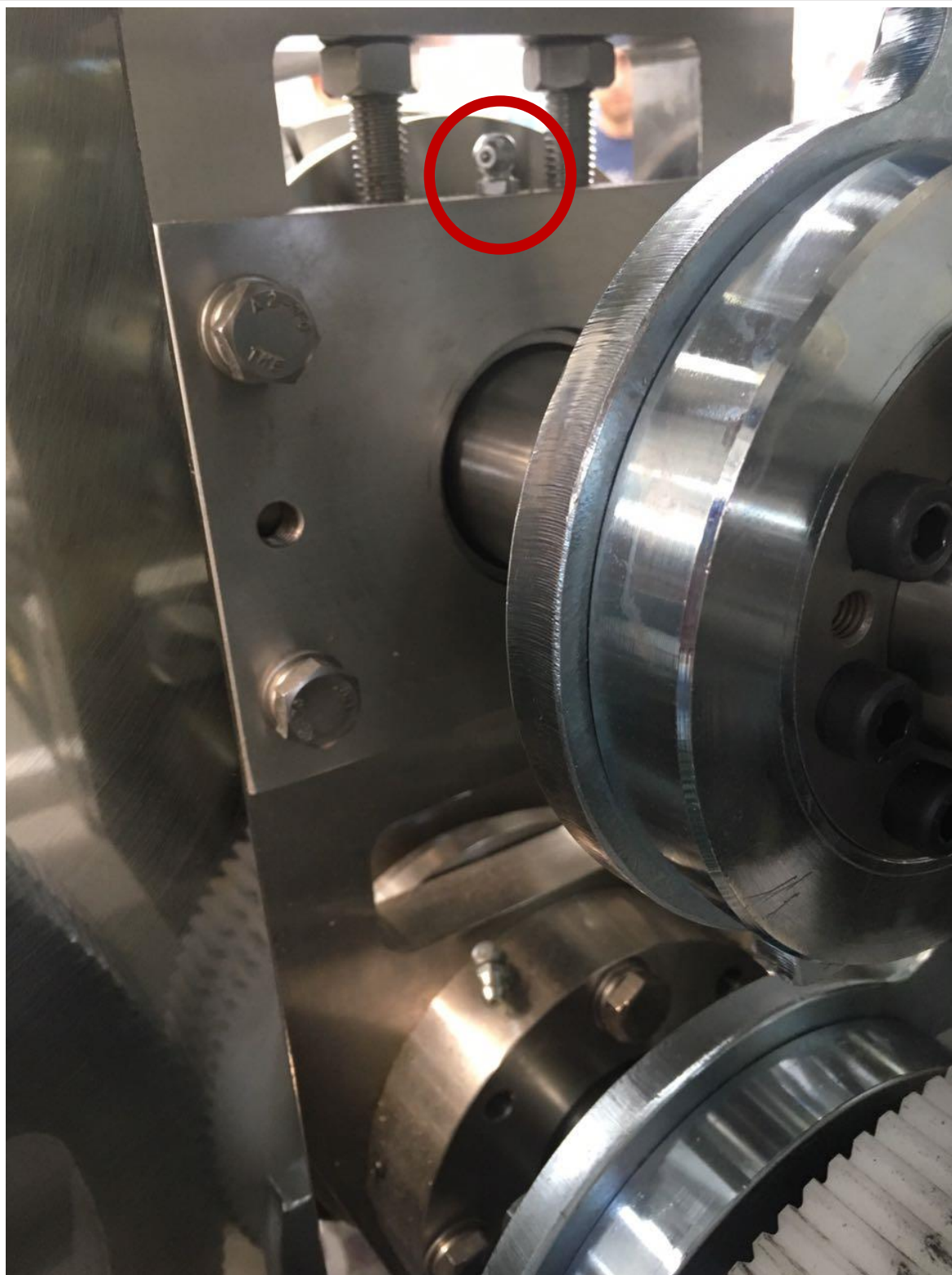


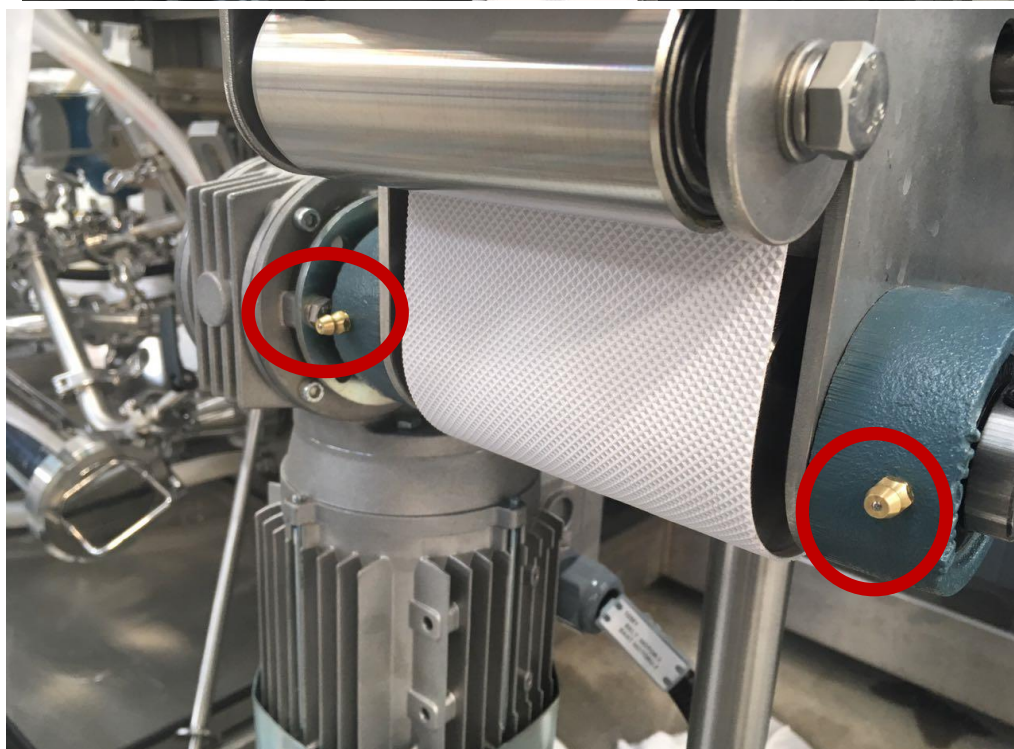
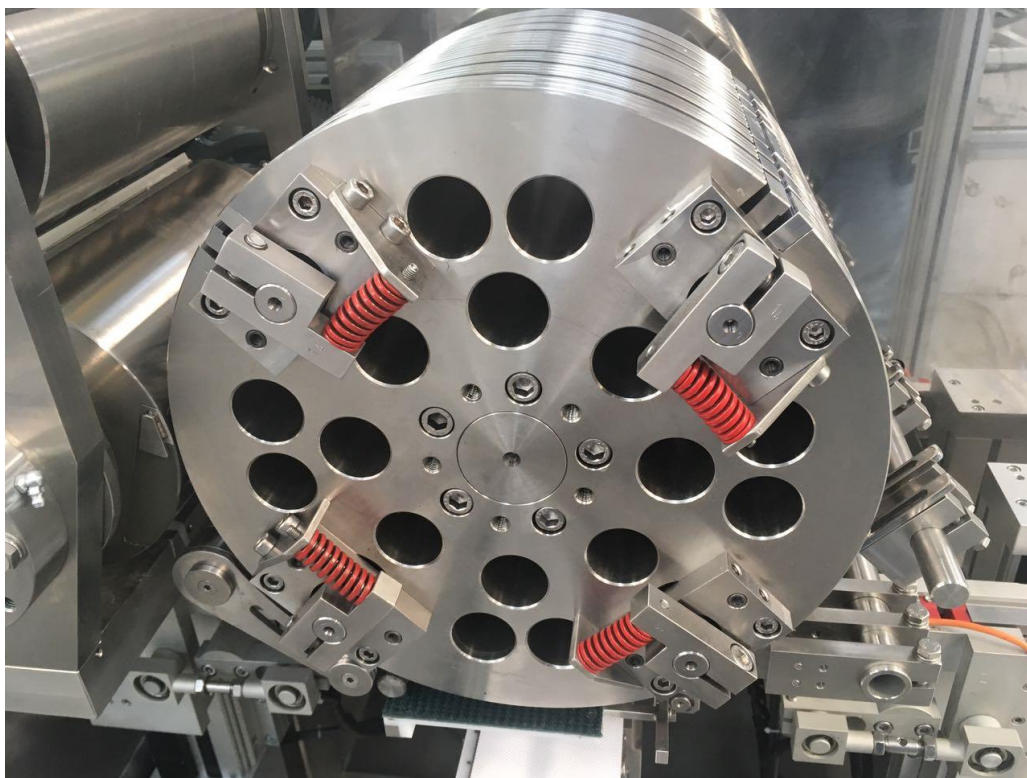
For the maintenance of the rest of parts, please read the user manuals of those parts that are sent to you.

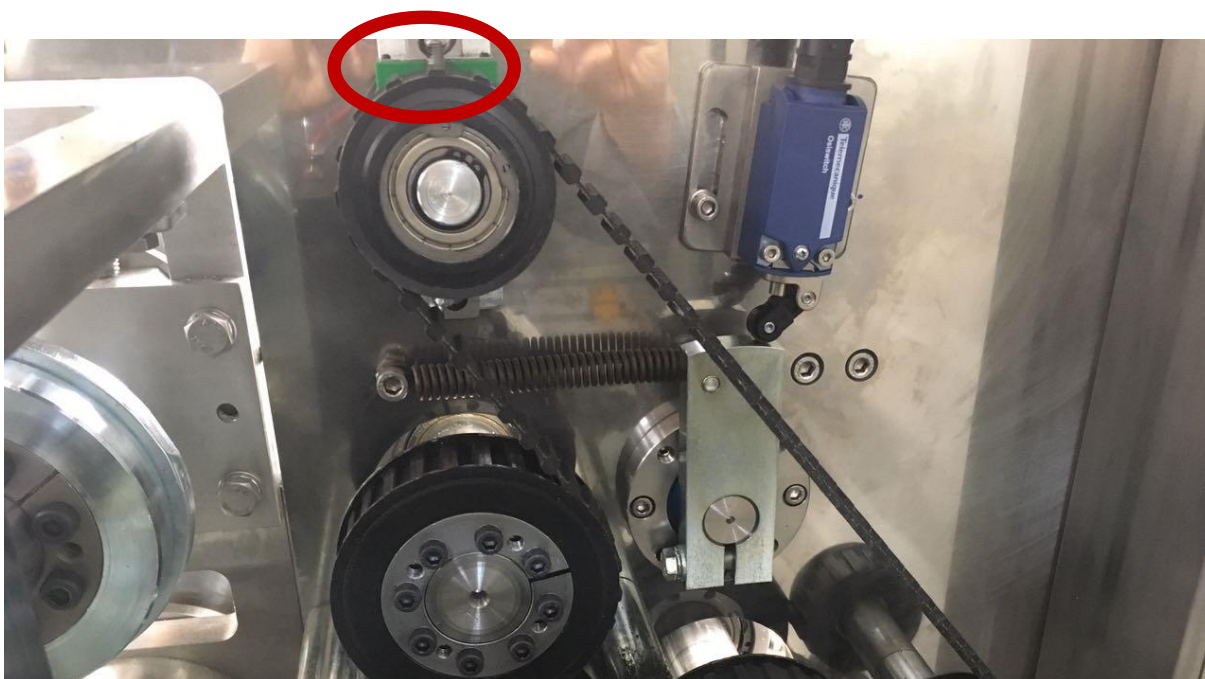
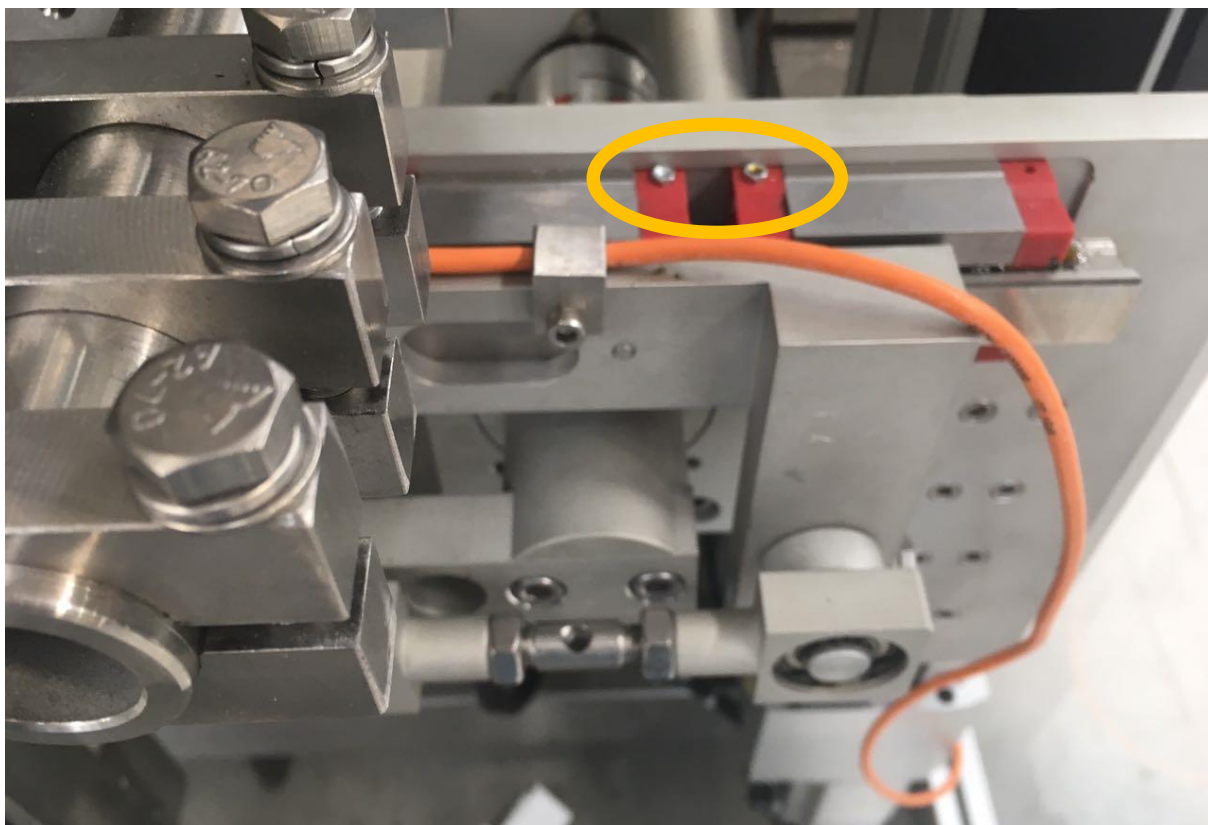
PERIODIC MAINTENANCE INSTRUCTIONS

- Turn the mechanisms by hand, check if there is any anomaly. Change the bearings of mechanisms that don't turn.
- Visually check the puller rolls.
- Grease all the linear sledges on the machine.
- Do general cleaning of the machine.
- While cleaning electric panel, use vacuum cleaner.
- Check the tightness of bolts on stacking and cutting systems.
- Clean the tank and the filter on the wetting installation.

MECHANIC GREASING AREAS









PRECAUTIONS REQUIRED TO TAKE BY OPERATOR



The operator must wear protective goggles before using the machine.



Accessories such as necklaces, rings, watches should not be worn.

It is necessary that the work clothes to be worn are not loose and that ID cards such as name badges can not be worn.

Non-slip footwear and rubber wristband clothes should be worn.



The operator shouldn't definitely use the machine with alcohol.



The operator should not operate the machine when the covers of the machine are open.

Safety switch and emergency stop connections and their locations must not be altered.



The operator must obey the general safety warnings on the machine and in the user manual.

PRECAUTIONS TO BE PERFORMED BEFORE MAINTENANCE

- The operator should take his own safety precautions.
- Accessories such as necklaces, rings, watches should not be worn.
- It is necessary that the work clothes to be worn are not loose and that ID cards such as name badges can not be worn.
- Non-slip footwear and rubber wristband clothes should be worn.
- Air and electricity must be cut to secure the machine.
- All mechanical adjustments must be recorded even if they change during maintenance.
- For maintaining, equipments, suitable for the area, must be selected and intervened.
- If a malfunction is detected during maintenance, contact the authorized personnel or the manufacturer.

PROBLEMS AND SOLUTIONS

<u>PROBLEMS</u>	<u>SOLUTIONS</u>
- In the line, the bobbin is getting wide at the backside during operation.	- Tighten the rear conveyor belt with the help of the belt tensioners. - Please close the distances between pulling rollers of the rear conveyors distances.
- There is excessive noise on the cutting unit.	- The papers are dry, wet them. - Cutting knife blunt, put on a spare blade or send it to whetting. - Extreme pressure on the cutting blade, pull it back.
- Cut is not complete or cutting quality is bad.	- Blade is blunt, please fit the spare part. - Move the cut blade forward.
- The strap that moves the cutting unit is loosening.	- The knife has difficulty in extreme pressure, pull back the knife. - Belt strain pulley loosened, tighten it.
- The size of the finished products is constantly different.	- The conveyor can not pull the product. Close the conveyor distance. - The strain is too much, please decrease it.
- Excessive solution on conveyors.	- Wetting rate is high, reduce the rate on the screen. - Wetting nozzles are not adjusted, correct nozzles. - Extreme pressure on pulling units, give some distances.

SAFETY DATASHEET FOR 70% ISOPROPHYL ALCOHOL-WATER
SOLUTION

Health	2
Fire	3
Reactivity	0
Personal Protection	E

Material Safety Data Sheet
Isopropyl Alcohol, 70% MSDS

Section 1: Chemical Product and Company Identification

Product Name: Isopropyl Alcohol, 70%

Catalog Codes: SLI1669

CAS#: Mixture.

RTECS: Not applicable.

TSCA: TSCA 8(b) inventory: Isopropyl alcohol; Water

CI#: Not available.

Synonym: 2-Propanol, 70%; Isopropanol, 70%; Isopropyl Rubbing Alcohol

Chemical Name: Not applicable.

Chemical Formula: Not applicable.

Contact Information:

Sciencelab.com, Inc.

14025 Smith Rd.

Houston, Texas 77396

US Sales: 1-800-901-7247

International Sales: 1-281-441-4400

Order Online: ScienceLab.comCHEMTREC (24HR Emergency Telephone), call:
1-800-424-9300

International CHEMTREC, call: 1-703-527-3887

For non-emergency assistance, call: 1-281-441-4400

Section 2: Composition and Information on Ingredients

Composition:

Name	CAS #	% by Weight
Isopropyl alcohol	67-63-0	70
Water	7732-18-5	30

Toxicological Data on Ingredients: Isopropyl alcohol: ORAL (LD50): Acute: 5045 mg/kg [Rat]. 3600 mg/kg [Mouse]. 6410 mg/kg [Rabbit]. DERMAL (LD50): Acute: 12800 mg/kg [Rabbit].

Section 3: Hazards Identification

Potential Acute Health Effects:

Hazardous in case of skin contact (irritant), of eye contact (irritant), of ingestion, . Slightly hazardous in case of skin contact (sensitizer, permeator). Non-corrosive for skin. Non-corrosive to the eyes. Non-corrosive for lungs.

Potential Chronic Health Effects:

CARCINOGENIC EFFECTS: Classified A4 (Not classifiable for human or animal.) by ACGIH, 3 (Not classifiable for human.) by IARC [Isopropyl alcohol]. MUTAGENIC EFFECTS: Not available. TERATOGENIC EFFECTS: Not available. DEVELOPMENTAL TOXICITY: Classified Reproductive system/toxin/female, Development toxin [POSSIBLE] [Isopropyl alcohol]. The substance may be toxic to kidneys, liver, skin, central nervous system (CNS). Repeated or prolonged exposure to the substance can produce target organs damage.

Section 4: First Aid Measures

Eye Contact:

Check for and remove any contact lenses. In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Cold water may be used. Get medical attention.

Skin Contact:

In case of contact, immediately flush skin with plenty of water. Cover the irritated skin with an emollient. Remove contaminated clothing and shoes. Cold water may be used. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention.

Serious Skin Contact:

Wash with a disinfectant soap and cover the contaminated skin with an anti-bacterial cream. Seek medical attention.

Inhalation:

If inhaled, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical attention if symptoms appear.

Serious Inhalation:

Evacuate the victim to a safe area as soon as possible. Loosen tight clothing such as a collar, tie, belt or waistband. If breathing is difficult, administer oxygen. If the victim is not breathing, perform mouth-to-mouth resuscitation. Seek medical attention.

Ingestion:

Do NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Loosen tight clothing such as a collar, tie, belt or waistband. Get medical attention if symptoms appear.

Serious Ingestion: Not available.

Section 5: Fire and Explosion Data

Flammability of the Product: Flammable.

Auto-Ignition Temperature: The lowest known value is 399°C (750.2°F) (Isopropyl alcohol).

Flash Points: CLOSED CUP: 18.3°C (64.9°F) - 24 deg. C (75 deg. F)

Flammable Limits: The greatest known range is LOWER: 2% UPPER: 12.7% (Isopropyl alcohol)

Products of Combustion: These products are carbon oxides (CO, CO₂).

Fire Hazards in Presence of Various Substances:

Highly flammable in presence of open flames and sparks, of heat. Flammable in presence of oxidizing materials. Non-flammable in presence of shocks

Explosion Hazards in Presence of Various Substances:

Slightly explosive in presence of open flames and sparks, of heat. Non-explosive in presence of shocks.

Fire Fighting Media and Instructions:

Flammable liquid, soluble or dispersed in water. SMALL FIRE: Use DRY chemical powder. LARGE FIRE: Use alcohol foam, water spray or fog.

Special Remarks on Fire Hazards:

Vapor may travel considerable distance to source of ignition and flash back. CAUTION: MAY BURN WITH NEAR INVISIBLE FLAME. Hydrogen peroxide sharply reduces the autoignition temperature of Isopropyl alcohol. After a delay, Isopropyl alcohol ignites on contact with dioxgenyl tetrafluoroborate, chromium trioxide, and potassium tert-butoxide. When heated to decomposition it emits acrid smoke and fumes. (Isopropyl alcohol)

Special Remarks on Explosion Hazards:

Secondary alcohols are readily autooxidized in contact with oxygen or air, forming ketones and hydrogen peroxide. It can become potentially explosive. It reacts with oxygen to form dangerously unstable peroxides which can concentrate and explode during distillation or evaporation. The presence of 2-butanone increases the reaction rate for peroxide formation. Explosive in the form of vapor when exposed to heat or flame. May form explosive mixtures with air. Isopropyl alcohol +

phosgene forms isopropyl chloroformate and hydrogen chloride. In the presence of iron salts, thermal decomposition can occur, which in some cases can become explosive. A homogeneous mixture of concentrated peroxides + isopropyl alcohol are capable of detonation by shock or heat. Barium perchlorate + isopropyl alcohol gives the highly explosive alkyl perchlorates. It forms explosive mixtures with trinitromethane and hydrogen peroxide. It produces a violent explosive reaction when heated with aluminum isopropoxide + crotonaldehyde. Mixtures of isopropyl alcohol + nitroform are explosive. (Isopropyl alcohol)

Section 6: Accidental Release Measures

Small Spill:

Dilute with water and mop up, or absorb with an inert dry material and place in an appropriate waste disposal container.

Large Spill:

Flammable liquid. Keep away from heat. Keep away from sources of ignition. Stop leak if without risk. Absorb with DRY earth, sand or other non-combustible material. Do not touch spilled material. Prevent entry into sewers, basements or confined areas; dike if needed. Be careful that the product is not present at a concentration level above TLV. Check TLV on the MSDS and with local authorities.

Section 7: Handling and Storage

Precautions:

Keep away from heat. Keep away from sources of ignition. Ground all equipment containing material. Do not ingest. Do not breathe gas/fumes/ vapor/spray. Wear suitable protective clothing. In case of insufficient ventilation, wear suitable respiratory equipment. If ingested, seek medical advice immediately and show the container or the label. Avoid contact with skin and eyes. Keep away from incompatibles such as oxidizing agents, acids.

Storage:

Store in a segregated and approved area. Keep container in a cool, well-ventilated area. Keep container tightly closed and sealed until ready for use. Avoid all possible sources of ignition (spark or flame).

Section 8: Exposure Controls/Personal Protection

Engineering Controls:

Provide exhaust ventilation or other engineering controls to keep the airborne concentrations of vapors below their respective threshold limit value. Ensure that eyewash stations and safety showers are proximal to the work-station location.

Personal Protection:

Safety glasses. Lab coat. Dust respirator. Be sure to use an approved/certified respirator or equivalent. Gloves (impervious).

Personal Protection in Case of a Large Spill:

Splash goggles. Full suit. Dust respirator. Boots. Gloves. A self contained breathing apparatus should be used to avoid inhalation of the product. Suggested protective clothing might not be sufficient; consult a specialist BEFORE handling this product.

Exposure Limits:

Isopropyl alcohol TWA: 983 STEL: 1230 (mg/m³) [Australia] TWA: 200 STEL: 400 (ppm) from ACGIH (TLV) [United States] [1999] TWA: 980 STEL: 1225 (mg/m³) from NIOSH TWA: 400 STEL: 500 (ppm) from NIOSH TWA: 400 STEL: 500 (ppm) [United Kingdom (UK)] TWA: 999 STEL: 1259 (mg/m³) [United Kingdom (UK)] TWA: 400 STEL: 500 (ppm) from OSHA (PEL) [United States] TWA: 980 STEL: 1225 (mg/m³) from OSHA (PEL) [United States] Consult local authorities for acceptable exposure limits.

Section 9: Physical and Chemical Properties

Physical state and appearance: Liquid.

Odor: Alcohol like.

Taste: Not available.

Molecular Weight: Not applicable.

Color: Clear Colorless.

pH (1% soln/water): Neutral.

Boiling Point: The lowest known value is 82.5°C (180.5°F) (Isopropyl alcohol). Weighted average: 87.75°C (189.9°F)

Melting Point: May start to solidify at -88.5°C (-127.3°F) based on data for: Isopropyl alcohol.

Critical Temperature: The lowest known value is 235°C (455°F) (Isopropyl alcohol).

Specific Gravity: Weighted average: 0.84 (Water = 1)

Vapor Pressure: The highest known value is 4.4 kPa (@ 20°C) (Isopropyl alcohol). Weighted average: 3.77 kPa (@ 20°C)

Vapor Density: The highest known value is 2.07 (Air = 1) (Isopropyl alcohol). Weighted average: 1.63 (Air = 1)

Volatility: Not available.

Odor Threshold: The highest known value is 22 ppm (Isopropyl alcohol)

Water/Oil Dist. Coeff.: The product is equally soluble in oil and water.

Ionicity (in Water): Not available.

Dispersion Properties: See solubility in water, methanol, diethyl ether, n-octanol, acetone.

Solubility: Easily soluble in cold water, hot water, methanol, diethyl ether, n-octanol, acetone.

Section 10: Stability and Reactivity Data

Stability: The product is stable.

Instability Temperature: Not available.

Conditions of Instability: Heat, flame, ignition sources, incompatible materials

Incompatibility with various substances: Reactive with oxidizing agents, acids, alkalis.

Corrosivity: Non-corrosive in presence of glass.

Special Remarks on Reactivity:

Reacts violently with hydrogen + palladium combination, nitroform, oleum, COCl₂, aluminum triisopropoxide, oxidants
Incompatible with acetaldehyde, chlorine, ethylene oxide, isocyanates, acids, alkaline earth, alkali metals, caustics, amines, crotonaldehyde, phosgene, ammonia. Isopropyl alcohol reacts with metallic aluminum at high temperatures. Isopropyl alcohol attacks some plastics, rubber, and coatings. Vigorous reaction with sodium dichromate + sulfuric acid. (Isopropyl alcohol)

Special Remarks on Corrosivity: Not available.

Polymerization: Will not occur.

Section 11: Toxicological Information

Routes of Entry: Absorbed through skin. Eye contact. Inhalation.

Toxicity to Animals:

Acute oral toxicity (LD50): 5143 mg/kg (Mouse) (Calculated value for the mixture). Acute dermal toxicity (LD50): 18286 mg/kg (Rabbit) (Calculated value for the mixture).

Chronic Effects on Humans:

CARCINOGENIC EFFECTS: Classified A4 (Not classifiable for human or animal.) by ACGIH, 3 (Not classifiable for human.) by IARC [Isopropyl alcohol]. DEVELOPMENTAL TOXICITY: Classified Reproductive system/toxin/female, Development toxin [POSSIBLE] [Isopropyl alcohol]. Contains material which may cause damage to the following organs: kidneys, liver, skin, central nervous system (CNS).

Other Toxic Effects on Humans:

Hazardous in case of skin contact (irritant), of ingestion, of inhalation. Slightly hazardous in case of skin contact (sensitizer, permeator).

Special Remarks on Toxicity to Animals: Not available.

Special Remarks on Chronic Effects on Humans:

May cause adverse reproductive/teratogenic effects (fertility, fetotoxicity, developmental abnormalities (developmental toxin)) based on animal studies. Detected in maternal milk in human. (Isopropyl alcohol)

Special Remarks on other Toxic Effects on Humans:

Acute Potential Health Effects: Skin: May cause mild skin irritation, and sensitization. Eyes: Can cause eye irritation.

Inhalation: Breathing in small amounts of this material during normal handling is not likely to cause harmful effects. However, breathing large amounts may be harmful and may affect the respiratory system and mucous membranes (irritation), behavior and brain (Central nervous system depression - headache, dizziness, drowsiness, stupor, incoordination, unconsciousness, coma and possible death), peripheral nerve and sensation, blood, urinary system, and liver. Ingestion: Swallowing small amounts during normal handling is not likely to cause harmful effects. Swallowing large amounts may be harmful. Swallowing large amounts may cause gastrointestinal tract irritation with nausea, vomiting and diarrhea, abdominal pain. It also may affect the urinary system, cardiovascular system, sense

Section 12: Ecological Information

Ecotoxicity: Not available.

BOD5 and COD: Not available.

Products of Biodegradation:

Possibly hazardous short term degradation products are not likely. However, long term degradation products may arise.

Toxicity of the Products of Biodegradation: The product itself and its products of degradation are not toxic.

Special Remarks on the Products of Biodegradation: Not available.

Section 13: Disposal Considerations**Waste Disposal:**

Waste must be disposed of in accordance with federal, state and local environmental control regulations.

Section 14: Transport Information

DOT Classification: CLASS 3: Flammable liquid.

Identification: : Isopropanol, solution (Isopropyl alcohol) UNNA: 1219 PG: II

Special Provisions for Transport: Not available.

Section 15: Other Regulatory Information**Federal and State Regulations:**

Connecticut hazardous material survey.: Isopropyl alcohol Illinois toxic substances disclosure to employee act: Isopropyl alcohol Rhode Island RTK hazardous substances: Isopropyl alcohol Pennsylvania RTK: Isopropyl alcohol Florida: Isopropyl alcohol Minnesota: Isopropyl alcohol Massachusetts RTK: Isopropyl alcohol New Jersey: Isopropyl alcohol New Jersey spill list: Isopropyl alcohol TSCA 8(b) inventory: Isopropyl alcohol; Water TSCA 4(a) final testing order: Isopropyl alcohol TSCA 8(a) IUR: Isopropyl alcohol TSCA 8(d) H and S data reporting: Isopropyl alcohol: Effective date: 12/15/86 Sunset Date: 12/15/96 TSCA 12(b) one time export: Isopropyl alcohol SARA 313 toxic chemical notification and release reporting: Isopropyl alcohol 70%

Other Regulations: OSHA: Hazardous by definition of Hazard Communication Standard (29 CFR 1910.1200).

Other Classifications:

WHMIS (Canada):

CLASS B-2: Flammable liquid with a flash point lower than 37.8°C (100°F). CLASS D-2B: Material causing other toxic effects (TOXIC).

DSCL (EEC):

R11- Highly flammable. R36- Irritating to eyes. S2- Keep out of the reach of children. S46- If swallowed, seek medical advice immediately and show this container or label.

HMIS (U.S.A.):

Health Hazard: 2

Fire Hazard: 3

Reactivity: 0

Personal Protection: E

National Fire Protection Association (U.S.A.):

Health: 1

Flammability: 3

Reactivity: 0

Specific hazard:

Protective Equipment:

Gloves (impervious). Lab coat. Dust respirator. Be sure to use an approved/certified respirator or equivalent. Wear appropriate respirator when ventilation is inadequate. Safety glasses.

Section 16: Other Information

References: Not available.

Other Special Considerations: Not available.

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